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OPTICS

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Diffraction 10

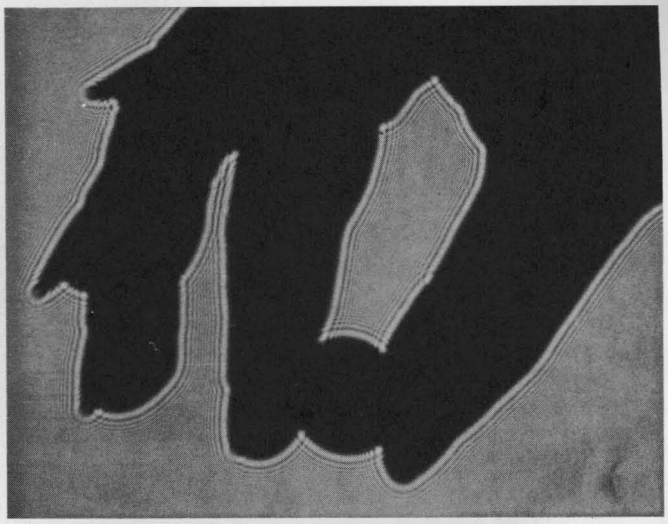


Fig. 10.1 The shadow of a hand holding a dime, cast directly on 4 × 5 Polaroid A.S.A. 3000 film using a He–Ne beam and no lenses. [Photo by E.H.]

10.1 PRELIMINARY CONSIDERATIONS

An opaque body placed midway between a screen and a point source casts an intricate shadow made up of bright and dark regions quite unlike anything one might expect from the tenets of geometrical optics (Fig. 10.1).* The work of Francesco Grimaldi in the sixteen hundreds was the first published detailed study of this *deviation of light from rectilinear propagation*, something he called *"diffractio"*. *The effect is a general characteristic of wave phenomena occurring whenever a portion of a wavefront, be it sound, a matter wave or light, is obstructed in some way*. If in the course of encountering an obstacle, either transparent or opaque, a region of the wavefront is altered in amplitude or phase, diffraction will occur.† The various segments of the wavefront which propagate beyond the obstacle interfere to cause the particular energy-density distribution referred to as the diffraction pattern. There is no significant physical distinction between *interference* and *diffraction*. It has however become somewhat customary, if not always appropriate, to speak of interference when considering the superposition of only a few waves and diffraction when treating a large number of waves. Even so, one refers to multiple-beam interference in one context and diffraction from a grating in another.

We might mention parenthetically that the wave theory, although the most natural, is not the only means for dealing with certain diffraction phenomena. For example, diffraction from a grating (Section 10.2.7) can be analyzed using a corpuscular quantum approach.* For our purposes however, the classical wave theory which provides the simplest effective formalism will more than suffice throughout this chapter.

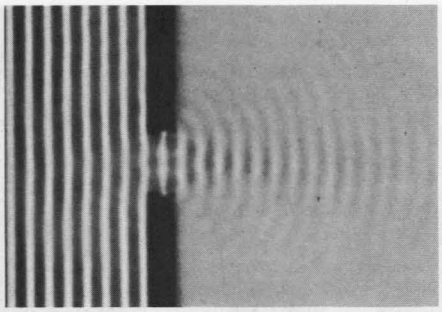
It should be emphasized that optical instruments make use of only a portion of the complete incident wavefront. Diffraction effects are accordingly of great significance in the detailed understanding of devices containing lenses, stops, source slits, mirrors, etc. If all defects in a lens system were removed, the ultimate sharpness of an image would be limited by diffraction.

As an initial approach to the problem, let's reconsider Huygens' principle (Section 4.2.1). Accordingly, each point on a wavefront can be envisaged as a source of secondary spherical wavelets. The progress through space of the wavefront or any portion thereof can then presumably be determined. At any particular time, the shape of the wave-

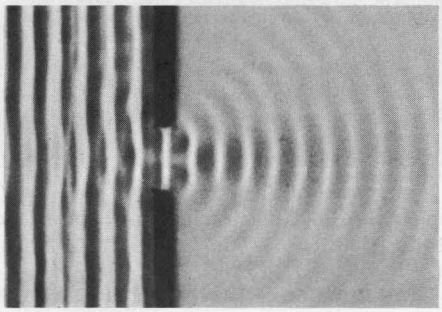
* The effect is easily seen but you need a fairly strong source. A high-intensity lamp shining through a small hole works well. If you look at the shadow pattern arising from a pencil under point-source illumination, you will see an unusual bright region bordering the edge and even a faintly illuminated band down the middle of the shadow. Take a close look at the shadow cast by your hand in direct sunlight.

† Diffraction associated with transparent obstacles is not usually considered, although if you have ever driven an automobile at night with a few rain droplets on your eye glasses you are no doubt quite familiar with the effect. If you have not, put a droplet of water or saliva on a glassplate, hold it very close to your eye, and look directly through it at a point source. You'll see bright and dark fringes.

* W. Duane, *Proc. Nat. Acad. Sci.* **9**, 158 (1923).



(a)



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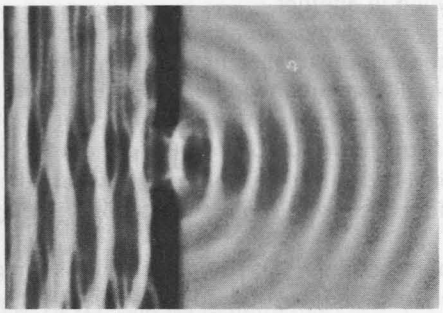


(c)

Fig. 10.2 Diffraction through an aperture with varying λ as seen in a ripple tank. [Photo courtesy PSSC *Physics*, D. C. Heath, Boston, 1960.]

front is supposed to be the envelope of the secondary wavelets (Fig. 4.3). The technique however ignores most of each secondary wavelet, retaining only that portion common to the envelope. As a result of this inadequacy, Huygens' principle is unable to account for the diffraction process. That this is indeed the case is borne out by everyday experience. Sound waves (e.g. $\nu = 500$ Hz, $\lambda \approx 68$ cm) easily "bend" around large objects like telephone poles and

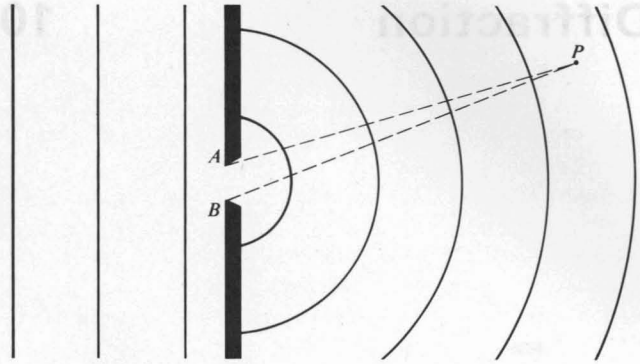


Fig. 10.3 Diffraction at a small aperture.

trees which on the contrary cast fairly distinct shadows when illuminated by light. Yet Huygens' principle is independent of any wavelength considerations and would predict the same wavefront configurations in both situations. The difficulty was resolved by Fresnel with his addition of the concept of interference. The corresponding *Huygens–Fresnel* principle states that *every unobstructed point of a wavefront, at a given instant in time, serves as a source of spherical secondary wavelets (of the same frequency as the primary wave). The amplitude of the optical field at any point beyond is the superposition of all of these wavelets (considering their amplitudes and relative phases)*. Applying these ideas on the very simplest qualitative level, refer to the ripple tank photographs of Fig. 10.2 and the illustration in Fig. 10.3. If each unobstructed point on the incoming plane wave acts as a coherent secondary source, the maximum optical path-length difference amongst these will be $\Lambda_{\max} = |\overline{AP} - \overline{BP}|$, corresponding to a source point at each edge of the aperture. But $\Lambda_{\max}$ is less than or equal to $\overline{AB}$, the latter being the case when P is on the screen. When $\lambda \gg \overline{AB}$, as in Fig. 10.3, it follows that $\lambda \gg \Lambda_{\max}$ and since the waves were initially in phase, they must all interfere constructively (to varying degrees) wherever P happens to be [see Fig. 10.2(c)]. The antithetic situation occurs when $\lambda \ll \overline{AB}$ as in Fig. 10.2(a). Now the area where $\lambda \gg \Lambda_{\max}$ is limited to a small region extending out directly in front of the aperture and it is only there that all of the wavelets will interfere constructively. Beyond this zone some of the wavelets can interfere destructively and the "shadow" begins. Keep in mind that the idealized *geometrical shadow* corresponds to $\lambda \to 0$.

The Huygens–Fresnel principle has some shortcomings (which we will examine later) in addition to the fact that the whole thing at this point is rather hypothetical. Gustav Kirchhoff developed a more rigorous theory based directly on the solution of the differential wave equation. Kirchhoff, although a contemporary of Maxwell, did his work prior to Hertz's demonstration (and the resulting popularization) of the propagation of electromagnetic waves in 1887. Accordingly, Kirchhoff employed the older elastic-solid theory of light. His refined analysis lent credence to the assumptions of Fresnel and led to an even more precise formulation of Huygens' principle as an exact consequence of the wave equation. Even so, the Kirchhoff theory is itself an approximation which is valid for sufficiently small wavelengths, i.e. when the diffracting apertures have dimensions which are large in comparison to λ. The difficulty arises from the fact that we require the solution of a partial differential equation which meets the boundary conditions imposed by the obstruction. This kind of rigorous solution is only obtainable in a very few special cases. Kirchhoff's theory however works fairly well even though it deals only with scalar waves and is insensitive to the fact that light is a transverse vector field.*

It should be stressed that the problem of determining an exact solution for a particular diffracting configuration is amongst the most troublesome to be dealt with in optics. The first such solution, utilizing the electromagnetic theory of light, was published by Arnold Johannes Wilhelm Sommerfeld (1868–1951) in 1896. But there the problem was physically somewhat unrealistic in that it involved an infinitely thin yet opaque, perfectly conducting, plane screen. The result was nonetheless extremely valuable, providing a good deal of insight into the fundamental processes involved.

Rigorous solutions of this sort do not exist even today for many of the configurations of practical interest. We will therefore, out of necessity, rely on the approximate treatments of Huygens–Fresnel and Kirchhoff. In recent times, microwave techniques have been employed to conveniently study features of the diffraction field which might otherwise be almost impossible to examine optically. The Kirchhoff theory has held up remarkably well under this kind of scrutiny.* In many cases, the simpler Huygens–Fresnel treatment will prove quite adequate for our purposes.

10.1.1 Opaque Obstructions

Diffraction may be envisioned as arising from the interaction of electromagnetic waves with some sort of physical obstruction. We would do well therefore to briefly re-examine the processes involved, i.e. what actually takes place within the material of the opaque object?

One possible description is that a screen may be considered to be a continuum, i.e. its microscopic structure may be neglected. For a nonabsorbing metal sheet (no joule heating, therefore infinite conductivity) we can write Maxwell's equations in the metal, and in the surrounding medium, and then match the two at the boundaries. Precise solutions can thus be obtained for very simple configurations (A. Sommerfeld). The reflected and diffracted waves then result from the current distribution within the sheet.

Now, examining the screen on a submicroscopic scale, imagine the electron cloud of each atom set into vibration by the electric field of the incident radiation. The classical model, which speaks of electron-oscillators vibrating and reemitting at the source frequency (Section 3.3.1) serves quite well so that we need not be concerned with the quantum-mechanical description. The amplitude and phase of a particular oscillator within the screen is determined by the local electric field surrounding it. This in turn is a superposition of the incident field and the fields of all of the other vibrating electrons. A large opaque screen with no apertures, be it made of black paper or aluminum foil, has one obvious effect: there is no optical field in the region beyond it. Electrons near the illuminated surface are driven into oscillation by the impinging light. They emit radiant energy which is ultimately reflected backward, or absorbed by the material in the form of heat, or both. In any case, the incident primary wave and the electron-oscillator fields superimpose in such a way as to yield zero light at any point beyond the screen. This might seem a remarkably special balance, but it actually is not. If the primary wave were not canceled completely it would propagate deeper into the material of the screen, exciting more electrons to radiate. This in turn would further weaken the primary wave until it ultimately vanished (if the screen were thick enough). Even an opaque material like silver, in the form of a sufficiently thin sheet, is transparent (recall the half-silvered mirror).

* A vectorial formulation of the scalar Kirchhoff theory is discussed in J. D. Jackson, *Classical Electrodynamics*, p. 283. Also see Sommerfeld, *Optics*, p. 325. You might as well take a look at B. B. Baker and E. T. Copson, *The Mathematical Theory of Huygens' Principle*, as a general reference to diffraction. None of these texts is easy reading.

* C. L. Andrews, *Am. J. Phys.* **19**, 250 (1951); S. Silver, *J. Opt. Soc. Am.* **52**, 131 (1962).

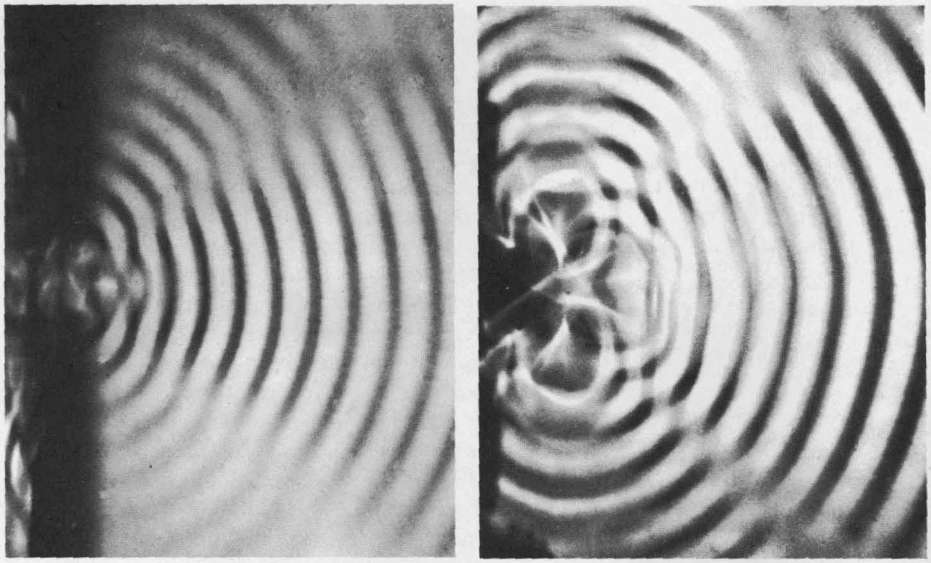


Fig. 10.4 Ripple-tank photos. In one case the waves are simply diffracted by a slit, in the other a series of equally spaced point sources span the aperture and generate a similar pattern. [Photos courtesy PSSC *Physics*, D.C. Heath, Boston, 1960.]

Now, remove a small disk-shaped segment from the center of the screen so that light streams through the aperture. The oscillators which uniformly cover it are removed along with the disk and so the remaining electrons within the screen are no longer affected by them. As a first and certainly approximate approach, *assume that the mutual interaction of the oscillators is essentially negligible*, i.e. the electrons in the screen are completely unaffected by the removal of the electrons in the disk. The field in the region beyond the aperture will then be that which existed prior to the removal of the disk, namely zero, minus the contribution from the disk alone. Except for the sign, it is as if the source and screen were taken away leaving only the oscillators on the disk, rather than vice versa. In other words, the diffraction field, in this approximation, can be pictured as arising exclusively from a set of fictitious noninteracting oscillators distributed uniformly over the region of the aperture. This, of course, is the essence of the Huygens–Fresnel principle.

We can expect, however, that rather than there being no interaction at all between electron-oscillators, there is a short-range effect, since the oscillator fields drop off with distance. In this physically more realistic view, the electrons within the vicinity of the aperture's edge are affected when the disk is removed. For large apertures, the number of oscillators in the disk is very much greater than those along the edge. In such cases, if the point of observation is far away, and in the forward direction, the Huygens–Fresnel principle should, and does, work well (Fig. 10.4). For very small apertures, or at points of observation in the vicinity of the aperture, edge effects become important and we can anticipate difficulties. Indeed, at a point within the aperture itself, the electron-oscillators on the edge are of the greatest significance because of their proximity. Yet these electrons were certainly not unaffected by the removal of the adjacent oscillators of the disk. Thus, the deviation from the Huygens–Fresnel principle should be appreciable.

10.1.2 Fraunhofer and Fresnel Diffraction

Imagine that we have an opaque shield, Σ, containing a single small aperture which is being illuminated by plane waves from a very distant point source, S. The plane of observation σ is a screen parallel with, and very close to, Σ. Under these conditions an image of the aperture is projected onto the screen which is clearly recognizable despite some slight fringing around its periphery. As the plane of observation moves further away from Σ, the image of the aperture, although still easily recognizable, becomes increasingly more structured as the fringes become more prominent. The phenomenon observed is known as *Fresnel* or *near-field* diffraction. Slowly moving the plane of observation out still further results in a continuous change in the fringes. At a very great distance from Σ the projected pattern will have spread out considerably, bearing little or no resemblance to the actual aperture. Thereafter moving σ essentially only

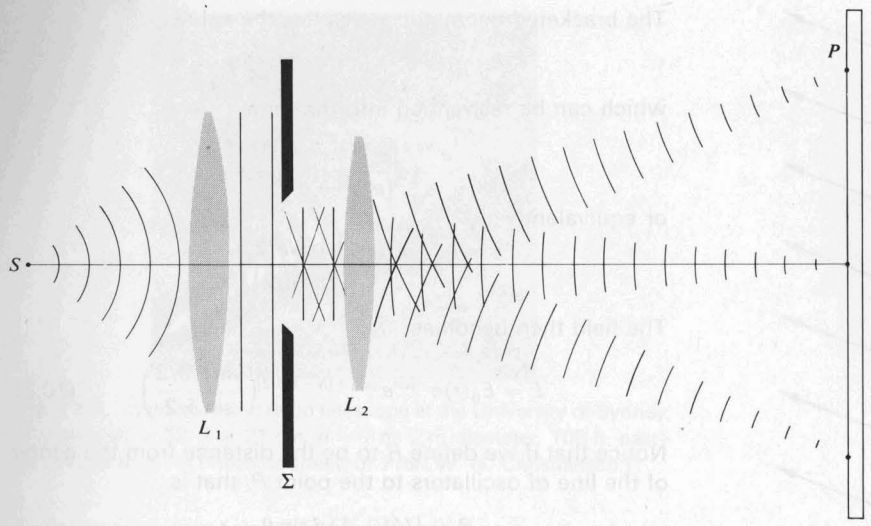


Fig. 10.5 Fraunhofer diffraction.

changes the size of the pattern and not its shape. This is *Fraunhofer* or *far-field* diffraction. If at that point we could sufficiently reduce the wavelength of the incoming radiation, the pattern would revert to the Fresnel case. If λ were decreased even more, so that it approached zero, the fringes would disappear and the image would take on the limiting shape of the aperture as predicted by geometrical optics. Returning to the original set up, if the point source were now moved toward Σ, spherical waves would impinge on the aperture and a Fresnel pattern would exist, even on a distant plane of observation.

In other words, consider a point source S, and a point of observation P, where both are very far from Σ and no lenses are present (Problem 10.1). As long as both the incoming and outgoing waves approach being planar (differing therefrom by a small fraction of a wavelength) over the extent of the diffracting apertures (or obstacles), Fraunhofer diffraction obtains. When S or P or both are too near Σ for the curvature of the wavefronts to be negligible, Fresnel diffraction prevails. The distinction will be made analytically more precise later when we consider the mathematical details.

A practical realization of the Fraunhofer condition, where both S and P are effectively at infinity, is achieved by using an equivalent arrangement to that of Fig. 10.5. The point source S is located at F_1, the principal focus of lens L_1, and the plane of observation is the second focal plane of L_2. In the terminology of geometrical optics the source plane and σ are conjugate planes.

These same ideas can be generalized to any lens system forming an image of an extended source or object (Problem 10.19).* Indeed, that image would be a Fraunhofer diffraction pattern. It is because of these important practical considerations, as well as its inherent simplicity, that we will examine Fraunhofer prior to Fresnel diffraction, even though it is a special case of the latter.

10.1.3 Several Coherent Oscillators

As a simple yet logical bridge between the studies of interference and diffraction, consider the arrangement of Fig. 10.6. The illustration depicts a linear array of N coherent point oscillators (or radiating antennas), which are each identical even to their polarization. For the moment, consider the oscillators to have no intrinsic phase difference, i.e. they each have the same epoch angle. The rays shown are all almost parallel, meeting at some very distant point P. If the spatial extent of the array is comparatively small, the separate wave amplitudes arriving at P will be essentially equal,

* A helium–neon laser can be set up to generate magnificent patterns without any auxiliary lenses but this needs plenty of space. If the plane of observation σ is very far from Σ, the diffraction pattern at some point P may be envisioned as arising from the superposition of plane waves coming from every point of the aperture and traveling in the direction of P. By using a large well-corrected lens (L_2) the diffracted waves can be focused on a nearby plane without substantially changing the shape of the pattern. Moreover, with the lens present the incident wave need not be planar so long as the optical path lengths for all rays, from the aperture to P, are essentially equal.

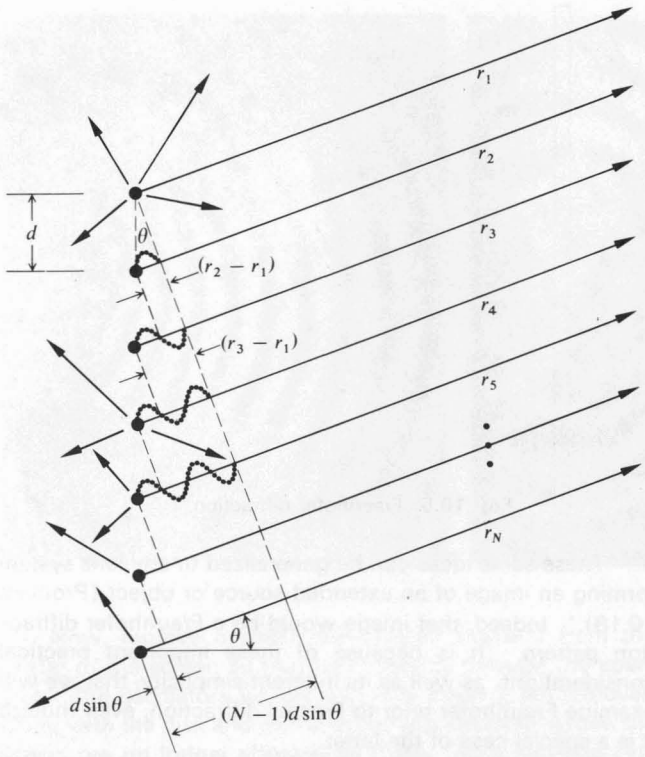


Fig. 10.6 A linear array of in-phase coherent oscillators. Note that at the angle shown $\delta = \pi$ while at $\theta = 0$ δ would be zero.

having traveled nearly equal distances, that is

$$E_0(r_1) = E_0(r_2) = \cdots = E_0(r_N) = E_0(r).$$

The sum of the interfering spherical wavelets yields an electric field at P, given by the real part of

$$E = E_0(r)e^{i(kr_1 - \omega t)} + E_0(r)e^{i(kr_2 - \omega t)} + \cdots + E_0(r)e^{i(kr_N - \omega t)}. \tag{10.1}$$

It should be clear, from Section 9.1, that we need not be concerned with the vector nature of the electric field for this configuration. Now then

$$E = E_0(r)e^{-i\omega t}e^{ikr_1}[1 + e^{ik(r_2 - r_1)} + e^{ik(r_3 - r_1)} + \cdots + e^{ik(r_N - r_1)}].$$

The phase difference between adjacent sources is obtained from the expression $\delta = k_0\Lambda$ and since $\Lambda = nd \sin \theta$, in a medium of index n, $\delta = kd \sin \theta$. Making use of Fig. 10.6, it follows that $\delta = k(r_2 - r_1)$, $2\delta = k(r_3 - r_1)$ etc. Thus the field at P may be written as

$$E = E_0(r)e^{-i\omega t}e^{ikr_1}[1 + (e^{i\delta}) + (e^{i\delta})^2 + (e^{i\delta})^3 + \cdots (e^{i\delta})^{N-1}]. \tag{10.2}$$

The bracketed geometric series has the value

$$(e^{i\delta N} - 1)/(e^{i\delta} - 1)$$

which can be rearranged into the form

$$\frac{e^{iN\delta/2}[e^{iN\delta/2} - e^{-iN\delta/2}]}{e^{i\delta/2}[e^{i\delta/2} - e^{-i\delta/2}]}$$

or equivalently

$$e^{i(N-1)\delta/2}\left[\frac{\sin N\delta/2}{\sin \delta/2}\right].$$

The field then becomes

$$E = E_0(r)e^{-i\omega t}e^{i[kr_1 + (N-1)\delta/2]}\left(\frac{\sin N\delta/2}{\sin \delta/2}\right). \tag{10.3}$$

Notice that if we define R to be the distance from the center of the line of oscillators to the point P, that is

$$R = \tfrac{1}{2}(N-1)d \sin \theta + r_1,$$

then Eq. (10.3) takes on the form

$$E = E_0(r)e^{i(kR - \omega t)}\left(\frac{\sin N\delta/2}{\sin \delta/2}\right). \tag{10.4}$$

Finally, then, the flux-density distribution within the diffraction pattern due to N coherent, identical, distant point sources in a linear array is proportional to $EE^*/2$ for complex E or

$$I = I_0\frac{\sin^2 (N\delta/2)}{\sin^2 (\delta/2)}, \tag{10.5}$$

where I_0 is the flux density from any single source arriving at P (see Problem 10.2 for a graphical derivation of the irradiance). For $N = 0$, $I = 0$, for $N = 1$, $I = I_0$, and for $N = 2$, $I = 4I_0 \cos^2 (\delta/2)$ in accord with Eq. (9.6). The functional dependence of I on θ is more apparent in the form

$$I = I_0\frac{\sin^2 [N(kd/2) \sin \theta]}{\sin^2 [(kd/2) \sin \theta]}. \tag{10.6}$$

The $\sin^2 [N(kd/2) \sin \theta]$ term undergoes rapid fluctuations, while the function modulating it, $\{\sin [(kd/2) \sin \theta]\}^{-2}$, varies relatively slowly. The combined expression gives rise to a series of sharp principal peaks separated by small subsidiary maxima. The principal maxima occur in directions θ_m such that $\delta = 2m\pi$ where $m = 0, \pm 1, \pm 2, \ldots$. Because $\delta = kd \sin \theta$

$$d \sin \theta_m = m\lambda. \tag{10.7}$$

Since $[\sin^2 N\delta/2]/[\sin^2 \delta/2] = N^2$ for $\delta = 2m\pi$ (from

Fig. 10.7 Interferometric radio telescope at the University of Sydney, Australia ($N = 32$, $\lambda = 21$ cm, $d = 7$ m, 2 m diameter, 700 ft. east–west base line). [Photo courtesy of Prof. W. N. Christiansen.]

L'Hôspital's rule) the principal maxima have values $N^2 I_0$. This is to be expected inasmuch as all of the oscillators are in phase at that orientation. The system will radiate a maximum in a direction perpendicular to the array ($m = 0$, $\theta_0 = 0$ and π). As θ increases, δ increases and I falls off to zero at $N\delta/2 = \pi$, its first minimum. Note that if $d < \lambda$ in Eq. (10.7), only the $m = 0$ or zero-order principal maximum exists. *If we were looking at an idealized line source of electron-oscillators separated by atomic distances, we could expect only that one principal maximum in the light field.*

The antenna array of Fig. 10.7 can then transmit radiation in the narrow beam or lobe corresponding to a principal maximum (the parabolic dishes shown reflect into the forward direction and the radiation pattern is no longer symmetrical around the common axis.) Suppose that we have a system in which we can introduce an intrinsic phase shift of ε between adjacent oscillators. In that case

$$\delta = kd \sin\theta + \varepsilon;$$

the various principal maxima will occur at new angles

$$d \sin\theta_m = m\lambda - \varepsilon/k.$$

Concentrating on the central maximum $m = 0$, its orientation θ_0 can be varied at will by merely adjusting the value of ε.

The principle of reversibility, which states that without absorption, wave motion is reversible, leads to the same field pattern for an antenna used as either a transmitter or receiver. The array, functioning as a radio telescope, can therefore be "pointed" by combining the output from the individual antennas with an appropriate phase shift, ε, introduced between each of them. For a given ε the output of the system corresponds to the signal impinging on the array from a specific direction in space.

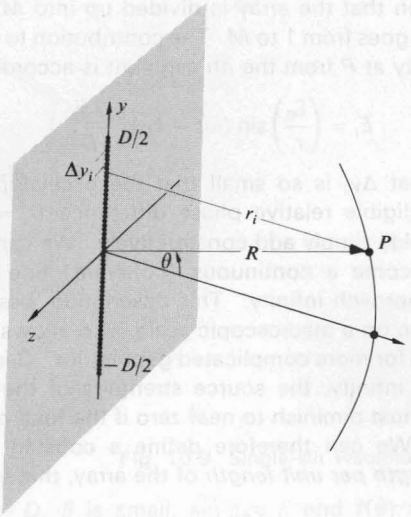


Fig. 10.8 A coherent line source.

Figure 10.7 is a photograph of the first multiple radio interferometer designed by W. N. Christiansen and built in Australia in 1951. It consists of 32 parabolic antennas, each 2 m in diameter, designed to function in phase at the wavelength of the 21 cm hydrogen emission line. The antennas are arranged along an east–west baseline with 7 m separating each one. This particular array utilizes the earth's rotation as the scanning mechanism.

Examine Fig. 10.8 which depicts an idealized line source of electron-oscillators (e.g., the secondary sources of the Huygens–Fresnel principle for a long slit whose width is much less than λ illuminated by plane waves). Each point emits a spherical wavelet which we write as

$$E = \left(\frac{\mathcal{E}_0}{r}\right) \sin(\omega t - kr)$$

explicitly indicating the inverse r-dependence of the amplitude. The quantity $\mathcal{E}_0$ is said to be the *source strength*. The present situation is distinct from that of Fig. 10.6 in that now the sources are very weak, their number, N, is tremendously large and the separation between them vanishingly small. A minute, but finite segment of the array Δy_i, will contain $\Delta y_i(N/D)$ sources where D is the entire length of the array.

Imagine then that the array is divided up into M such segments, i.e., i goes from 1 to M. The contribution to the electric field intensity at P from the ith segment is accordingly

$$E_i = \left(\frac{\mathcal{E}_0}{r_i}\right) \sin(\omega t - kr_i) \left(\frac{N\Delta y_i}{D}\right)$$

provided that Δy_i is so small that the oscillators within it have a negligible relative phase difference (r_i = constant) and their fields simply add constructively. We can cause the array to become a continuous (coherent) line source by letting N approach infinity. This description, besides being fairly realistic on a macroscopic scale, also allows the use of the calculus for more complicated geometries. Certainly as N approaches infinity, the source strengths of the individual oscillators must diminish to near zero if the total output is to be finite. We can therefore define a constant $\mathcal{E}_L$ as the *source strength per unit length* of the array, that is

$$\mathcal{E}_L \equiv \frac{1}{D} \lim_{N \to \infty} (\mathcal{E}_0 N). \tag{10.8}$$

The net field at P from all M segments is

$$E = \sum_{i=1}^{M} \frac{\mathcal{E}_L}{r_i} \sin(\omega t - kr_i)\, \Delta y_i.$$

For a continuous line source Δy_i can become infinitesimal ($M \to \infty$) and the summation is then transformed into a definite integral

$$E = \mathcal{E}_L \int_{-D/2}^{+D/2} \frac{\sin(\omega t - kr)}{r}\, dy, \tag{10.9}$$

where $r = r(y)$. The approximations used to evaluate Eq. (10.9) must depend on the position of P with respect to the array and will therefore make the distinction between Fraunhofer and Fresnel diffraction. The coherent *optical* line source does not now exist as a physical entity but we will make good use of it as a mathematical device.

10.2 FRAUNHOFER DIFFRACTION

10.2.1 The Single Slit

Return to Fig. 10.8 where now the point of observation is very distant from the coherent line source and $R \gg D$. Under these circumstances $r(y)$ never deviates appreciably from its midpoint value R so that the quantity $(\mathcal{E}_L/R)$ at P is essentially constant for all elements dy. It follows from Eq. (10.9) that the field at P due to the differential segment of the source dy is

$$dE = \frac{\mathcal{E}_L}{R} \sin(\omega t - kr)\, dy, \tag{10.10}$$

where $(\mathcal{E}_L/R)\, dy$ is the amplitude of the wave. Notice that the phase is very much more sensitive to variations in $r(y)$ than is the amplitude so that we will have to be more careful about introducing approximations into it. We can expand $r(y)$, in precisely the same manner as was done in Problem (9.4), to get it as an explicit function of y, thus

$$r = R - y \sin\theta + (y^2/2R) \cos^2\theta + \cdots, \tag{10.11}$$

where θ is measured from the xz-plane. The third term can be ignored so long as its contribution to the phase is insignificant even when $y = \pm D/2$, i.e. $(\pi D^2/4\lambda R) \cos^2\theta$ must be negligible. This will be true for all values of θ when R is adequately large and we again have the Fraunhofer condition. The distance r is then linear in y. Substituting into Eq. (10.10) and integrating leads to

$$E = \frac{\mathcal{E}_L}{R} \int_{-D/2}^{+D/2} \sin[\omega t - k(R - y \sin\theta)]\, dy, \tag{10.12}$$

and finally

$$E = \frac{\mathcal{E}_L D}{R} \frac{\sin[(kD/2) \sin\theta]}{(kD/2) \sin\theta} \sin(\omega t - kR). \tag{10.13}$$

To simplify the appearance of things let

$$\beta \equiv (kD/2) \sin\theta \tag{10.14}$$

so that

$$E = \frac{\mathcal{E}_L D}{R} \left(\frac{\sin\beta}{\beta}\right) \sin(\omega t - kR). \tag{10.15}$$

The quantity most readily measured is the irradiance (forgetting the constants) $I(\theta) = \langle E^2 \rangle$ or

$$I(\theta) = \frac{1}{2} \left(\frac{\mathcal{E}_L D}{R}\right)^2 \left(\frac{\sin\beta}{\beta}\right)^2, \tag{10.16}$$

where $\langle \sin^2(\omega t - kR) \rangle = \frac{1}{2}$. When $\theta = 0$, $\sin\beta/\beta = 1$ and $I(\theta) = I(0)$ which corresponds to the *principal maximum*. *The irradiance resulting from an idealized coherent line source in the Fraunhofer approximation* is then

$$I(\theta) = I(0) \left(\frac{\sin\beta}{\beta}\right)^2 \tag{10.17}$$

or using the *sinc function* (Section 7.9, and Table 1 of the Appendix)

$$I(\theta) = I(0)\, \mathrm{sinc}^2\, \beta.$$

There is symmetry about the y-axis and this expression holds for θ measured in any plane containing that axis.

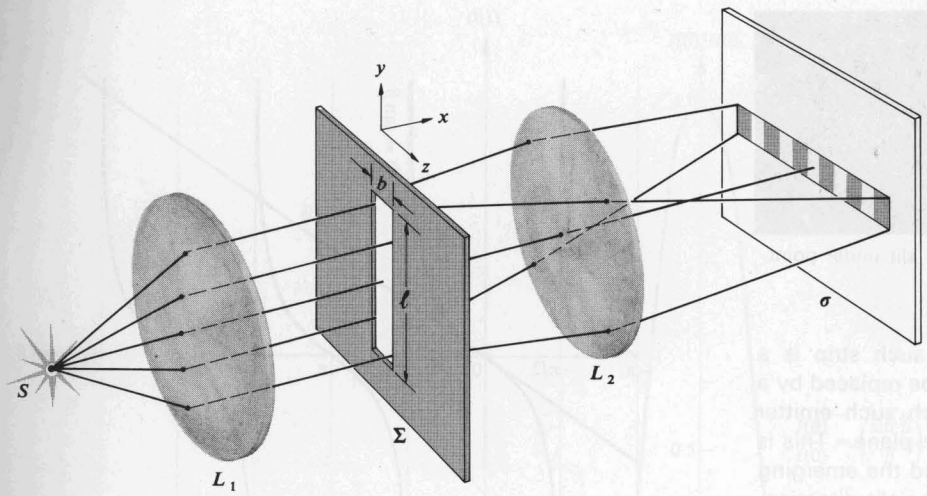


Fig. 10.9 Single-slit Fraunhofer diffraction.

Notice that since $\beta = (\pi D/\lambda) \sin\theta$, when $D \gg \lambda$, the irradiance drops extremely rapidly as θ deviates from zero. This arises from the fact that β becomes very large for large values of length D (a cm or so when using light). The phase of the line source is equivalent, by way of Eq. (10.15), to that of a point source located at the center of the array, a distance R from P. Finally then, a relatively long coherent line source ($D \gg \lambda$) can be envisioned as a single point emitter radiating predominantly in the $\theta = 0$ direction, i.e. its emission resembles a circular wave in the xz-plane. In contrast, notice that if $\lambda \gg D$, β is small, $\sin\beta \approx \beta$ and $I(\theta) \approx I(0)$. The irradiance is then constant for *all* θ and the line source resembles a point source emitting spherical waves.

We can now turn our attention to the problem of Fraunhofer diffraction by a slit or elongated narrow rectangular hole (Fig. 10.9). An aperture of this sort might typically have a width of several hundred λ and a length of a few centimeters. The usual procedure to follow in the analysis is to divide the slit into a series of long differential strips (dz by ℓ), parallel to the y-axis, as shown in Fig. 10.10. We

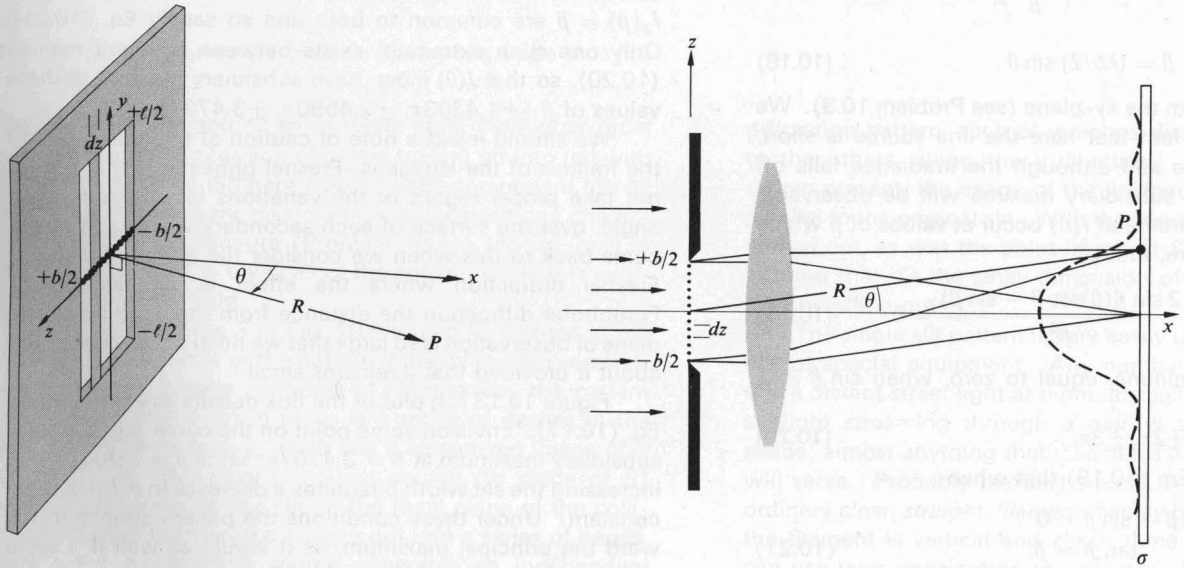


Fig. 10.10 Single-slit geometry.

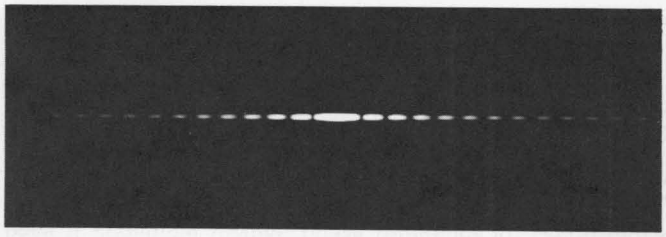


Fig. 10.11 Diffraction pattern of a single vertical slit under point-source illumination.

immediately recognize however that each such strip is a long coherent line source and can therefore be replaced by a point emitter on the z-axis. In effect, each such emitter radiates a circular wave in the ($y = 0$ or) xz-plane. This is certainly reasonable, since the slit is long and the emerging wavefronts are practically unobstructed in the slit direction. There will therefore be very little diffraction parallel to the edges of the slit. The problem has, as such, been reduced to that of finding the field in the xz-plane due to an infinite number of point sources extending across the width of the slit along the z-axis. We then need only evaluate the integral of the contribution dE from each element dz in the Fraunhofer approximation. But once again, this is equivalent to a coherent line source so that the complete solution for the slit is, as we have seen,

$$I(\theta) = I(0)\left(\frac{\sin\beta}{\beta}\right)^2 \qquad [10.17]$$

provided that

$$\beta = (kb/2)\sin\theta \qquad (10.18)$$

and θ is measured from the xy-plane (see Problem 10.3). We call attention to the fact that here the line source is short, $D = b$, β is not large and although the irradiance falls off rapidly, higher-order subsidiary maxima will be observable (Fig. 10.11). The extrema of $I(\theta)$ occur at values of β which cause $dI/d\beta$ to be zero, that is

$$\frac{dI}{d\beta} = I(0)\frac{2\sin\beta(\beta\cos\beta - \sin\beta)}{\beta^3} = 0. \qquad (10.19)$$

The irradiance has minima, equal to zero, when $\sin\beta = 0$, whereupon

$$\beta = \pm\pi, \pm 2\pi, \pm 3\pi, \ldots . \qquad (10.20)$$

It also follows from Eq. (10.19) that when

$$\beta\cos\beta - \sin\beta = 0$$
$$\tan\beta = \beta. \qquad (10.21)$$

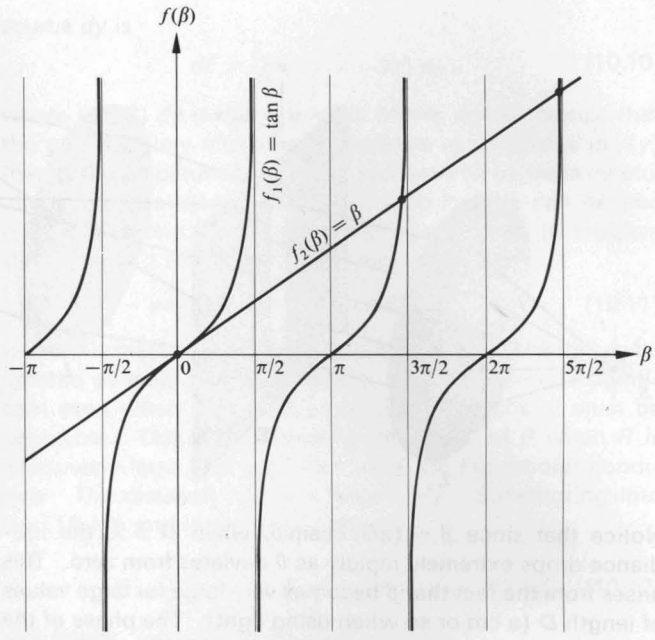


Fig. 10.12 The points of intersection of the two curves are the solutions of Eq. (10.21).

The solutions to this transcendental equation can be gotten graphically, as shown in Fig. 10.12. The points of intersection of the curves $f_1(\beta) = \tan\beta$, with the straight line $f_2(\beta) = \beta$ are common to both and so satisfy Eq. (10.21). Only one such extremum exists between adjacent minima (10.20), so that $I(\theta)$ must have subsidiary maxima at these values of β ($\pm 1.4303\pi$, $\pm 2.4590\pi$, $\pm 3.4707\pi$, ...).

We should inject a note of caution at this point: one of the frailties of the Huygens–Fresnel principle is that it does not take proper regard of the variations in amplitude, with angle, over the surface of each secondary wavelet. We will come back to this when we consider the *obliquity factor* in Fresnel diffraction where the effect is significant. In Fraunhofer diffraction the distance from the aperture to the plane of observation is so large that we need not be concerned about it provided that θ remains small.

Figure 10.13 is a plot of the flux density as expressed by Eq. (10.17). Envision some point on the curve, e.g. the third subsidiary maximum at $\beta = 3.4707\pi$; since $\beta = (\pi b/\lambda)\sin\theta$, increasing the slit width b requires a decrease in θ if β is to be constant. Under these conditions the pattern shrinks in toward the principal maximum, as it would as well if λ were

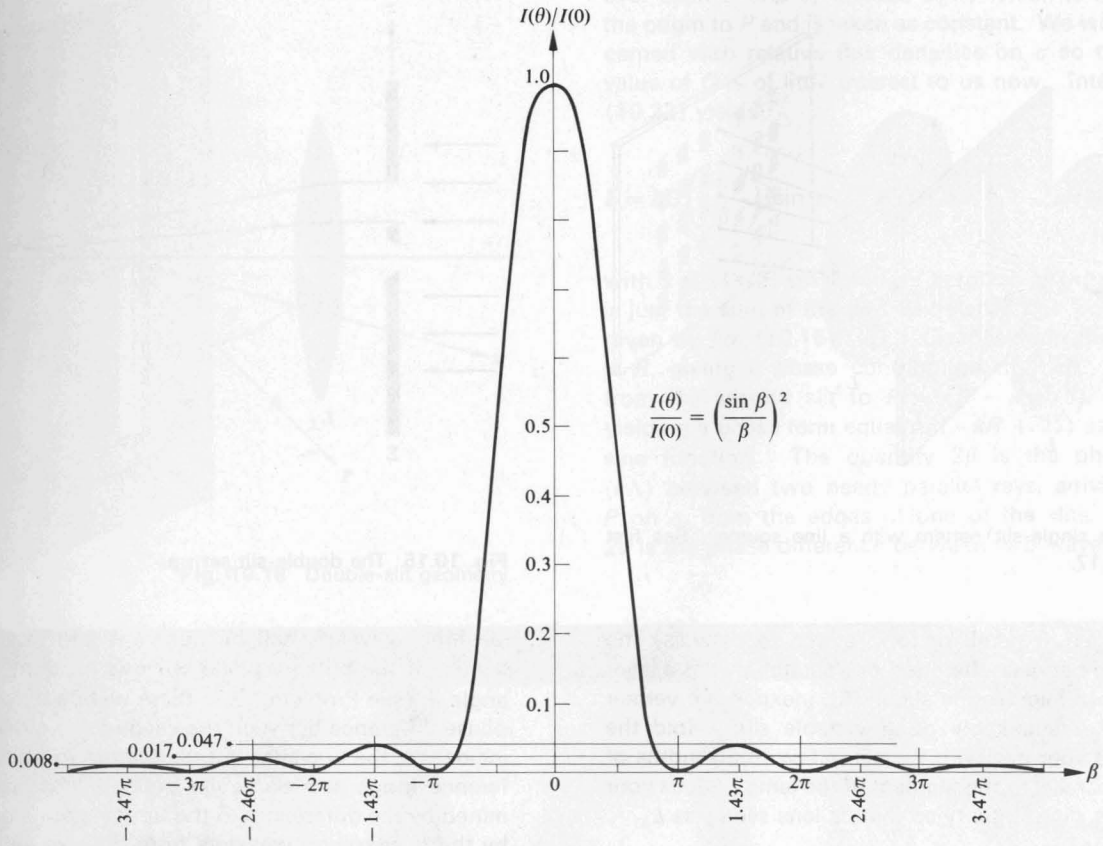


Fig. 10.13 The Fraunhofer diffraction pattern of a single slit.

decreased. If the source emits white light, the higher-order maxima show a succession of colors trailing off into red with increasing θ. Each different colored light-component has its minima and subsidiary maxima at angular positions characteristic of that wavelength (Problem 10.4). Indeed, only in the region about $\theta = 0$ will all of the constituent colors overlap to yield white light.

The point source S in Fig. 10.9 would be imaged at the position of the center of the pattern if the diffracting screen Σ were removed. Under this sort of illumination, the pattern produced with the slit in place is a series of dashes in the yz-plane of the screen σ, much like a spread-out image of S (Fig. 10.11). An incoherent line source (in place of S) positioned parallel to the slit, in the focal plane of the collimator L_1, will broaden the pattern out into a series of bands. Any point on the line source generates an independent diffraction pattern, each of which is displaced, with respect to the others, along the y-direction. With no diffracting screen present, the image of the line source would be a line parallel to the original slit. With the screen in place the line is spread out, as was the point image of S (Fig. 10.14). Keep in mind that it's the small dimension of the slit which does the spreading out.

The single slit pattern is very easily observed without the use of special equipment. Any number of sources will do, e.g. a distant street light at night, a small incandescent lamp, sunlight streaming through a narrow space in a window shade; almost anything that resembles a point or line source will serve. Probably the best source for our purposes is an ordinary clear, *straight-filament* display bulb (the kind where the filament is vertical and about three inches long). You can use your imagination to generate all sorts of single slit

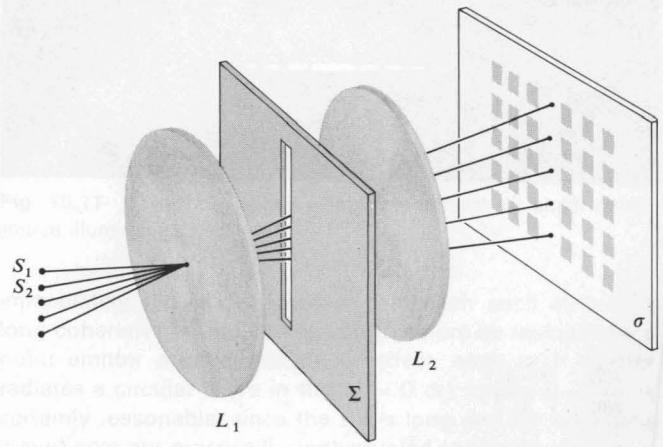


Fig. 10.14 The single-slit pattern with a line source See first photo of Fig. 10.17.

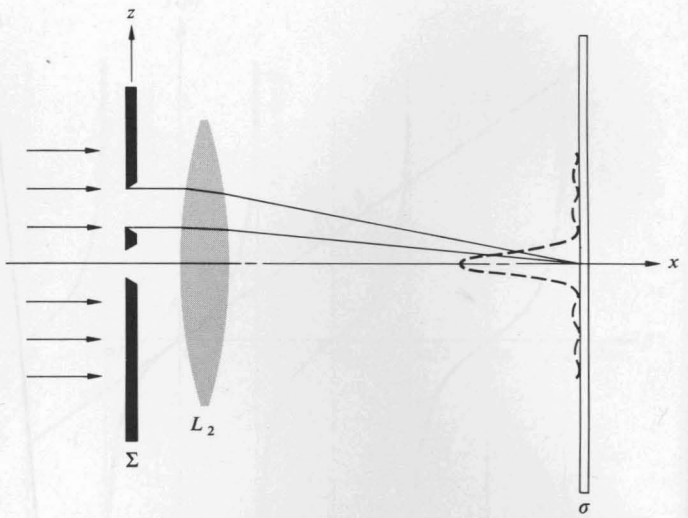


Fig. 10.15 The double-slit set-up.

arrangements, e.g. a comb or fork rotated to decrease the projected space between the tines, or a scratch across a layer of india ink on a microscope slide. An inexpensive vernier caliper makes a remarkably good variable slit. Hold the caliper close to your eye with the slit, a few thousandths of an inch wide, parallel to the filament of the lamp. Focus your eye beyond the slit at infinity so that its lens serves as L_2.

10.2.2 The Double Slit

It might at first seem from Fig. 10.10 that the location of the principal maximum is always to be in line with the center of the diffracting aperture; this however is not generally true. The diffraction pattern is actually centered about the axis of the lens and has exactly the same shape and location regardless of the slit's position, as long as its orientation is unchanged and the approximations are valid (Fig. 10.15). All waves traveling parallel to the lens axis converge on the second focal point of L_2, this then is the image of S, and the center of the diffraction pattern. Suppose now that we have two long slits of width b and center-to-center separation a (Fig. 10.16). Each aperture, by itself, would generate the same single-slit diffraction pattern on the viewing screen σ. At any point on σ, the contributions from the two slits overlap and even though each must be essentially equal in amplitude, they may well differ significantly in phase. Since the same primary wave excites the secondary sources at each slit, the resulting wavelets will be coherent and interference must occur. If the primary plane wave is incident on Σ at some angle θ_i (see Problem 10.3) there will be a constant relative phase difference between the secondary sources. At normal incidence, the wavelets are all emitted in phase. The interference fringe at a particular point of observation is determined by the differences in the optical path lengths traversed by the overlapping wavelets from the two slits. As we will see, the consequent flux-density distribution (Fig. 10.17) is a combination of a rapidly varying double-slit interference system, modulated by a single-slit diffraction pattern.

To obtain an expression for the optical disturbance at a point on σ, we have only to slightly reformulate the single-slit analysis. Each of the two apertures is divided into differential strips (dz by ℓ), which in turn behave as an infinite number of point sources aligned along the z-axis. The total contribution to the electric field, in the Fraunhofer approximation (10.12), is then

$$E = C\int_{-b/2}^{b/2} F(z)\,dz + C\int_{a-b/2}^{a+b/2} F(z)\,dz, \tag{10.22}$$

where $F(z) = \sin[\omega t - k(R - z\sin\theta)]$. The constant-amplitude factor C is the secondary source strength per unit length along the z-axis (assumed to be independent of z

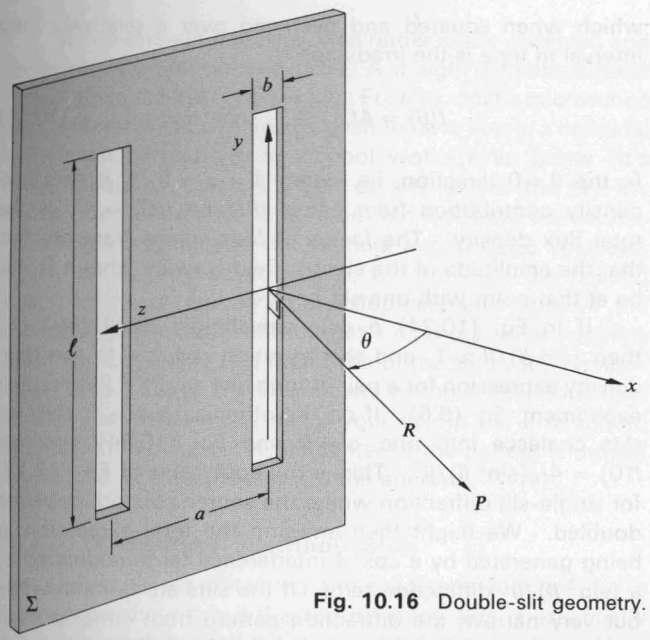


Fig. 10.16 Double-slit geometry.

over each aperture) divided by R which is measured from the origin to P and is taken as constant. We will only be concerned with relative flux densities on σ so that the actual value of C is of little interest to us now. Integration of Eq. (10.22) yields

$$E = bC\left(\frac{\sin\beta}{\beta}\right)[\sin(\omega t - kR) + \sin(\omega t - kR + 2\alpha)] \quad (10.23)$$

with $\alpha \equiv (ka/2)\sin\theta$ and as before $\beta \equiv (kb/2)\sin\theta$. This is just the sum of the two fields at P, one from each slit, as given by Eq. (10.15). The distance from the first slit to P is R, giving a phase contribution of $-kR$. The distance from the second slit to P is $(R - a\sin\theta)$ or $(R - 2\alpha/k)$ yielding a phase term equal to $(-kR + 2\alpha)$ as in the second sine function. The quantity 2β is the phase difference $(k\Lambda)$ between two nearly parallel rays, arriving at a point P on σ, from the edges of one of the slits. The quantity 2α is the phase difference between two waves arriving at P,

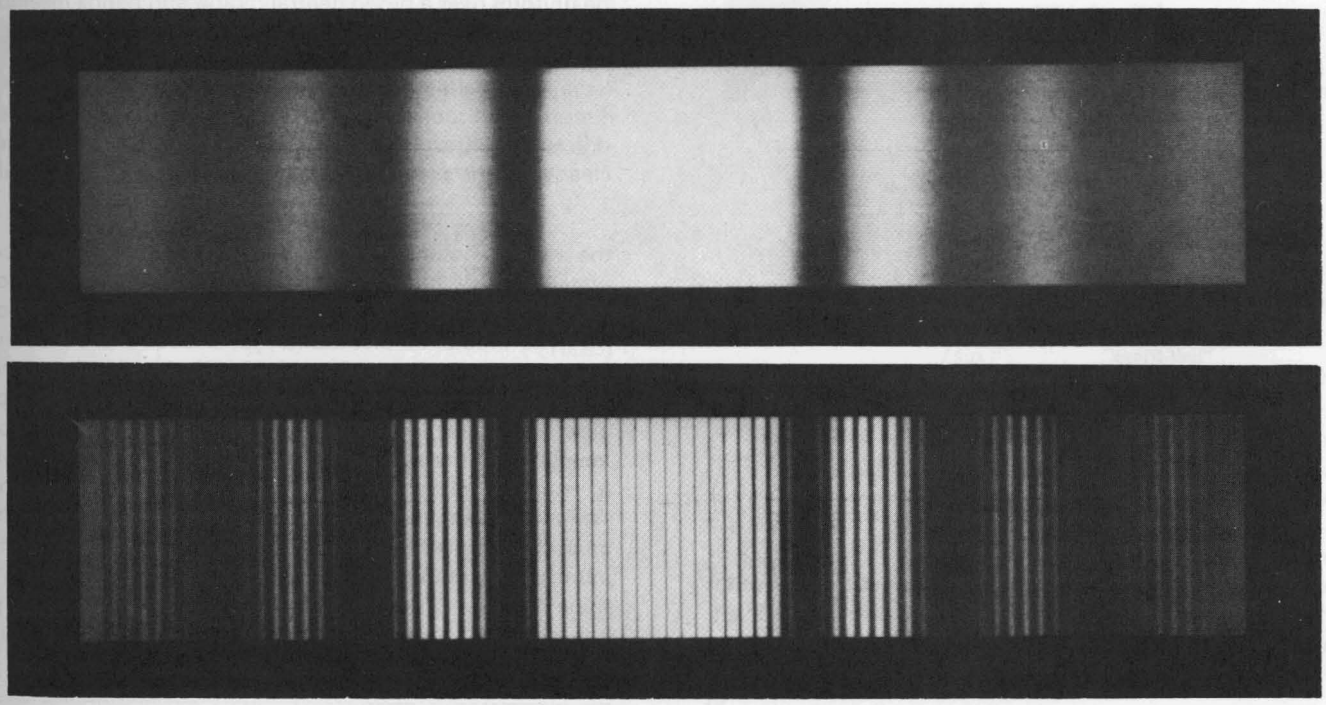


Fig. 10.17 Single- and double-slit Fraunhofer patterns. The faint cross hatching arises entirely in the printing process. [Photos courtesy M. Cagnet, M. Francon, and J. C. Thrierr: *Atlas optischer Erscheinungen*, Berlin–Heidelberg–New York: Springer, 1962.]

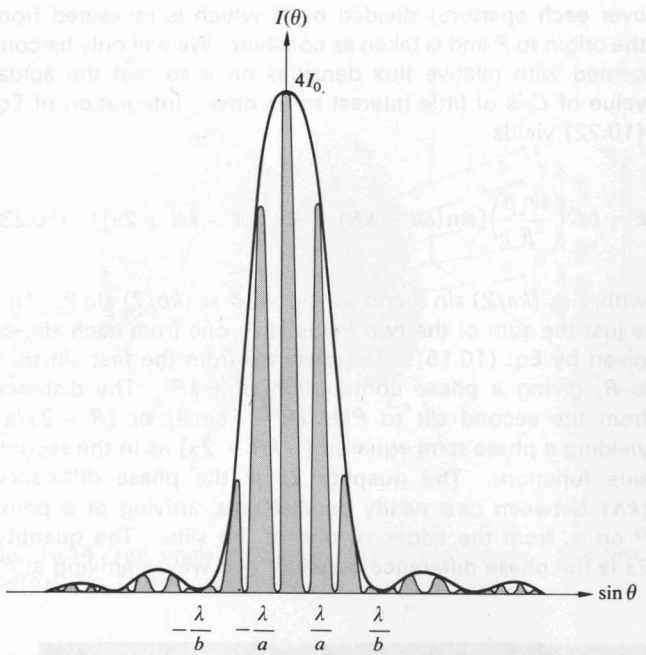


Fig. 10.18 A double slit pattern ($a = 3b$).

one having originated at any point in the first slit, the other coming from the corresponding point in the second slit. Simplifying Eq. (10.23) a bit further, it becomes

$$E = 2bC\left(\frac{\sin\beta}{\beta}\right)\cos\alpha\sin(\omega t - kR + \alpha),$$

which when squared and averaged over a relatively long interval in time is the irradiance

$$I(\theta) = 4I_0\left(\frac{\sin^2\beta}{\beta^2}\right)\cos^2\alpha. \tag{10.24}$$

In the $\theta = 0$ direction, i.e. when $\beta = \alpha = 0$, I_0 is the flux-density contribution from either slit and $I(0) = 4I_0$ is the total flux density. The factor of four comes from the fact that the amplitude of the electric field is twice what it would be at that point with one slit covered.

If in Eq. (10.24) b gets vanishingly small ($kb \ll 1$), then $(\sin\beta)/\beta \approx 1$, and the equation reduces to the flux-density expression for a pair of long line sources, i.e. Young's experiment, Eq. (9.6). If on the other hand $a = 0$. the two slits coalesce into one, $\alpha = 0$ and Eq. (10.24) becomes $I(0) = 4I_0(\sin^2\beta)/\beta^2$. This is the equivalent of Eq. (10.17) for single-slit diffraction where the source strength has been doubled. We might then envision the total expression as being generated by a $\cos^2\alpha$ interference term modulated by a $(\sin^2\beta)/\beta^2$ diffraction term. If the slits are finite in width but very narrow, the diffraction pattern from either slit will be uniform over a broad central region and bands resembling the idealized Young's fringes will appear within that region. At angular positions (θ-values) where

$$\beta = \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$$

diffraction effects are such that no light reaches σ, and clearly none is available for interference. At points on σ where

$$\alpha = \pm\pi/2, \pm 3\pi/2, \pm 5\pi/2, \ldots$$

the various contributions to the electric field will be completely out of phase and cancel, regardless of the actual amount of light made available from the diffraction process.

The irradiance distribution for a double-slit Fraunhofer pattern is illustrated in Fig. 10.18. Notice that it is a combination of Fig. 9.6 and 10.13. The curve is for the particular case when $a = 3b$, i.e. when $\alpha = 3\beta$. You can get a rough idea of what the pattern will look like since if $a = mb$, where m is any number, there will be $2m$ bright fringes (counting "fractional fringes" as well)* within the central diffraction peak (Problem 10.5). It may occur that an interference maximum and a diffraction minimum (zero) correspond to the same θ-value. In that case no light is available at that precise position to partake in the interference process, and the suppressed peak is said to be a *missing order*.

* Notice that m need not be an integer. Moreover if m is an integer there will be "half-fringes" as shown in Fig. 10.18(b).

The double-slit pattern is also rather easily observed and the seeing is well worth the effort. A straight filament, tubular bulb is again the best line source. For slits, coat a microscope slide with india ink or if you happen to have some; a colloidal suspension of graphite in alcohol works even better (it's more opaque). Scratch a pair of slits across the dry ink with a razor blade and stand about ten feet from the source. Hold the slits parallel to the filament and close to your eye, which, when focused at infinity, will serve as the needed lens. Interpose red or blue cellophane and observe the change in the width of the fringes. Find out what happens when you cover one and then both of the slits with a microscope slide. Move the slits slowly in the z-direction; then holding them stationary, move your eye in the z-direction; verify that the position of the center of the pattern is indeed determined by the lens and not the aperture.

10.2.3 Diffraction by Many Slits

The procedure for obtaining the irradiance function for a monochromatic wave diffracted by many slits is essentially the same as that used when considering two slits. Here again, the limits of integration must be appropriately altered. Consider the case of N long, parallel, narrow slits each of width b and center-to-center separation a, as illustrated in Fig. 10.19. With the origin of the coordinate system once more at the center of the first slit, the total optical disturbance at a point on the screen σ is given by

$$E = C\int_{-b/2}^{b/2} F(z)\,dz + C\int_{a-b/2}^{a+b/2} F(z)\,dz + C\int_{2a-b/2}^{2a+b/2} F(z)\,dz + \cdots + C\int_{(N-1)a-b/2}^{(N-1)a+b/2} F(z)\,dz \tag{10.25}$$

where as before, $F(z) = \sin[\omega t - k(R - z\sin\theta)]$. This applies to the Fraunhofer condition, so that the aperture configuration must be such that all of the slits are close to the origin and the approximation (10.11)

$$r = R - z\sin\theta \tag{10.26}$$

applies over the entire array. The contribution from the jth slit (where the first one is numbered zero), obtained by evaluating only that one integral in Eq. (10.25), is then

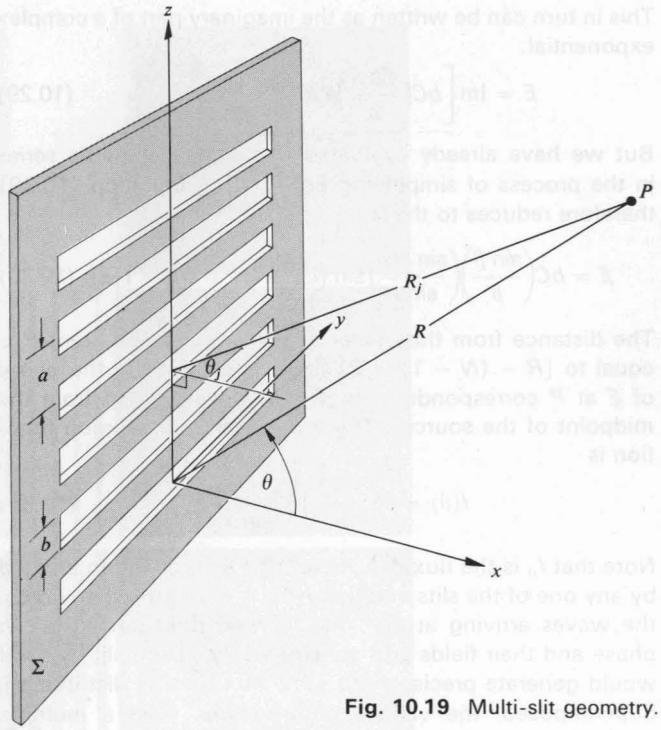


Fig. 10.19 Multi-slit geometry.

$$E_j = \frac{C}{k\sin\theta}[\sin(\omega t - kR)\sin(kz\sin\theta) - \cos(\omega t - kR)\cos(kz\sin\theta)]_{ja-b/2}^{ja+b/2}$$

provided that we require $\theta_j \approx \theta$. After some manipulation this becomes

$$E_j = bC\left(\frac{\sin\beta}{\beta}\right)\sin(\omega t - kR + 2\alpha j) \tag{10.27}$$

recalling that $\beta = (kb/2)\sin\theta$ and $\alpha = (ka/2)\sin\theta$. Notice that this is equivalent to the expression for a line source (10.15) or of course, a single slit, where in accord with Eq. 10.26 and Fig. 10.19, $R_j = R - ja\sin\theta$ so that $-kR + 2\alpha j = -kR_j$. The total optical disturbance as given by Eq. (10.25) is simply the sum of the contributions from each of the slits, that is

$$E = \sum_{j=0}^{N-1} E_j$$

or

$$E = \sum_{j=0}^{N-1} bC\left(\frac{\sin\beta}{\beta}\right)\sin(\omega t - kR + 2\alpha j). \tag{10.28}$$

This in turn can be written as the imaginary part of a complex exponential:

$$E = \mathrm{Im}\left[bC\left(\frac{\sin\beta}{\beta}\right)e^{i(\omega t - kR)}\sum_{j=0}^{N-1}(e^{i2\alpha})^j\right]. \qquad (10.29)$$

But we have already evaluated this same geometric series in the process of simplifying Eq. (10.2). Equation (10.29) therefore reduces to the form

$$E = bC\left(\frac{\sin\beta}{\beta}\right)\left(\frac{\sin N\alpha}{\sin\alpha}\right)\sin\left[\omega t - kR + (N-1)\alpha\right]. \qquad (10.30)$$

The distance from the center of the array to the point P is equal to $[R - (N-1)(a/2)\sin\theta]$ and therefore the phase of E at P corresponds to that of a wave emitted from the midpoint of the source. The flux-density distribution function is

$$I(\theta) = I_0\left(\frac{\sin\beta}{\beta}\right)^2\left(\frac{\sin N\alpha}{\sin\alpha}\right)^2. \qquad (10.31)$$

Note that I_0 is the flux density in the $\theta = 0$ direction emitted by any one of the slits and that $I(0) = N^2 I_0$. In other words, the waves arriving at P in the forward direction are all in phase and their fields add constructively. Each slit by itself would generate precisely the same flux-density distribution. Superimposed, the various contributions yield a multiple wave interference system modulated by the single-slit diffraction envelope. If the width of each aperture were shrunk to zero, Eq. (10.31) would become the flux-density expression (10.6) for a linear coherent array of oscillators. As in that earlier treatment, (10.17), *principal maxima* occur when $(\sin N\alpha/\sin\alpha) = N$, i.e. when

$$\alpha = 0, \pm\pi, \pm 2\pi, \ldots$$

or equivalently since $\alpha = (ka/2)\sin\theta$

$$a\sin\theta_m = m\lambda \qquad (10.32)$$

with $m = 0, \pm 1, \pm 2, \ldots$. This is quite general and gives rise to the same θ-locations for these maxima regardless of the value of $N \geq 2$. Minima, of zero flux-density, exist whenever $(\sin N\alpha/\sin\alpha)^2 = 0$ or when

$$\alpha = \pm\frac{\pi}{N}, \pm\frac{2\pi}{N}, \pm\frac{3\pi}{N}, \cdots, \pm\frac{(N-1)\pi}{N}, \pm\frac{(N+1)\pi}{N}, \cdots. \qquad (10.33)$$

Between consecutive principal maxima (i.e. over the range in α of π) there will therefore be $N - 1$ minima. And of course between each pair of minima there will have to be a *subsidiary maximum*. The term $(\sin N\alpha/\sin\alpha)^2$, which we can think of as embodying the interference effects, has a rapidly varying numerator and a slowly varying denominator. The subsidiary maxima are therefore located approximately at points where $\sin N\alpha$ has its greatest value, namely

$$\alpha = \pm\frac{3\pi}{2N}, \pm\frac{5\pi}{2N}, \cdots \qquad (10.34)$$

The $N - 2$ *subsidiary maxima* between consecutive principal maxima are clearly visible in Fig. 10.20. We can get some idea of the flux density at these peaks by rewriting Eq. (10.31) as

$$I(\theta) = \frac{I(0)}{N^2}\left(\frac{\sin\beta}{\beta}\right)^2\left(\frac{\sin N\alpha}{\sin\alpha}\right)^2, \qquad (10.35)$$

where at the points of interest $|\sin N\alpha| = 1$. For *large* N, α is small and $\sin^2\alpha \approx \alpha^2$. At the first subsidiary peak $\alpha = 3\pi/2N$, in which case

$$I \approx I(0)\left(\frac{\sin\beta}{\beta}\right)^2\left(\frac{2}{3\pi}\right)^2 \qquad (10.36)$$

and the flux density has dropped to about $\frac{1}{22}$ of that of the adjacent principal maximum (see Problem 10.6). Since $(\sin\beta)/\beta$ for small β varies slowly it will not differ from one appreciably, close to the zeroth-order principal maximum, so that $I/I(0) \approx \frac{1}{22}$. This flux-density ratio for the next secondary peak is down to $\frac{1}{62}$ and it continues to decrease as α approaches a value halfway between the principal maxima. At that symmetry point, $\alpha \approx \pi/2$, $\sin\alpha \approx 1$ and the flux-density ratio has its lowest value of approximately $1/N^2$. Thereafter $\alpha > \pi/2$ and the flux densities of the subsidiary maxima begin to increase.

Try duplicating Fig. 10.20 using a tubular bulb and homemade slits. You'll probably have difficulty clearly seeing the subsidiary maxima, with the effect that the only perceptible difference between the double and multiple-slit patterns may be an apparent broadening in the dark regions between principal maxima. As in Fig. 10.20, the dark regions will become wider than the bright bands as N increases and the secondary peaks fade out. If we consider each principal maximum to be bounded in width by two adjacent zeroes, then each will extend over a length in θ, $(\sin\theta \approx \theta)$ of approximately $2\lambda/Na$. As N increases, the principal maxima maintain their relative spacing (λ/a) while becoming increasingly more narrow. Figure 10.21 shows the case of six slits where $a = 4b$.

The multiple-slit interference term in Eq. 10.35 has the form $(\sin^2 N\alpha)/N^2\sin^2\alpha$; thus, for large N, $(N^2\sin^2\alpha)^{-1}$

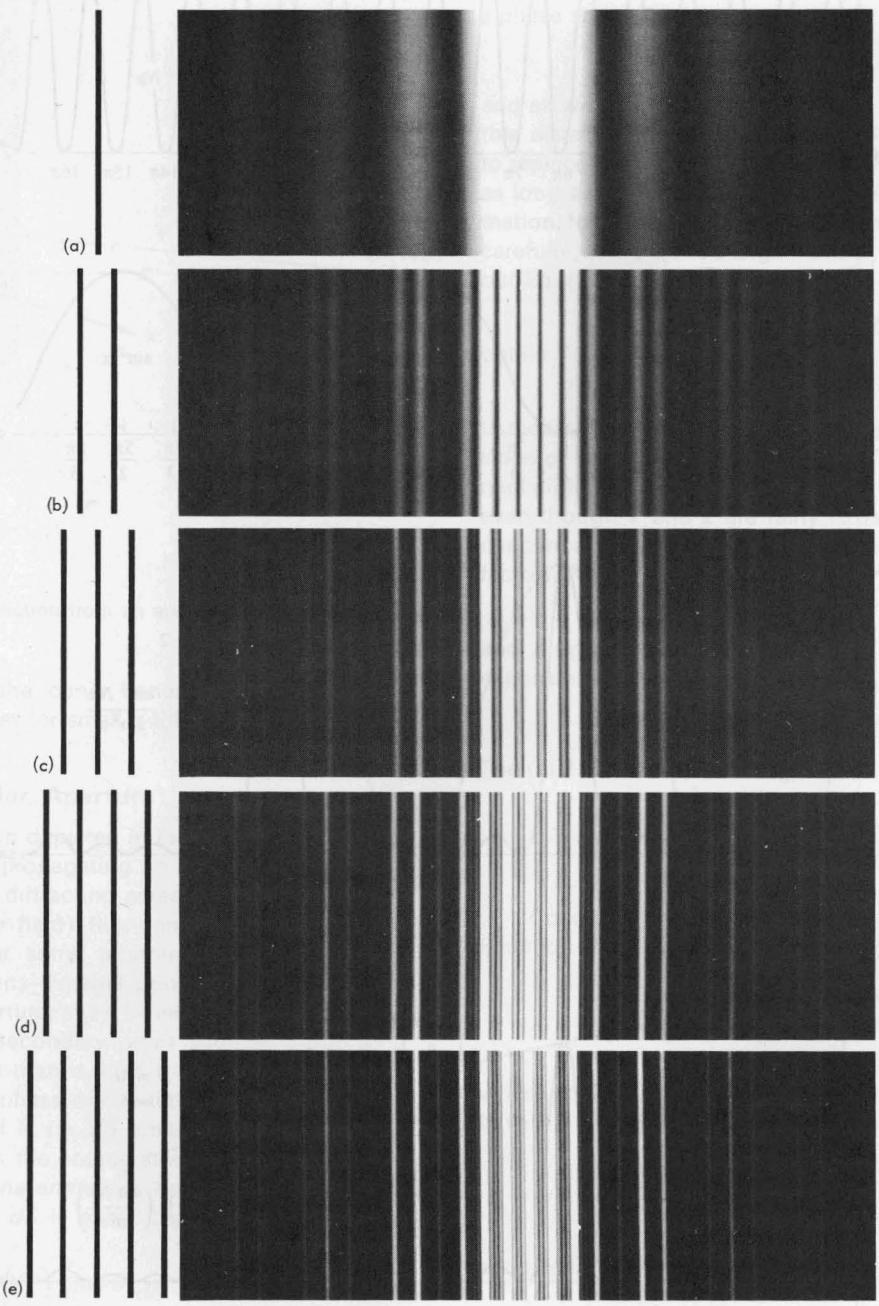


Fig. 10.20 Diffraction patterns for slit systems shown at left.

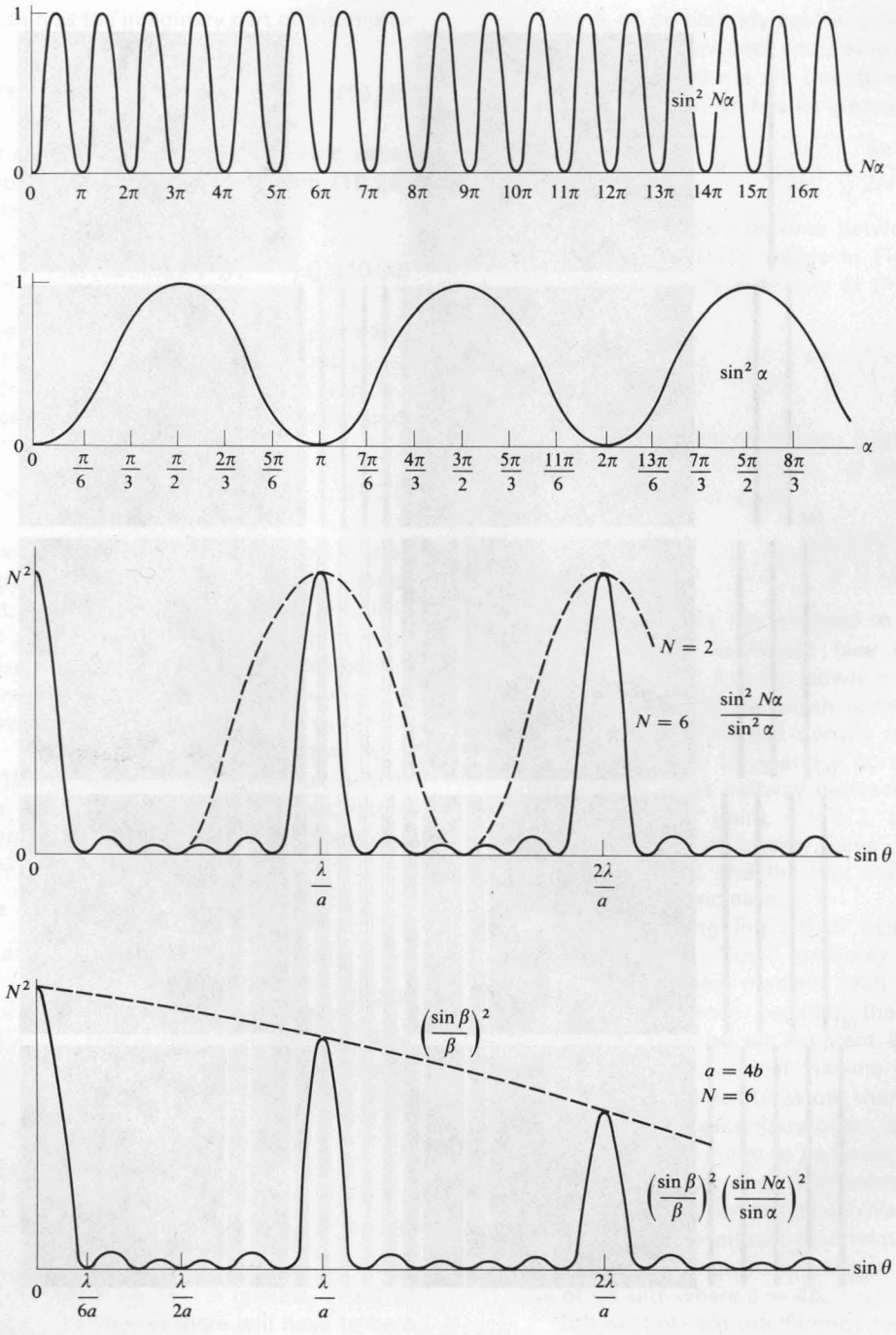


Fig. 10.21 Multiple-slit pattern ($a = 4b$, $N = 6$).

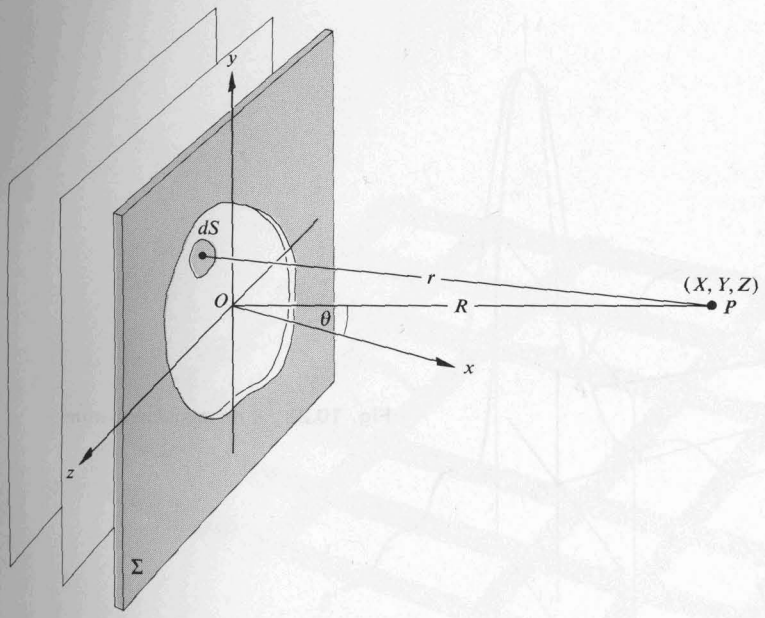


Fig. 10.22 Fraunhofer diffraction from an aperture.

may be envisioned as the curve beneath which $\sin^2 N\alpha$ rapidly varies. Notice that for small α this interference term looks like $\text{sinc}^2 N\alpha$.

10.2.4 The Rectangular Aperture

Consider the configuration depicted in Fig. 10.22. A monochromatic plane wave propagating in the x-direction is incident on the opaque diffracting screen Σ. We wish to find the consequent (far-field) flux-density distribution in space or equivalently at some arbitrary distant point P. According to the Huygens–Fresnel principle, a differential area dS, within the aperture, may be envisioned as being covered with coherent secondary point sources. But dS is much smaller in extent than is λ, so that all of the contributions at P remain in phase and interfere constructively. This is true regardless of θ, i.e. dS emits a spherical wave (Problem 10.7). If $\mathcal{E}_A$ is the source strength per unit area, *assumed constant over the entire aperture*, then the optical disturbance at P due to dS is either the real or imaginary part of

$$dE = \left(\frac{\mathcal{E}_A}{r}\right) e^{i(\omega t - kr)}\, dS. \tag{10.37}$$

The choice is yours and depends only on whether you like sine or cosine waves, there being no difference except for a phase shift. The distance from dS to P is

$$r = [X^2 + (Y - y)^2 + (Z - z)^2]^{1/2} \tag{10.38}$$

and as we have seen the Fraunhofer condition occurs when this distance approaches infinity. As before, it will suffice to replace r by the distance $\overline{OP}$, i.e. R, in the amplitude term as long as the aperture is relatively small. But the approximation for r in the phase needs to be treated a bit more carefully; $k = 2\pi/\lambda$ is a large number. To that end we expand out Eq. (10.38), and by making use of

$$R = [X^2 + Y^2 + Z^2]^{1/2} \tag{10.39}$$

obtain

$$r = R[1 + (y^2 + z^2)/R^2 - 2(Yy + Zz)/R^2]^{1/2} \tag{10.40}$$

In the far-field case R is very large in comparison to the dimensions of the aperture and the $(y^2 + z^2)/R^2$ term is certainly negligible. Since P is very far from Σ, θ can still be kept small, even though Y and Z are fairly large and this mitigates any concern about the directionality of the emitters (the obliquity factor). Now

$$r = R[1 - 2(Yy + Zz)/R^2]^{1/2}$$

and dropping all but the first two terms in the binomial expansion, we have

$$r = R[1 - (Yy + Zz)/R^2].$$

The total disturbance arriving at P is

$$E = \frac{\mathcal{E}_A e^{i(\omega t - kR)}}{R} \iint_{\text{Aperture}} e^{ik(Yy + Zz)/R}\, dS. \tag{10.41}$$

Consider the specific configuration shown in Fig. 10.23. Equation (10.41) can now be written as

$$E = \frac{\mathcal{E}_A e^{i(\omega t - kR)}}{R} \int_{-b/2}^{+b/2} e^{ikYy/R}\, dy \int_{-a/2}^{+a/2} e^{ikZz/R}\, dz.$$

where $dS = dy\, dz$. With $\beta' \equiv kbY/2R$ and $\alpha' \equiv kaZ/2R$, we have

$$\int_{-b/2}^{+b/2} e^{ikYy/R}\, dy = b\left(\frac{e^{i\beta'} - e^{-i\beta'}}{2i\beta'}\right) = b\left(\frac{\sin\beta'}{\beta'}\right)$$

and similarly

$$\int_{-a/2}^{+a/2} e^{ikZz/R}\, dz = a\left(\frac{e^{i\alpha'} - e^{-i\alpha'}}{2i\alpha'}\right) = a\left(\frac{\sin\alpha'}{\alpha'}\right),$$

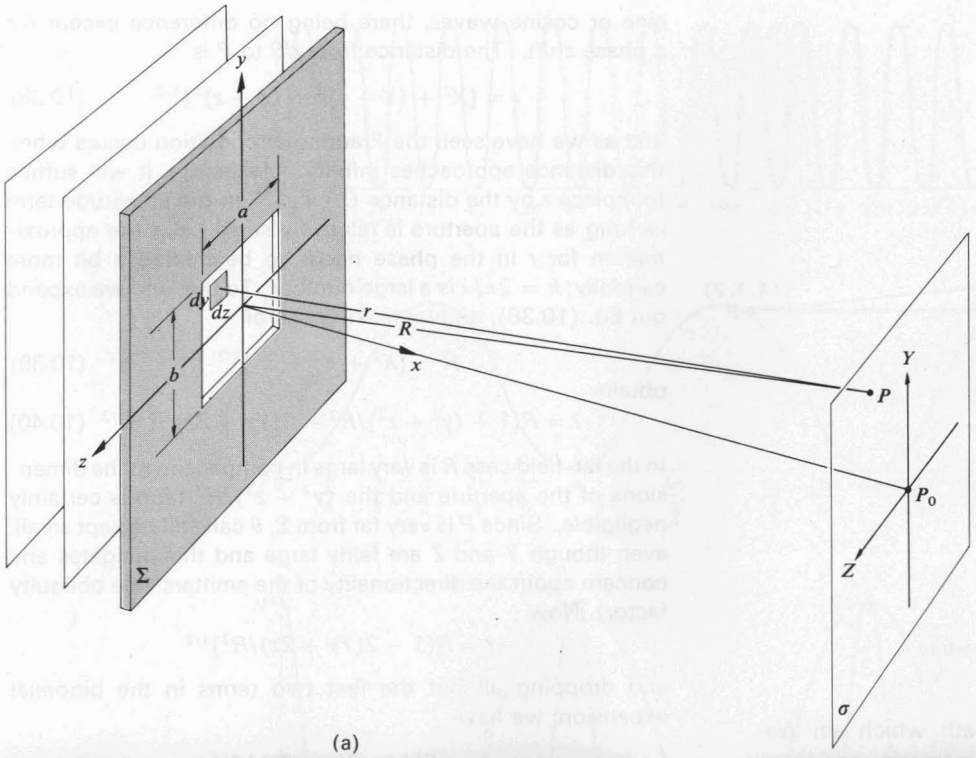


Fig. 10.23 A rectangular aperture.

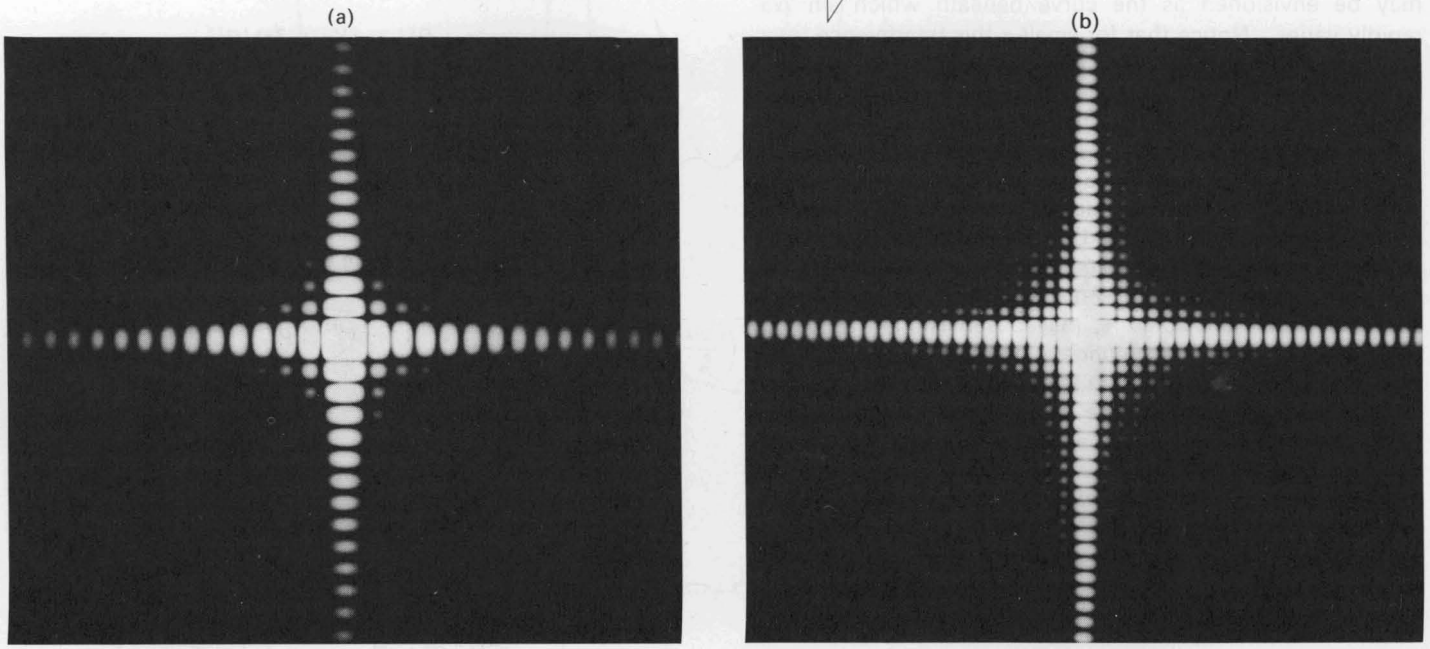


Fig. 10.24 (a) Fraunhofer pattern of a square aperture. (b) The same pattern further exposed to bring out some of the faint terms. [Photos by E.H.]

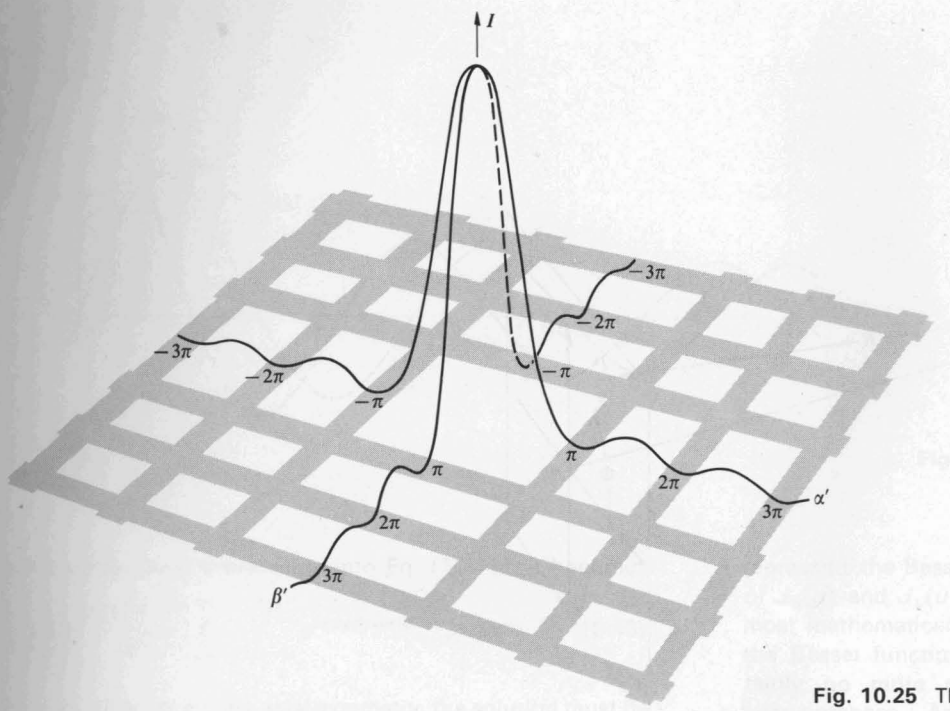


Fig. 10.25 The irradiance distribution for a square aperture.

so that

$$E = \frac{A\,\mathcal{E}_A e^{i(\omega t - kR)}}{R}\left(\frac{\sin\alpha'}{\alpha'}\right)\left(\frac{\sin\beta'}{\beta'}\right), \tag{10.42}$$

where A is the area of the aperture. Since $I = \langle(\mathrm{Re}\,E)^2\rangle$

$$I(Y,Z) = I(0)\left(\frac{\sin\alpha'}{\alpha'}\right)^2\left(\frac{\sin\beta'}{\beta'}\right)^2 \tag{10.43}$$

where $I(0)$ is the irradiance at P_0 i.e. at $Y = 0$, $Z = 0$ (see Fig. 10.24). At values of Y and Z such that $\alpha' = 0$ or $\beta' = 0$, $I(Y,Z)$ assumes the familiar shape of Fig. 10.13. When β' or α' are nonzero integer multiples of π or equivalently when Y and Z are nonzero integer multiples of $\lambda R/b$ and $\lambda R/a$ respectively, $I(Y,Z) = 0$ and we have a rectangular grid of nodal lines as indicated in Fig. 10.25. Notice that the pattern in the Y-, Z-directions varies *inversely* with the y-, z-aperture dimensions. A horizontal, rectangular opening will produce a pattern with a verticle rectangle at its center.

Along the β'-axis, $\alpha' = 0$ and the subsidiary maxima are located approximately halfway between zeroes, i.e. at $\beta'_m = \pm 3\pi/2, \pm 5\pi/2, \pm 7\pi/2, \ldots$. At each subsidiary maximum $\sin\beta'_m = 1$, and of course, along the β'-axis since $\alpha' = 0$, $(\sin\alpha')/\alpha' = 1$ so that the relative irradiances are approximated simply by

$$\frac{I}{I(0)} = \frac{1}{\beta'^2_m}. \tag{10.44}$$

Similarly along the α'-axis

$$\frac{I}{I(0)} = \frac{1}{\alpha'^2_m}. \tag{10.45}$$

The flux-density ratio* drops off rather rapidly from 1 to $\frac{1}{22}$ to $\frac{1}{62}$ to $\frac{1}{122}$, etc. Even so, the off-axis secondary peaks are

* These particular photographs were taken during an undergraduate laboratory session. A 1.5 mW helium–neon laser was used as a plane-wave source. The apparatus was set up in a long darkened room and the pattern was cast directly on 4 × 5 Polaroid (ASA 3000) film. The film was located at about 30 ft from a small aperture so that no focusing lens was needed. The shutter, placed directly in front of the laser, was a student-contrived cardboard guillotine arrangement and therefore no exposure times are available. Any camera shutter (a single-lens reflex with the lens removed and the back open) will serve, but the cardboard one was more fun.

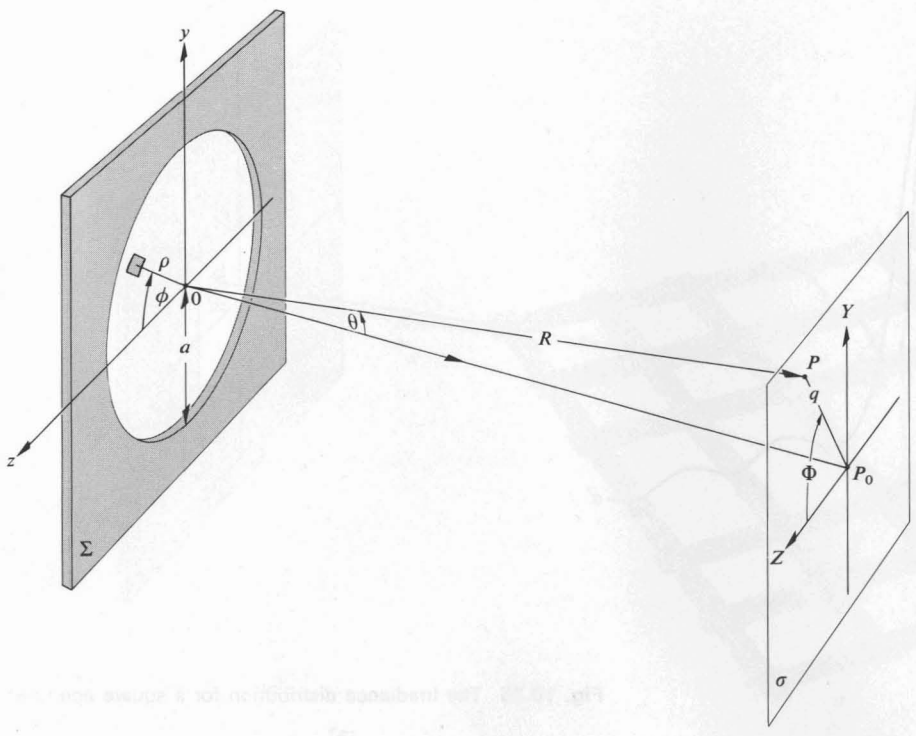


Fig. 10.26 Circular aperture geometry.

still smaller, e.g. the four corner peaks (whose coordinates correspond to appropriate combinations of $\beta' = \pm 3\pi/2$ and $\alpha' = \pm 3\pi/2$) nearest to the central maximum, each have relative irradiances of $(\frac{1}{22})^2$.

10.2.5 The Circular Aperture

Fraunhofer diffraction at a circular aperture is an effect of very great practical significance in the study of optical instrumentation. Envision a typical arrangement; plane waves impinging on a screen Σ containing a circular aperture and the consequent far-field diffraction pattern spread across a distant observing screen σ. By using a focusing lens L_2, σ can be brought in close to the aperture without changing the pattern. Now, if L_2 is positioned within and exactly fills the diffracting opening in Σ, the form of the pattern is essentially unaltered. The light wave reaching Σ is cropped so that only a circular segment propagates through L_2 to form an image in the focal plane. But quite obviously this is the same process that takes place in the eye, a telescope, microscope or camera lens. The image of a distant point source as formed by a perfectly aberration-free converging lens, is never a point, but rather some sort of diffraction pattern. We are essentially collecting only a fraction of the incident wavefront and cannot therefore hope to form a perfect image. As shown in the last section, the expression for the optical disturbance at P, arising from an arbitrary aperture in the far-field case, is

$$E = \frac{\mathcal{E}_A e^{i(\omega t - kR)}}{R} \iint_{\text{Aperture}} e^{ik(Yy + Zz)/R}\, dS. \qquad [10.41]$$

For a circular opening, symmetry would suggest introducing spherical polar coordinates in both the plane of the aperture and the plane of observation, as shown in Fig. 10.26. Therefore, let

$$z = \rho \cos\phi \qquad y = \rho \sin\phi$$

$$Z = q \cos\Phi \qquad Y = q \sin\Phi$$

and so the differential element of area is now

$$dS = \rho\, d\rho\, d\phi.$$

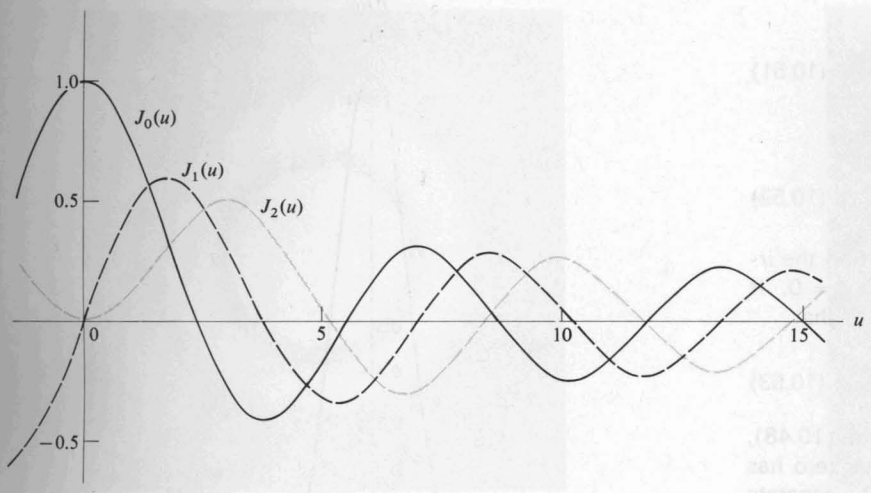


Fig. 10.27 Bessel functions.

Substituting these expressions into Eq. (10.41), it becomes

$$E = \frac{\mathcal{E}_A e^{i(\omega t - kR)}}{R} \int_{\rho=0}^{a} \int_{\phi=0}^{2\pi} e^{i(k\rho q/R)\cos(\phi - \Phi)} \rho \, d\rho \, d\phi \qquad (10.46)$$

Because of the complete axial symmetry, the solution must be independent of Φ. We might just as well solve Eq. (10.46) with $\Phi = 0$ as with any other value, thereby simplifying things slightly.

The portion of the double integral associated with the variable ϕ,

$$\int_0^{2\pi} e^{i(k\rho q/R)\cos\phi} \, d\phi,$$

is one that arises quite frequently in the mathematics of physics. It is a unique function in that it cannot be reduced to any of the more common forms, such as the various hyperbolic, exponential or trigonometric functions and indeed, with the exception of these, it is perhaps the most often encountered. The quantity

$$J_0(u) = \frac{1}{2\pi} \int_0^{2\pi} e^{iu\cos v} \, dv \qquad (10.47)$$

is known as the *Bessel function* (of the first kind) of order zero. More generally

$$J_m(u) = \frac{i^{-m}}{2\pi} \int_0^{2\pi} e^{i(mv + u\cos v)} \, dv \qquad (10.48)$$

represents the Bessel function of order m. Numerical values of $J_0(u)$ and $J_1(u)$ are tabulated for a large range of u in most mathematical handbooks. Just like sine and cosine, the Bessel functions have series expansions and are certainly no more esoteric than these familiar childhood acquaintances. As seen in Fig. 10.27, $J_0(u)$ and $J_1(u)$ are slowly decreasing oscillatory functions that do nothing particularly dramatic.

Equation (10.46) can be rewritten as

$$E = \frac{\mathcal{E}_A e^{i(\omega t - kR)}}{R} 2\pi \int_0^a J_0(k\rho q/R)\rho \, d\rho. \qquad (10.49)$$

Another general property of Bessel functions, referred to as a recurrence relation, is

$$\frac{d}{du}\left[u^m J_m(u)\right] = u^m J_{m-1}(u).$$

When $m = 1$, this clearly leads to

$$\int_o^u u' J_0(u') \, du' = u J_1(u) \qquad (10.50)$$

with u' just serving as a dummy variable. If we now return to the integral in Eq. (10.49) and change the variable such that $w = k\rho q/R$, then $d\rho = (R/kq)\, dw$ and

$$\int_{\rho=0}^{\rho=a} J_0(k\rho q/R)\rho \, d\rho = (R/kq)^2 \int_{w=0}^{w=kaq/R} J_0(w) w \, dw.$$

Making use of Eq. (10.50), we get

$$E(t) = \frac{\mathcal{E}_A e^{i(\omega t - kR)}}{R} 2\pi a^2 (R/kaq) J_1(kaq/R). \quad (10.51)$$

The irradiance at point P is $\langle (\text{Re } E)^2 \rangle$ or $\frac{1}{2}EE^*$, that is

$$I = \frac{2\mathcal{E}_A^2 A^2}{R^2}\left[\frac{J_1(kaq/R)}{kaq/R}\right]^2, \quad (10.52)$$

where A is the area of the circular opening. To find the irradiance at the center of the pattern, i.e. at P_0, set $q = 0$. It follows from the above recurrence relation ($m = 1$) that

$$J_0(u) = \frac{d}{du}J_1(u) + \frac{J_1(u)}{u}. \quad (10.53)$$

From Eq. (10.47) we see that $J_0(0) = 1$ and from Eq. (10.48), $J_1(0) = 0$. The ratio of $J_1(u)/u$ as u approaches zero has the same limit (L'Hôspital's rule) as the ratio of the separate derivatives of its numerator and denominator, namely $dJ_1(u)/du$ over one. But that means that the right-hand side of Eq. (10.53) is twice that limiting value, so that $J_1(u)/u = \frac{1}{2}$ at $u = 0$. The irradiance at P_0 is therefore

$$I(0) = \frac{\mathcal{E}_A^2 A^2}{2R^2}, \quad (10.54)$$

which is the same result obtained for the rectangular opening (10.43). If R can be presumed essentially constant over the pattern, we can write

$$I = I(0)\left[\frac{2J_1(kaq/R)}{kaq/R}\right]^2. \quad (10.55)$$

Since $\sin\theta = q/R$, the irradiance can be written as a function of θ:

$$I(\theta) = I(0)\left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2, \quad (10.56)$$

and as such, is plotted in Fig. 10.28. Because of the axial symmetry, the towering central maximum corresponds to a high-irradiance circular spot known as the *Airy disk*. It was Sir George Biddell Airy (1801–92) Astronomer Royal of England, who first derived Eq. (10.56). The central disk is surrounded by a dark ring which corresponds to the first zero of the function $J_1(u)$. From the standard tables $J_1(u) = 0$ when $u = 3.83$ i.e. $kaq/R = 3.83$. The radius q_1 drawn to the center of this first dark ring can be thought of as the extent of the Airy disk. It is given by

$$q_1 = 1.22\frac{R\lambda}{2a}. \quad (10.57)$$

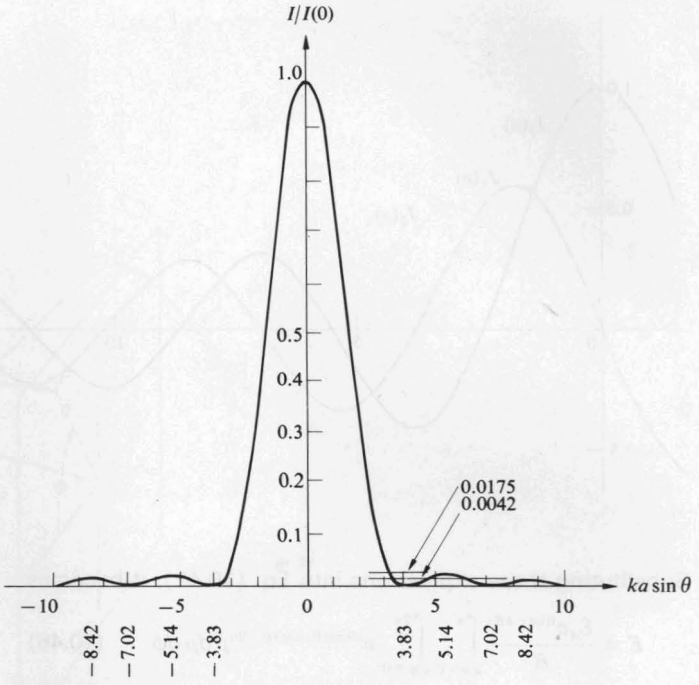


Fig. 10.28 The Airy pattern.

For a lens focused on the screen σ, the focal length $f \approx R$ and so

$$q_1 \approx 1.22\frac{f\lambda}{D}, \quad (10.58)$$

where D is the aperture diameter i.e. $D = 2a$. (The *diameter* of the Airy disk in the visible spectrum is *very roughly* equal to the $f/\#$ of the lens in millionths of a meter.) As shown in Figs. 10.29 to 10.31, q_1 varies inversely with the hole's diameter. As D approaches λ, the Airy disk can be very large indeed and the circular aperture begins to resemble a point source of spherical waves.

The higher-order zeroes occur at values of kaq/R equal to 7.02, 10.17, etc. The secondary maxima are located where u satisfies the condition

$$\frac{d}{du}\left[\frac{J_1(u)}{u}\right] = 0,$$

which is equivalent to $J_2(u) = 0$. From the tables then, these secondary peaks occur when kaq/R equals 5.14, 8.42, 11.6, etc.; whereupon $I/I(0)$ drops from one to 0.0175, 0.0042 and 0.0016 respectively.

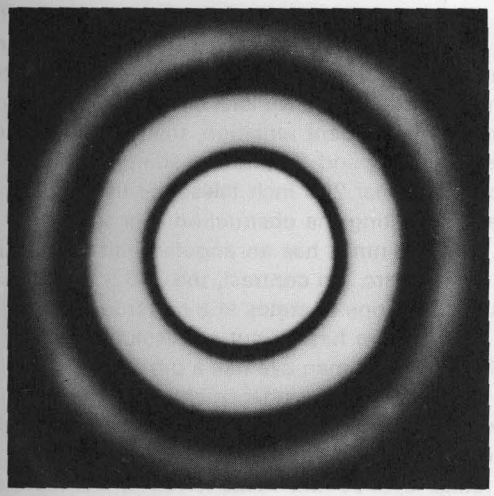


Fig. 10.29 Airy rings (0.5 mm hole diameter). [Photo by E.H.]

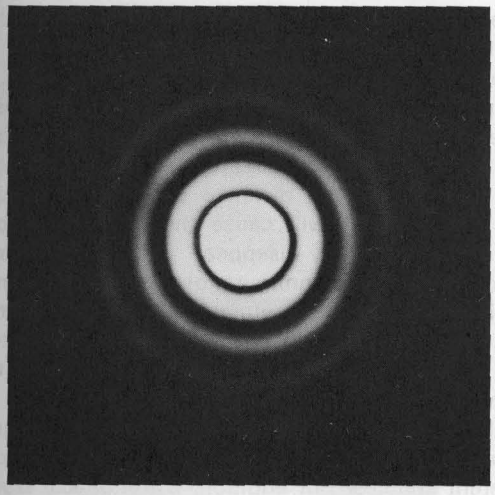
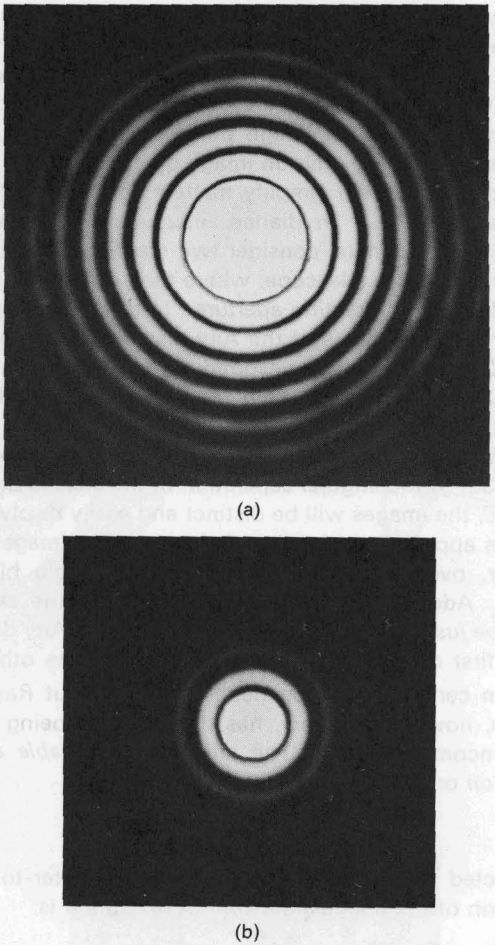


Fig. 10.30 Airy rings (1.0 mm hole diameter). [Photo by E.H.]

Circular apertures are preferable to rectangular ones as far as lens shapes go, since the circle's irradiance curve is broader around the central peak and drops off more rapidly thereafter. Exactly what fraction of the total light energy incident on σ is confined to within the various maxima is a question of interest, but one somewhat too involved to solve here.* On integrating the irradiance over a particular region

* See Born and Wolf, *Principles of Optics*, p. 398 or the very fine elementary text by Towne, *Wave Phenomena*, p. 464.



(a)

(b)

Fig. 10.31 (a) Airy rings—long exposure (1.5 mm hole diameter). (b) Central Airy disc—short exposure with the same aperture. [Photos by E.H.]

of the pattern, one finds that 84% of the light arrives within the Airy disk and 91% within the bounds of the second dark ring.

10.2.6 Resolution of Imaging Systems

Imagine that we have some sort of lens system which forms an image of an extended object. If the object is self-luminous it is likely that we can regard it as made up of an array of incoherent sources. On the other hand an object seen in reflected light will surely display some phase correlation between its various scattering points. When the point sources

are in fact incoherent, the lens system will form an image of the object, which consists of a distribution of partially overlapping, yet independent, Airy patterns. In the very finest lenses, where aberrations have been made negligible, the spreading out of each image point due to diffraction represents the ultimate limit on image quality.

Suppose that we simplify matters somewhat and only examine two equal irradiance, incoherent, distant point sources. For example, consider two stars seen through the objective lens of a telescope, where the entrance pupil corresponds to the diffracting aperture. In the previous section we saw that the radius of the Airy disk was given by $q_1 = 1.22f\lambda/D$. If $\Delta\theta$ is the corresponding angular measure then $\Delta\theta = 1.22\lambda/D$, inasmuch as $q_1/f = \sin\Delta\theta \approx \Delta\theta$. The Airy disk for each star will be spread out over an angular half width $\Delta\theta$ about its geometrical image point, as shown in Fig. 10.32. If the angular separation of the stars is $\Delta\varphi$ and if $\Delta\varphi \gg \Delta\theta$, the images will be distinct and easily resolved. As the stars approach each other their respective images come together, overlap and commingle into a single blend of fringes. Adopting Lord Rayleigh's criterion, the stars are said to be *just resolved* when the center of one Airy disk falls on the first minimum of the Airy pattern of the other star. (We can certainly do a bit better than this, but Rayleigh's criterion, however arbitrary, has the virtue of being particularly uncomplicated.*) The *minimum resolvable angular separation* or *angular limit of resolution* is

$$(\Delta\varphi)_{\min} = \Delta\theta = 1.22\lambda/D \tag{10.59}$$

as depicted in Fig. 10.33. If $\Delta\ell$ is the center-to-center separation of the images, the *limit of resolution* is

$$(\Delta\ell)_{\min} = 1.22f\lambda/D. \tag{10.60}$$

The *resolving power* for an image-forming system is generally defined as either $1/(\Delta\varphi)_{\min}$ or $1/(\Delta\ell)_{\min}$.

If the smallest resolvable separation between images is to be reduced, i.e. if the resolving power is to be increased, the wavelength, for instance, might be made smaller. Using ultraviolet rather than visible light in microscopy allows for the perception of finer detail. The electron microscope utilizes equivalent wavelengths of about 10^{-4} to 10^{-5} that of light. This makes it possible to examine objects which would otherwise be completely obscured by diffraction effects in the visible spectrum. On the other hand, the resolving power of a telescope can be increased by increasing the diameter of the objective lens or mirror. Besides collecting more of the incident radiation, this would also result in a smaller Airy disk and therefore a sharper, brighter image. The Mount Palomar 200 inch telescope has a mirror 5 m in diameter (neglecting the obstruction of a small region at its center). At 550 nm it has an angular limit of resolution of 2.6×10^{-2} s of arc. In contrast, the 250 ft diameter Jodrell Bank radio telescope operates at a rather long 21 cm wavelength. It therefore has a limit of resolution of only about 700 s of arc. The human eye has a pupil diameter which, of course, varies. Taking it under bright conditions to be about 2 mm, with $\lambda = 550$ nm, $(\Delta\varphi)_{\min}$ turns out to be roughly one minute of arc. With a focal length of about 20 mm, $(\Delta\ell)_{\min}$ on the retina is 6700 nm. This is roughly twice the mean spacing between receptors. The human eye should therefore be able to resolve two points, an inch apart, at a distance of some 100 yards. You will probably not be able to do quite that well; one part in one thousand is more likely.

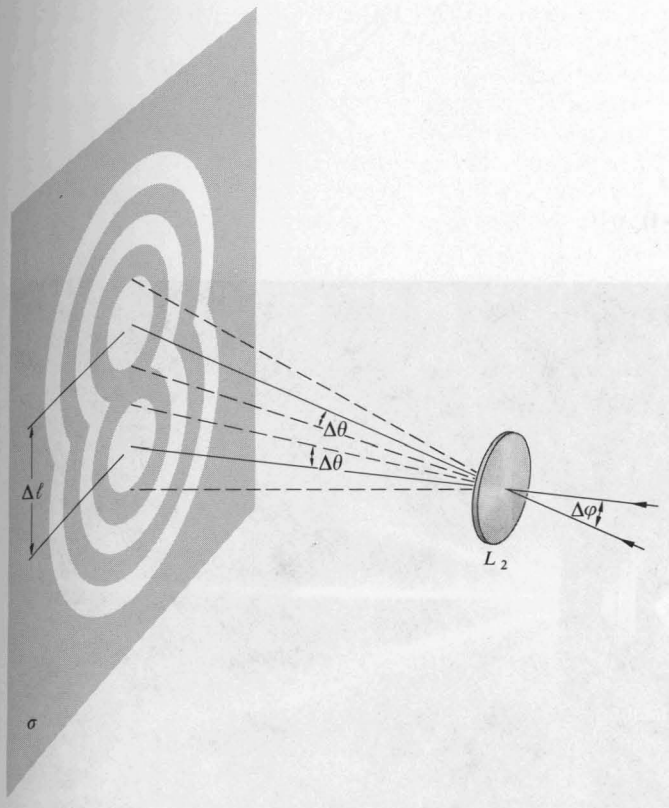
We should mention, before leaving this section, that a more appropriate criterion for resolving power has been proposed by C. Sparrow. Recall that at the Rayleigh limit there is a central minimum or saddle point between adjacent peaks. Decreasing the distance between the two point sources even further will cause the central dip to grow shallower and ultimately disappear. The angular separation corresponding to that configuration is Sparrow's limit. As shown in Fig. 10.40 the resultant maximum has a broad flat top, i.e. at the origin, which is the center of the peak, the second derivative of the irradiance function is zero; there is no change in slope.

Unlike the Rayleigh rule which rather tacitly assumes incoherence, the Sparrow condition can readily be generalized to coherent sources. In addition, astronomical studies of equal-brightness stars have shown that Sparrow's criterion is by far the more realistic.

10.2.7 The Diffraction Grating

A repetitive array of diffracting elements, either apertures or obstacles, which has the effect of producing periodic alterations in the phase, amplitude, or both, of an emergent wave is said to be a *diffraction grating*. One of the simplest such arrangements is the multiple-slit configuration of Section 10.2.3. It seems to have been invented by the American astronomer David Rittenhouse in about 1785. Some years later Joseph von Fraunhofer quite independently

* In Rayleigh's own words: "This rule is convenient on account of its simplicity and it is sufficiently accurate in view of the necessary uncertainty as to what exactly is meant by resolution." See Section 9.8.2 for further discussion.



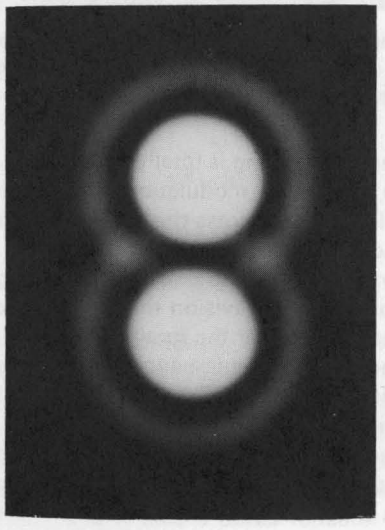
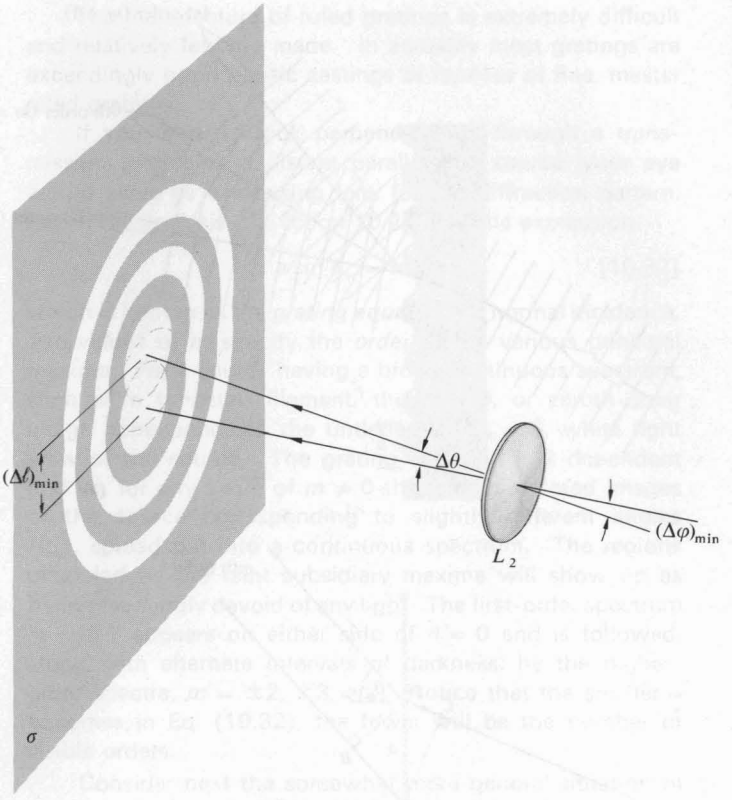


Fig. 10.32 Overlapping images.



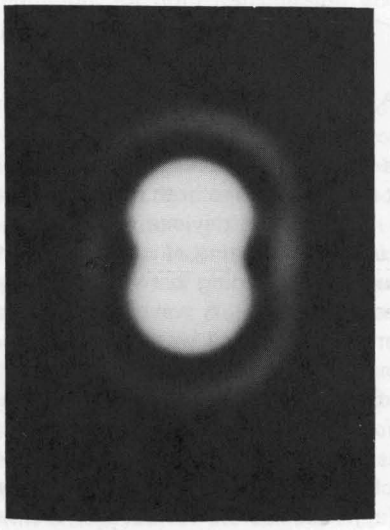


Fig. 10.33 Overlapping images.

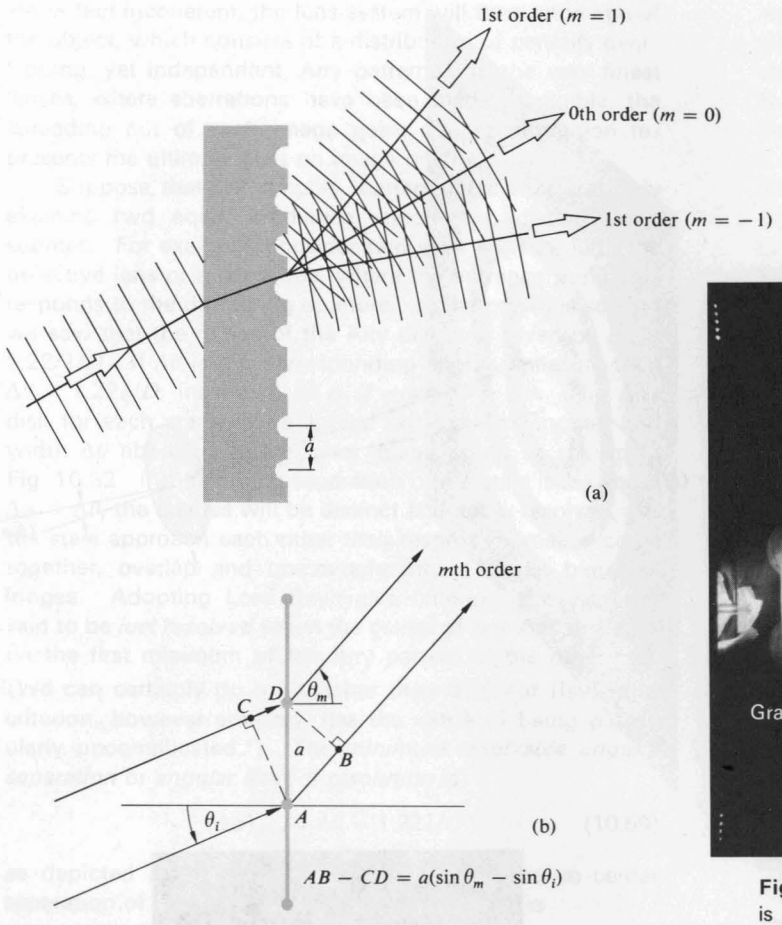


Fig. 10.34 A transmission grating.

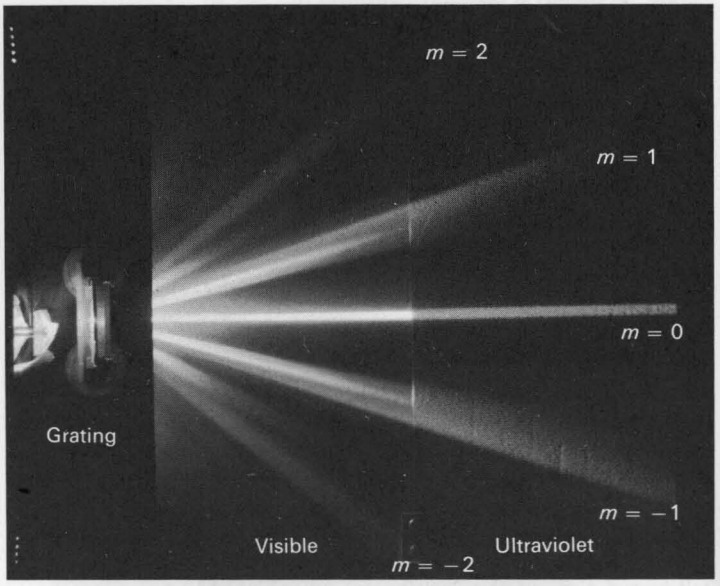


Fig. 10.35 Light passing through a grating. The region on the left is the visible spectrum, that on the right the ultraviolet. [Photo courtesy Klinger Scientific Apparatus Corp.]

rediscovered the principle, and went on to make a number of important contributions to both the theory and technology of gratings. The earliest devices were indeed multiple-slit assemblies, usually consisting of a grid of fine wire or thread wound about and extending between two parallel screws which served as spacers. A wavefront, in passing through such a system, is confronted by alternate opaque and transparent regions, so that it undergoes a modulation in *amplitude*. Accordingly, a multiple-slit configuration is said to be a *transmission amplitude grating*. Another more common form of transmission grating is made by ruling or scratching parallel notches into the surface of a flat, clear glass plate [Fig. 10.34(a)]. Each scratch serves as a source of scattered light and together they form a regular array of parallel line sources. When the grating is totally transparent, so that there is negligible amplitude modulation, the regular variations in the optical thickness across the grating yield a modulation in *phase* and we have what is known as a *transmission phase grating* (Fig. 10.35). In the Huygens–Fresnel representation you can envision the wavelets to be radiated with different phases over the grating surface. An emerging wavefront therefore contains periodic variations in its shape rather than its amplitude. This in turn is equivalent to an angular distribution of constituent plane waves.

On reflection from this kind of grating, light scattered by the various periodic surface features will arrive at some point P with a definite phase relationship. The consequent interference pattern generated after reflection is quite

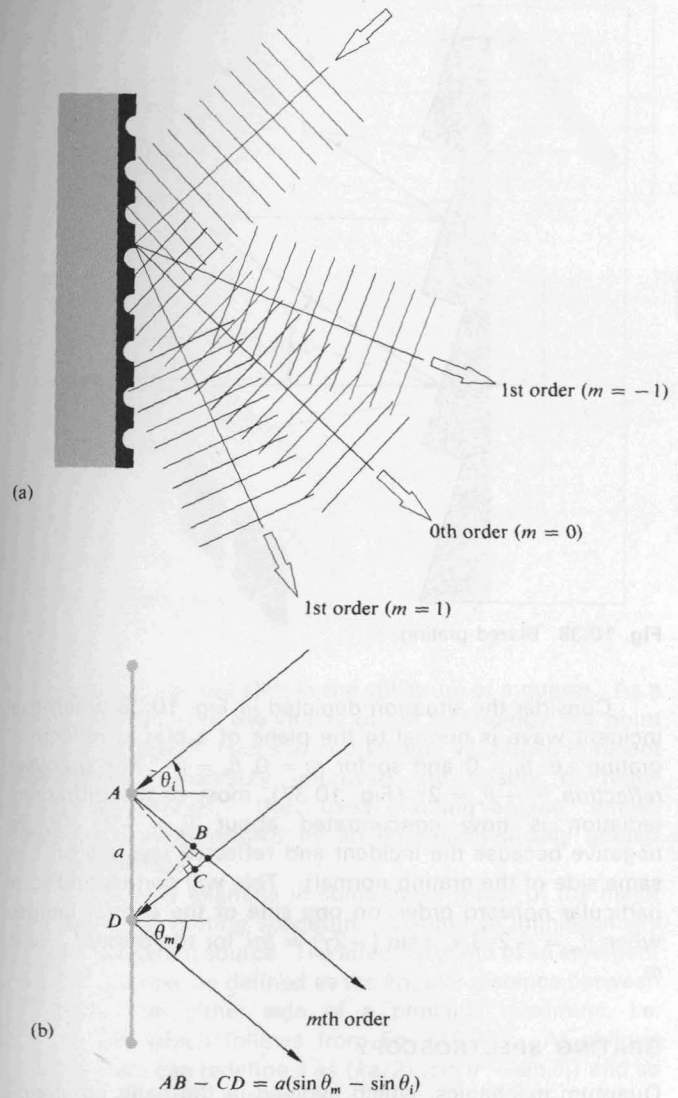


Fig. 10.36 A reflection grating.

similar to that arising from transmission. Gratings designed specifically to function in this fashion are known as *reflection phase gratings* (Fig. 10.36). Contemporary gratings of this sort are generally ruled in thin films of aluminum which have been evaporated onto optically flat glass blanks. The aluminum, being fairly soft, results in less wear on the diamond ruling tool and is also a better reflector in the ultraviolet region.

The manufacture of ruled gratings is extremely difficult and relatively few are made. In actuality most gratings are exceedingly good plastic castings or *replicas* of fine, master ruled gratings.

If you were to look perpendicularly through a transmission grating at a distant parallel line source, your eye would serve as a focusing lens for the diffraction pattern. Recall the analysis of Section 10.2.3 and the expression

$$a \sin\theta_m = m\lambda, \qquad [10.32]$$

which is known as the *grating equation* for normal incidence. The values of m specify the *order* of the various principal maxima. For a source having a broad continuous spectrum, such as a tungsten filament, the $m = 0$, or zeroth-order image corresponds to the undeflected, $\theta_0 = 0$, white light view of the source. The grating equation is λ dependent and so, for any value of $m \neq 0$ the various colored images of the source corresponding to slightly different angles (θ_m), spread out into a continuous spectrum. The regions occupied by the faint subsidiary maxima will show up as bands seemingly devoid of any light. The first-order spectrum $m = \pm 1$ appears on either side of $\theta = 0$ and is followed, along with alternate intervals of darkness, by the higher-order spectra, $m = \pm 2, \pm 3$, etc. Notice that the smaller a becomes in Eq. (10.32), the fewer will be the number of visible orders.

Consider next the somewhat more general situation of oblique incidence as depicted in Figs. 10.34 and 10.36. The grating equation, for both transmission and reflection, becomes

$$a(\sin\theta_m - \sin\theta_i) = m\lambda. \qquad (10.61)$$

This expression applies equally well regardless of the refractive index of the transmission grating itself (Problem 10.8). One of the main disadvantages of the devices examined thus far, and in fact the reason for their present obsolescence, is that they spread the available light energy out over a number of low-irradiance spectral orders. For a grating like that shown in Fig. 10.36 most of the incident light undergoes *specular reflection* as if from a plane mirror. It follows from the grating equation that $\theta_m = \theta_i$ corresponds to the zeroth order, $m = 0$. All of this light is essentially wasted, at least for spectroscopic purposes, since the constituent wavelengths overlap.

In an article in the *Encyclopaedia Britannica* of 1888 Lord Rayleigh suggested that it was at least theoretically possible to shift energy out of the useless zeroth order into one of the higher-order spectra. So motivated, Robert

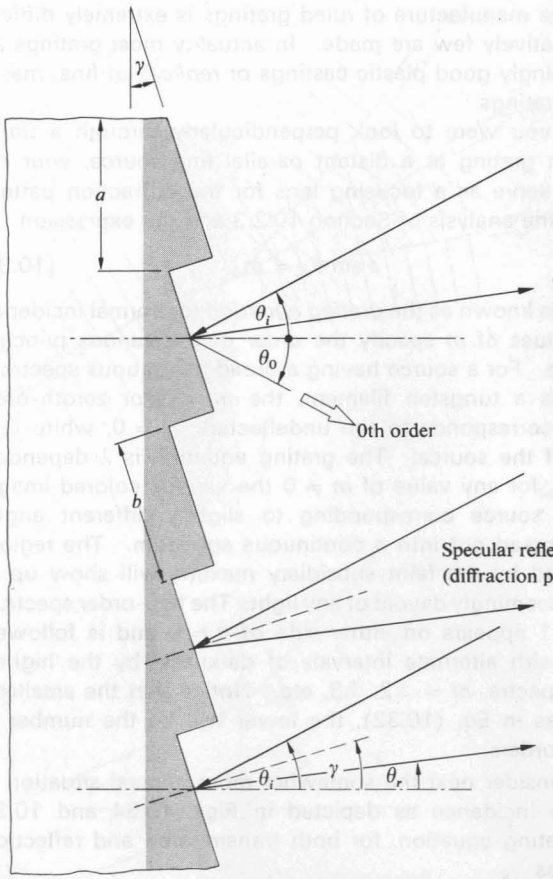


Fig. 10.37 Section of a blazed reflection phase grating.

Williams Wood (1868–1955) succeeded in 1910 in ruling grooves having a controlled shape, as shown in Fig. 10.37. Most modern gratings are of this shaped or *blazed* variety. The angular positions of the nonzero orders, θ_m-values, are determined by a, λ and, of more immediate interest, θ_i. But θ_i and θ_m are measured from the normal to the grating plane and not with respect to the individual groove surfaces. On the other hand, the location of the peak in the single facet diffraction pattern corresponds to *specular reflection* off that face, of each groove. It is governed by the *blaze angle* γ and can be varied independently of θ_m. This is somewhat analogous to the antenna array of Section 10.1.3 where we were able to control the spatial position of the interference pattern (10.6) by adjusting the relative phase shift between sources without actually changing their orientations.

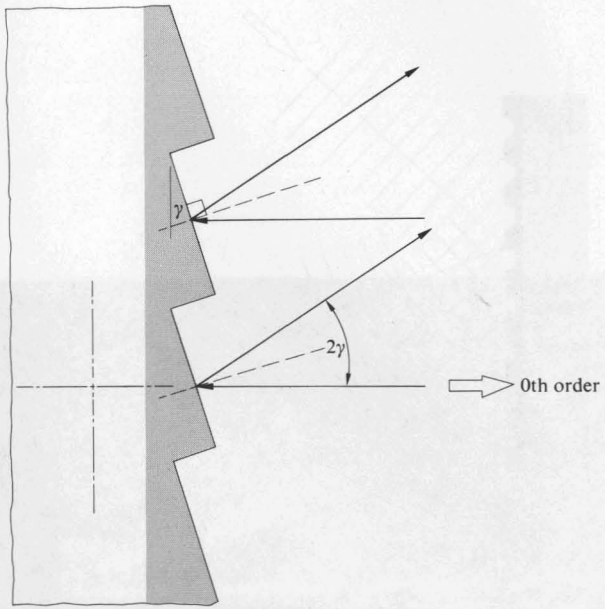


Fig. 10.38 Blazed grating.

Consider the situation depicted in Fig. 10.38 when the incident wave is normal to the plane of a blazed reflection grating i.e. $\theta_i = 0$ and so for $m = 0$, $\theta_0 = 0$. For *specular reflection* $\theta_i - \theta_r = 2\gamma$ (Fig. 10.37), most of the diffracted radiation is now concentrated about $\theta_r = -2\gamma$ (θ_r is negative because the incident and reflected rays are on the same side of the grating normal). This will correspond to a particular nonzero order, on one side of the central image, when $\theta_m = -2\gamma$ i.e. $a \sin(-2\gamma) = m\lambda$ for the desired λ and m.

GRATING SPECTROSCOPY

Quantum mechanics, which evolved in the early nineteen-twenties, had its initial thrust in the area of atomic physics. Predictions were made concerning the detailed structure of the hydrogen atom as manifested by its emitted radiation, and spectroscopy provided the vital proving ground. The need for larger and better gratings became apparent. Grating spectrometers, used over the range from soft x-rays to the far infrared, have enjoyed continued interest. In the hands of the astrophysicist, or rocket borne, they are yielding information concerning the very origins of the universe, information as varied as the temperature of a star, the rotation

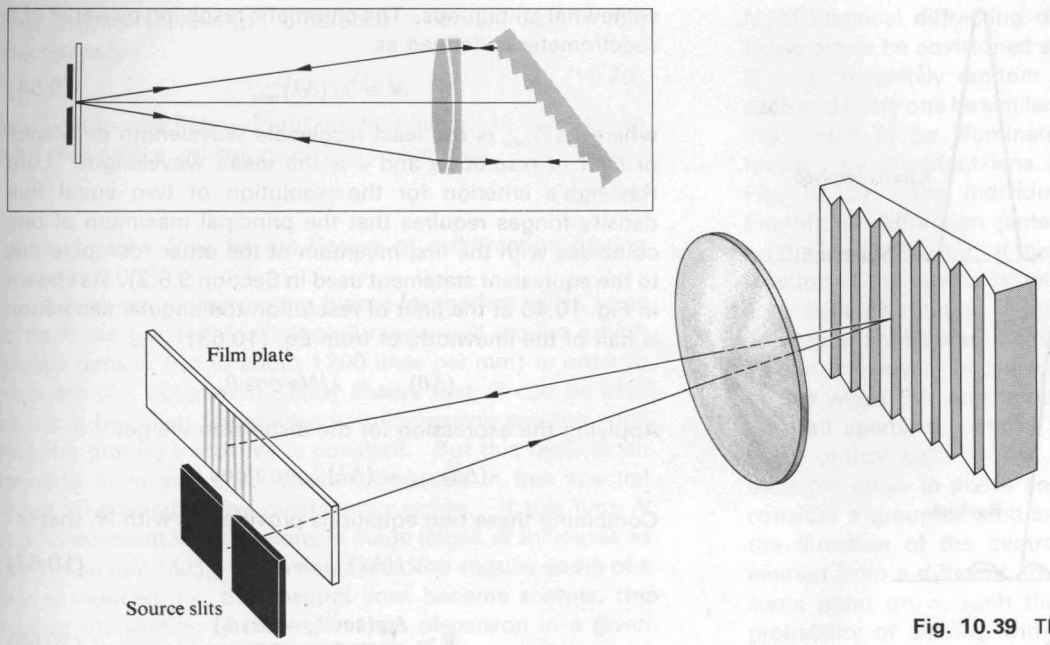


Fig. 10.39 The Littrow autocollimation mounting.

of a galaxy or the red shift in the spectrum of a quasar. As a further indication of the vitality of the technology, we point out that George R. Harrison and George W. Stroke in recent times have remarkably improved the quality of high-resolution gratings. They used a ruling engine* whose operation was controlled by an interferometrically guided servomechanism.

Let us now examine in some detail a few of the major features of the grating spectrum. Assume an infinitesimally narrow incoherent source. The effective width of an emergent spectral line may be defined as the angular distance between the zeroes on either side of a principal maximum, i.e. $\Delta\alpha = 2\pi/N$, which follows from Eq. (10.33). At oblique incidence we can redefine α as $(ka/2)(\sin\theta - \sin\theta_i)$ and so a *finite* change in α is given by

$$\Delta\alpha = (ka/2)\cos\theta(\Delta\theta) = 2\pi/N, \tag{10.62}$$

where the angle of incidence is constant, i.e. $\Delta\theta_i = 0$. Thus even when the incident light is monochromatic

$$\Delta\theta = 2\lambda/(Na\cos\theta_m) \tag{10.63}$$

is the *angular width of a line*, due to *instrumental broadening*. Interestingly enough, the angular linewidth varies inversely with the width of the grating itself, Na. Another quantity of import is the difference in angular position corresponding to a difference in wavelength. The *angular dispersion*, as in the case of a prism, is defined as

$$\mathscr{D} \equiv d\theta/d\lambda. \tag{10.64}$$

Differentiating the grating equation yields

$$\mathscr{D} = m/a\cos\theta_m. \tag{10.65}$$

This means that the angular separation between two different frequency lines will increase as the order increases.

Blazed plane gratings with nearly rectangular grooves are most often mounted so that the incident propagation vector is almost normal to either one of the groove faces. This is the condition of *autocollimation*, in which θ_i and θ_m are on the same side of the normal and $\gamma \approx \theta_i \approx -\theta_m$ (see Fig. 10.39) whereupon

$$\mathscr{D}_{\text{auto}} = 2\tan\theta_i/\lambda \tag{10.66}$$

which is independent of a.

When the wavelength difference between two lines is small enough so that they overlap, the resultant peak becomes

* For more details about these marvelous machines see A. R. Ingalls, *Sci. Amer.* **186,** 45 (1952) or the article by E. W. Palmer and J. F. Verrill, *Contemp. Phys.* **9,** 257 (1968).

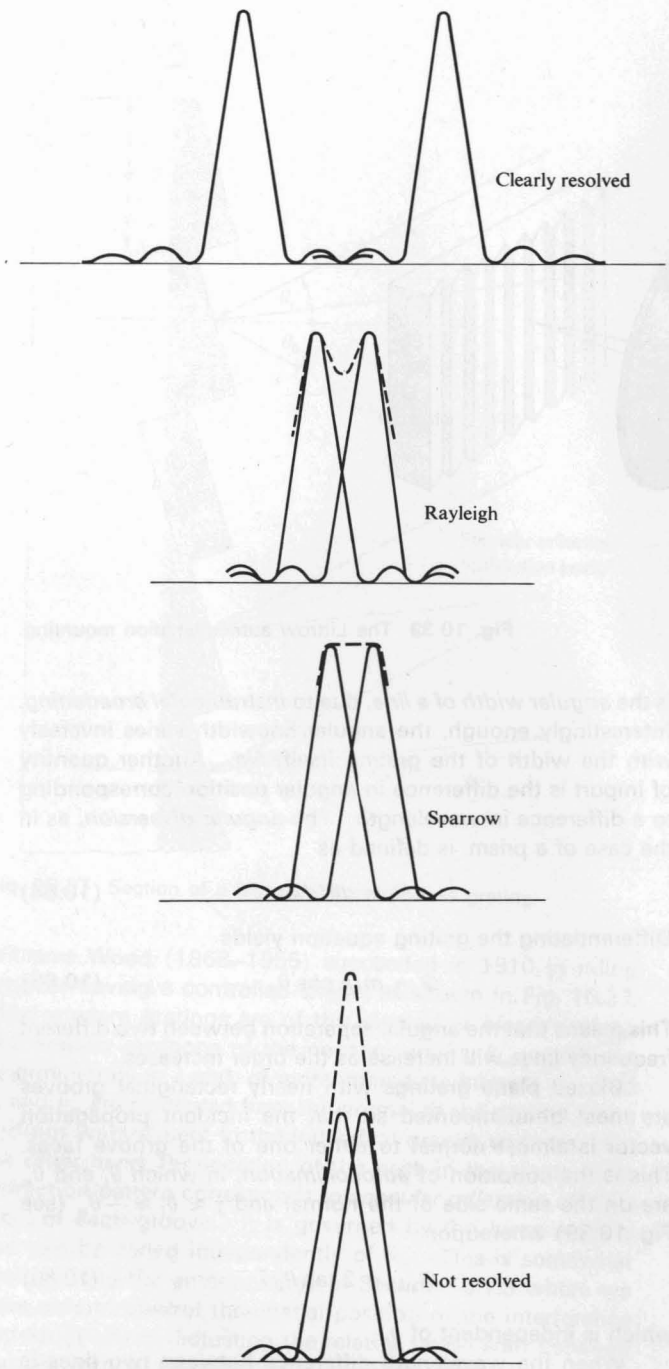


Fig. 10.40 Overlapping point images.

somewhat ambiguous. The chromatic resolving power $\mathscr{R}$ of a spectrometer is defined as

$$\mathscr{R} \equiv \lambda/(\Delta\lambda)_{\min} \qquad [9.54]$$

where $(\Delta\lambda)_{\min}$ is the least resolvable wavelength difference or *limit of resolution* and λ is the mean wavelength. Lord Rayleigh's criterion for the resolution of two equal flux density fringes requires that the principal maximum of one coincides with the first minimum of the other (compare this to the equivalent statement used in Section 9.8.2). As shown in Fig. 10.40 at the limit of resolution the angular separation is half of the linewidth, or from Eq. (10.63)

$$(\Delta\theta)_{\min} = \lambda/Na \cos\theta_m.$$

Applying the expression for the dispersion we get

$$(\Delta\theta)_{\min} = (\Delta\lambda)_{\min} m/a \cos\theta_m.$$

Combining these two equations provides us with $\mathscr{R}$, that is

$$\lambda/(\Delta\lambda)_{\min} = mN \qquad (10.67)$$

or

$$\mathscr{R} = \frac{Na(\sin\theta_m - \sin\theta_i)}{\lambda}. \qquad (10.68)$$

The resolving power is a function of the grating width Na, the angle of incidence and λ. A grating 6 inches wide and containing 15,000 lines per inch will have a total of 9×10^4 lines and a resolving power, in the second order, of 1.8×10^5. In the vicinity of 540 nm the grating could resolve a wavelength difference of 0.003 nm. Notice that the resolving power cannot exceed $2Na/\lambda$, which occurs when $\theta_i = -\theta_m = 90°$. The largest values of $\mathscr{R}$ are obtained when the grating is used in autocollimation, whereupon

$$\mathscr{R}_{\text{auto}} = \frac{2Na \sin\theta_i}{\lambda} \qquad (10.69)$$

and again θ_i and θ_m are both on the same side of the normal. For one of Harrison's 260 mm wide blazed gratings at about 75° in a Littrow mount, with $\lambda = 500$ nm, the resolving power just exceeds 10^6.

We now need to consider the problem of overlapping of orders. The grating equation makes it quite clear that a line of 600 nm in the first order will have precisely the same position, θ_m, as a 300 nm line in the second order or a 200 nm line when $m = 3$. If two lines of wavelength λ and $(\lambda + \Delta\lambda)$ in successive orders $(m + 1)$ and m just coincide then

$$a(\sin\theta_m - \sin\theta_i) = (m + 1)\lambda = m(\lambda + \Delta\lambda).$$

That precise wavelength difference is known as the *free spectral range*

$$(\Delta\lambda)_{\text{fsr}} = \lambda/m \tag{10.70}$$

as it was for the Fabry–Perot interferometer. In comparison with that device, whose resolving power was

$$\mathscr{R} = \mathscr{F}m \tag{9.54}$$

we might take N to be the finesse of a diffraction grating (Problem 10.9).

A high-resolution grating blazed for the first order, so as to have the greatest free spectral range, will require a high groove density (up to about 1200 lines per mm) in order to maintain $\mathscr{R}$. Equation (10.68) shows that $\mathscr{R}$ can be kept constant by ruling fewer lines with increasing spacing such that the grating width Na is constant. But this requires an increase in m and a subsequent decrease in free spectral range, characterized by overlapping orders. If this time N is held constant while a alone is made larger, $\mathscr{R}$ increases as does m so that $(\Delta\lambda)_{\text{fsr}}$ again decreases; the angular width of a line is reduced, i.e. the spectral lines become sharper, the coarser the grating becomes, but the dispersion in a given order diminishes, with the effect that the lines in that spectrum approach each other.

Thus far we have considered a particular type of periodic array, namely the *line grating*. A good deal more information is available in the literature* concerning their shapes, mountings, uses etc.

There are a few unlikely household items which can be used as crude gratings, along with a small light source. The grooved surface of a phonograph record works nicely near grazing incidence. And surprisingly enough, under the same conditions an ordinary fine-toothed comb will separate out the constituent wavelengths of white light. This occurs in exactly the same fashion as it would with a more orthodox reflection grating. In a letter to a friend dated May 12, 1673 James Gregory pointed out that sunlight passing through a feather would produce a colored pattern and he asked that his observations be conveyed to Mr. Newton. If you've got one, a feather makes a nice transmission grating.

TWO- AND THREE-DIMENSIONAL GRATINGS

Suppose that the diffracting screen Σ contains a large number, N, of identical diffracting objects (apertures or obstacles). These are to be envisioned as distributed over the surface of Σ in a completely random manner. We also require that each and every one be similarly oriented. Imagine the diffracting screen to be illuminated by plane waves which are focused by a perfect lens L_2, after emerging from Σ (see Fig. 10.15). The individual apertures generate identical Fraunhofer diffraction patterns. All of these then overlap on the image plane σ. If there is no regular periodicity in the location of the apertures, we cannot anticipate anything but a random distribution in the relative phases of the waves arriving at an arbitrary point P on σ. We have to be rather careful, however, because there is one exception which occurs when P is on the central axis, i.e. $P = P_0$. All rays, from all apertures, parallel to the central axis will traverse equal optical path lengths before reaching P_0. They will therefore arrive in phase and interfere constructively. Now consider a group of arbitrarily directed parallel rays (not in the direction of the central axis), each one of which is emitted from a different aperture. These will be focused at some point on σ, such that each ray will have an equal probability of arriving with any phase between 0 and 2π. Interference effects amongst these rays should therefore, on the average, negate each other* (Section 7.1.1). We can expect to see the result of N exactly overlapping and *effectively noninteracting* Fraunhofer patterns. The flux density in a particular region on σ will therefore be N times the individual aperture flux density diffracted in that direction. In addition, a bright spot will exist at the very center of the pattern where the flux density is N^2 times that of the individual apertures. If, for example, the screen contains N rectangular holes [Fig. 10.41(a)], the resultant pattern [Fig. 10.41(b)] will resemble Fig. 10.24.† Similarly, an array of circular holes depicted in Fig. 10.41(c) will produce the diffraction rings of Fig. 10.41(d).

As the number of apertures increases, there will be a tendency for the central spot to become so bright as to obscure the rest of the pattern. Note as well that the above considerations apply when all of the apertures are illuminated completely coherently. In actuality, the diffracted flux-

* See F. Kneubühl, "Diffraction Grating Spectroscopy," *Appl. Opt.* **8**, 505 (1969), R. S. Longhurst, *Geometrical and Physical Optics* and the extensive article by G. W. Stroke in the *Encyclopedia of Physics*, Vol. 29, Edited by S. Flügge, p. 426.

* For a statistical treatment, consult J. M. Stone, *Radiation and Optics*, p. 146. Make sure to examine Section 11.3.3(ii) herein, i.e., *The Array Theorem*.

† The radial fiber structure evident in Fig. 10.41(b) arises from the statistical nature of the process. See e.g. Sommerfeld, *Optics*, p. 194. Also take a look at "Diffraction Plates for Classroom Demonstrations" by R. B. Hoover, *Am. J. Phys.* **37**, 871 (1969).

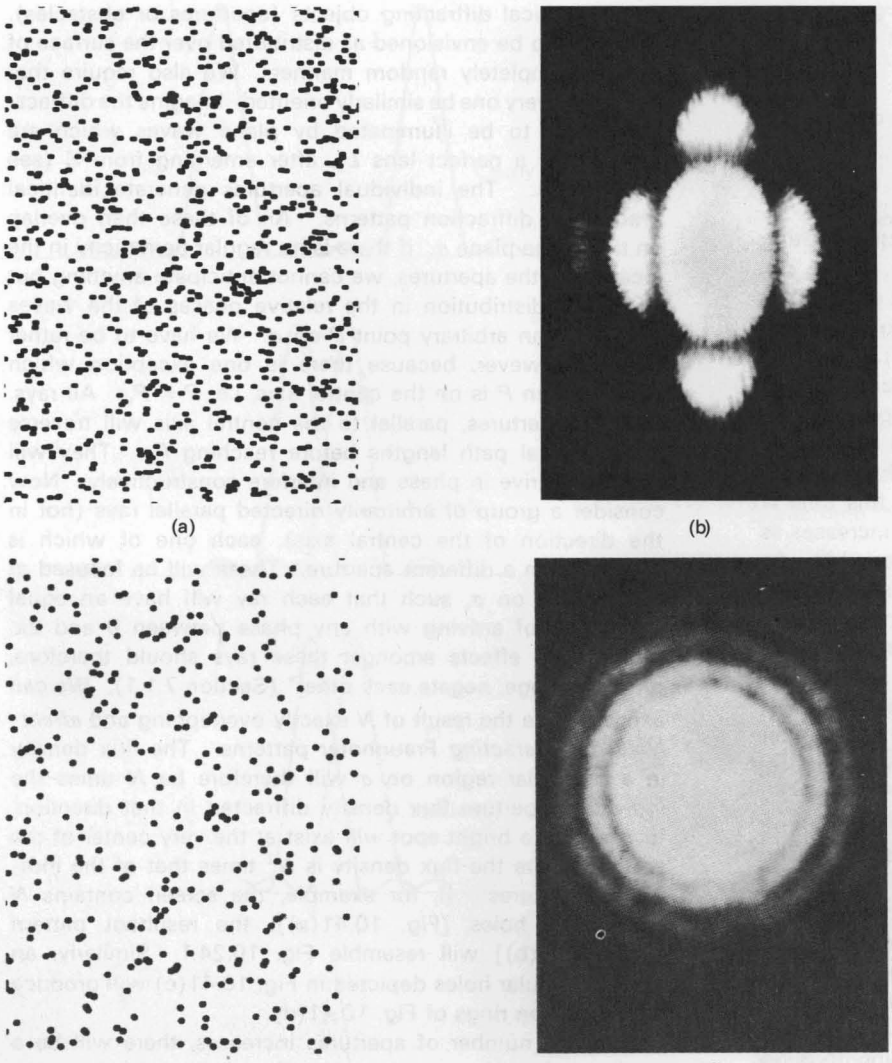


Fig. 10.41 (a) A random array of rectangular apertures. (b) The resulting white-light Fraunhofer pattern. (c) A random array of circular apertures. (d) The resulting white-light Fraunhofer pattern. [Photos courtesy The Ealing Corporation and Richard B. Hoover.]

density distribution will be determined by the degree of coherence (see Chapter 12). The pattern will run the gamut from no interference with completely incoherent light, to the case discussed above for completely coherent illumination (Problem 10.20).

The same kind of effects arise from what we might call a two dimensional *phase*-grating. For example, the halo or corona often seen about the sun or moon results from diffraction by random droplets of water vapor, i.e. cloud particles. If you would like to duplicate the effect, rub a very thin film of talcum powder on a microscope slide and then fog it up with your breath. Look at a white light point source. You should see a pattern of clear, concentric, colored rings (10.56) surrounding a white central disk. If you just see a white blur, you don't have a distribution of roughly equal sized droplets; have another try at the talcum. Strikingly beautiful patterns approximating concentric ring systems can be seen through an ordinary *mesh* nylon stocking. If

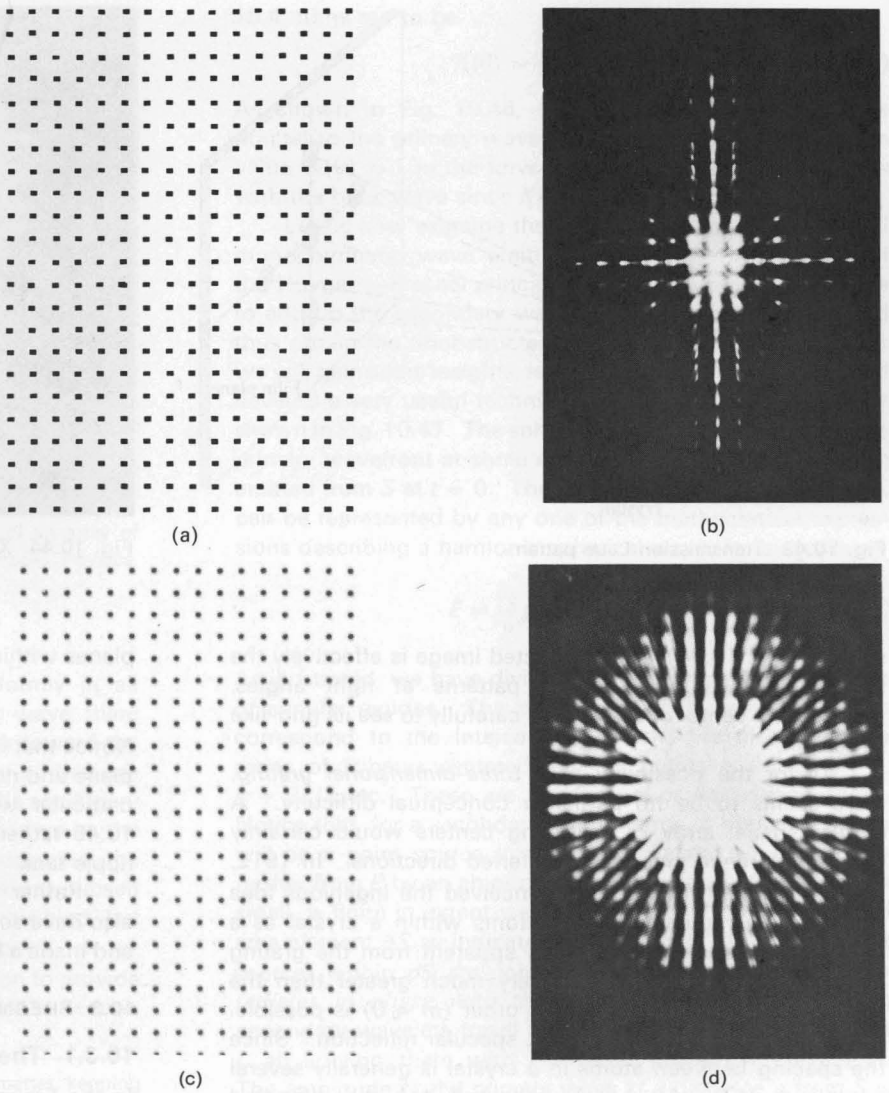


Fig. 10.42 (a) An ordered array of rectangular apertures. (b) The resulting white-light Fraunhofer pattern. (c) An ordered array of circular apertures. (d) The resulting white-light Fraunhofer pattern. [Photos courtesy Richard B. Hoover.]

you are fortunate enough to have mercury-vapor street lights, you'll have no trouble seeing all of their constituent visible spectral frequencies. (If not, block out most of a fluorescent lamp, leaving something resembling a small source.) Notice the increased symmetry as you increase the number of layers of nylon. Incidentally, this is precisely the way Rittenhouse, the inventor of the grating, became interested in the problem, only he used a silk handkerchief.

Consider the case of a *regular* two-dimensional array of diffracting elements (Fig. 10.42) under normally incident plane wave illumination. Each small element behaves as a coherent source. And because of the regular periodicity of the lattice of emitters, each emergent wave bears a fixed phase relation to the others. There will now be certain directions in which constructive interference prevails. Obviously, these occur when the distances from each diffracting element to P are such that the waves are nearly in phase at arrival. The phenomenon can be observed by looking at a point source through a piece of *square woven*, thin cloth (like nylon curtain material) or the fine metal mesh of a tea

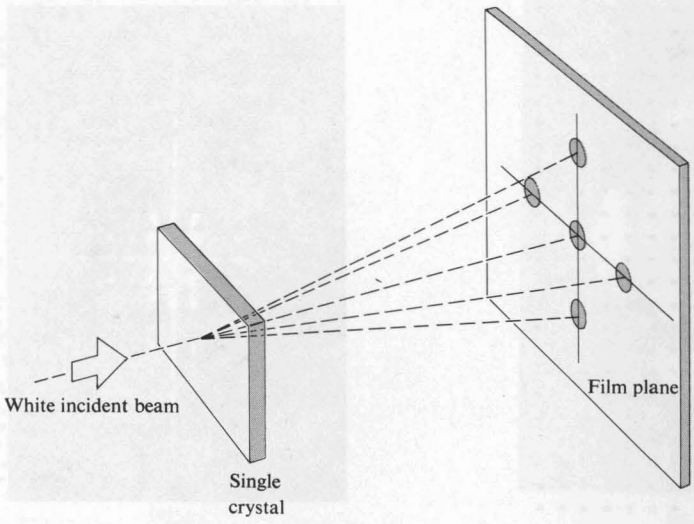


Fig. 10.43 Transmission Laue pattern.

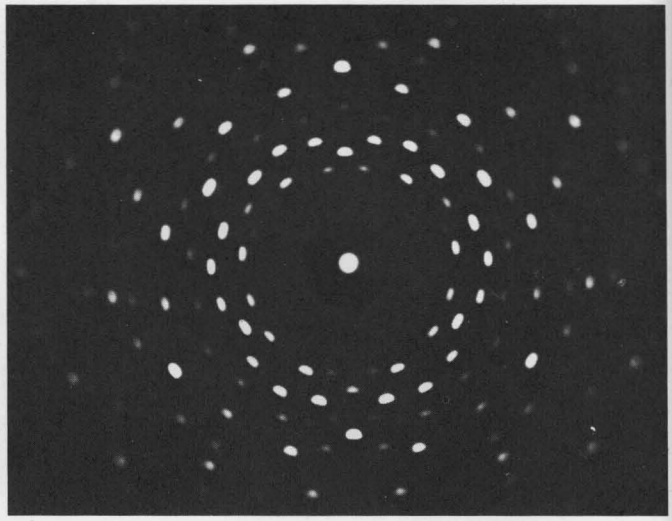


Fig. 10.44 X-ray diffraction pattern for quartz (SiO_2).

strainer (Fig. 10.84). The diffracted image is effectively the superposition of two grating patterns at right angles. Examine the center of the pattern carefully to see its grid-like structure.

As for the possibility of a *three-dimensional grating*, there seems to be no particular conceptual difficulty. A regular spatial array of scattering centers would certainly yield interference maxima in preferred directions. In 1912, Max van Laue (1879–1960) conceived the ingenious idea of using the regularly spaced atoms within a crystal as a three-dimensional grating. It is apparent from the grating equation (10.61) that if λ is very much greater than the grating spacing, only the zeroth order ($m = 0$) is possible. This is equivalent to $\theta_0 = \theta_i$, i.e. specular reflection. Since the spacing between atoms in a crystal is generally several angstroms ($1\text{Å} = 10^{-1}$ nm), light can only be diffracted in the zeroth order.

Von Laue's solution to the problem was to probe the lattice, not with light, but with x-rays whose wavelengths were comparable to the interatomic distances (Fig. 10.43). A narrow beam of white radiation (the broad continuous frequency range emitted by an x-ray tube) was directed onto a thin single crystal. The film plate (Fig. 10.44) revealed a Fraunhofer pattern consisting of an array of precisely located spots. These sites of constructive interference occurred whenever the angle between the beam and a set of atomic planes within the crystal obeyed Bragg's law:

$$2d \sin \theta = m\lambda. \tag{10.71}$$

Notice that in x-ray work θ is traditionally measured from the plane and not the normal to it. Each set of planes diffracts a particular wavelength into a particular direction. Figure 10.45 rather strikingly shows the analogous behavior in a ripple tank.

Rather than reducing λ to the x-ray range, we could also have scaled everything up by a factor of about a billion and made a lattice of metal balls as a grating for microwaves.

10.3 FRESNEL DIFFRACTION

10.3.1 The Free Propagation of a Spherical Wave

In the Fraunhofer configuration, the diffracting system was relatively small, while the point of observation was very distant. Under these circumstances a few potentially problematic features of the Huygens–Fresnel principle could be completely passed over without concern. But we are now dealing with the near-field region which extends right up to the diffracting element itself and any such approximations would be inappropriate. We therefore return to the Huygens–Fresnel principle in order to reexamine it more closely. At any instant, every point on the primary wavefront is envisioned as a continuous emitter of spherical secondary

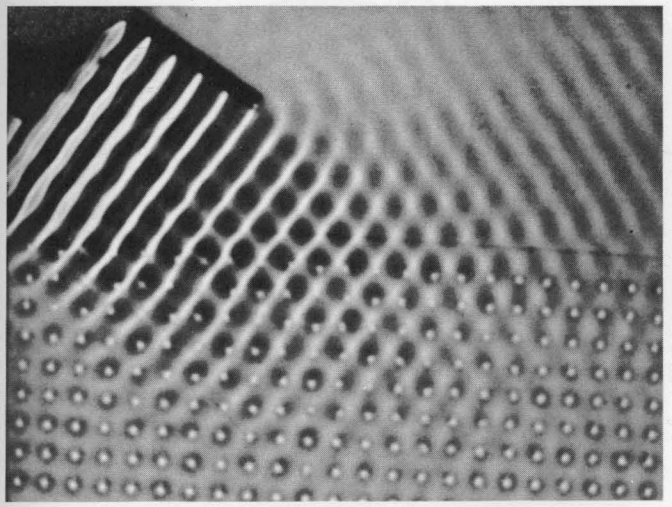


Fig. 10.45 Water waves in a ripple tank reflecting off an array of pegs acting as point scatterers. [Photo courtesy PSSC *Physics*, D. C. Heath, Boston, 1960.]

wavelets. But if each wavelet radiated uniformly in all directions, in addition to generating an ongoing wave, there would also appear a reverse wave traveling back toward the source. No such wave is found experimentally so that we must somehow modify the radiation pattern of the secondary emitters. We now introduce a function $K(\theta)$, known as the *obliquity* or *inclination factor*, in order to describe the directionality of the secondary emissions. Fresnel himself recognized the need to introduce a quantity of this kind, but he did little more than conjecture about its form.* It remained for the more analytic Kirchhoff formulation to provide an actual expression for $K(\theta)$ which, as we will see in Section 10.4, turns out to be

$$K(\theta) = \tfrac{1}{2}(1 + \cos\theta). \tag{10.72}$$

As shown in Fig. 10.46, θ is the angle made with the normal to the primary wavefront, **k**. This has its maximum value, $K(0) = 1$ in the forward direction and also dispenses with the back wave since $K(\pi) = 0$.

Let us now examine the free propagation of a spherical monochromatic wave emitted from a *point source S*. If the Huygens–Fresnel principle is correct, we should be able to add up the secondary wavelets arriving at a point P and thus obtain the unobstructed primary wave. In the process we will gain some insights, recognize a few shortcomings and develop a very useful technique. Consider the construction shown in Fig. 10.47. The spherical surface corresponds to the primary wavefront at some arbitrary time t' after it has been emitted from S at $t = 0$. The disturbance, having a radius ρ, can be represented by any one of the mathematical expressions describing a harmonic spherical wave, for example,

$$E = \frac{\mathcal{E}_0}{\rho}\cos(\omega t' - k\rho). \tag{10.73}$$

As illustrated, we have divided the wavefront into a number of annular regions. The boundaries of the various regions correspond to the intersections of the wavefront with a series of spheres centered at P of radius $r_0 + \lambda/2$, $r_0 + \lambda$, $r_0 + 3\lambda/2$ etc. These are the *Fresnel* or *half-period zones*. Notice that, for a secondary point source in one zone, there will be a point source in the next adjacent zone which is further from P by an amount $\lambda/2$. Since each zone, although small, is finite in extent we define a ring-shaped differential area element dS as indicated in Fig. 10.48. All of the point sources within dS are coherent and *we assume that each radiates in phase with the primary wave* (10.73). The secondary wavelets travel a distance r to reach P, at a time t, all arriving there with the same phase, $\omega t - k(\rho + r)$. The amplitude of the primary wave at a distance ρ from S is $\mathcal{E}_0/\rho$. We assume, accordingly, that the source strength per unit area $\mathcal{E}_A$ of the secondary emitters on dS is proportional to $\mathcal{E}_0/\rho$ by way of a constant Q, i.e. $\mathcal{E}_A = Q\mathcal{E}_0/\rho$. The contribution to the optical disturbance at P from the secondary sources on dS is, therefore,

$$dE = K\frac{\mathcal{E}_A}{r}\cos[\omega t - k(\rho + r)]\,dS. \tag{10.74}$$

The obliquity factor must vary slowly and may be presumed constant over a single Fresnel zone. To get dS as a function

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* It is interesting to read Fresnel's own words on the matter, keeping in mind that he was talking about light as an elastic vibration of the ether.

> Since the impulse communicated to every part of the primitive wave was directed along the normal, the motion which each tends to impress upon the ether ought to be more intense in this direction than in any other; and the rays which would emanate from it, if acting alone, would be less and less intense as they deviated more and more from this direction.
>
> The investigation of the law according to which their intensity varies about each center of disturbance is doubtless a very difficult matter; . . .

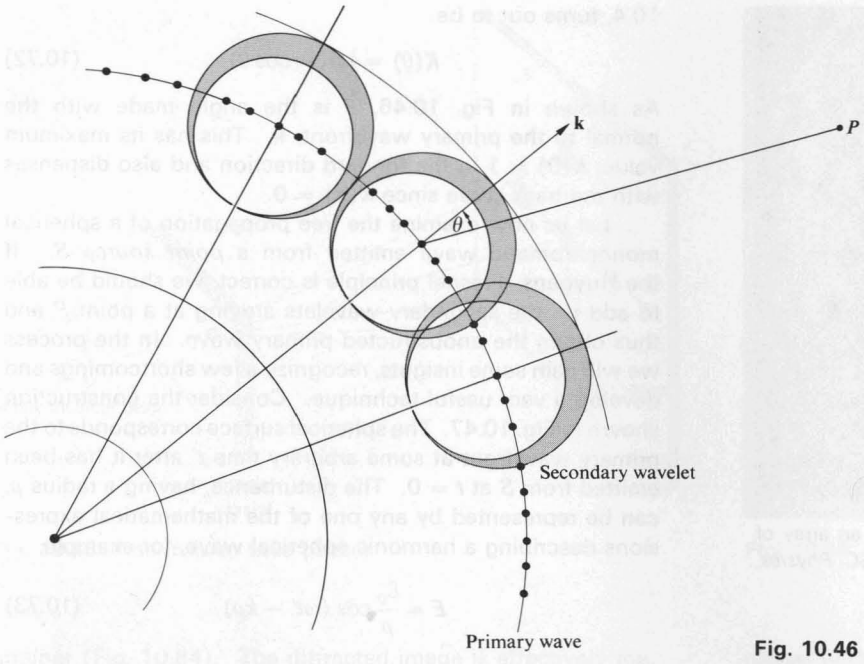


Fig. 10.46 Secondary wavelets.

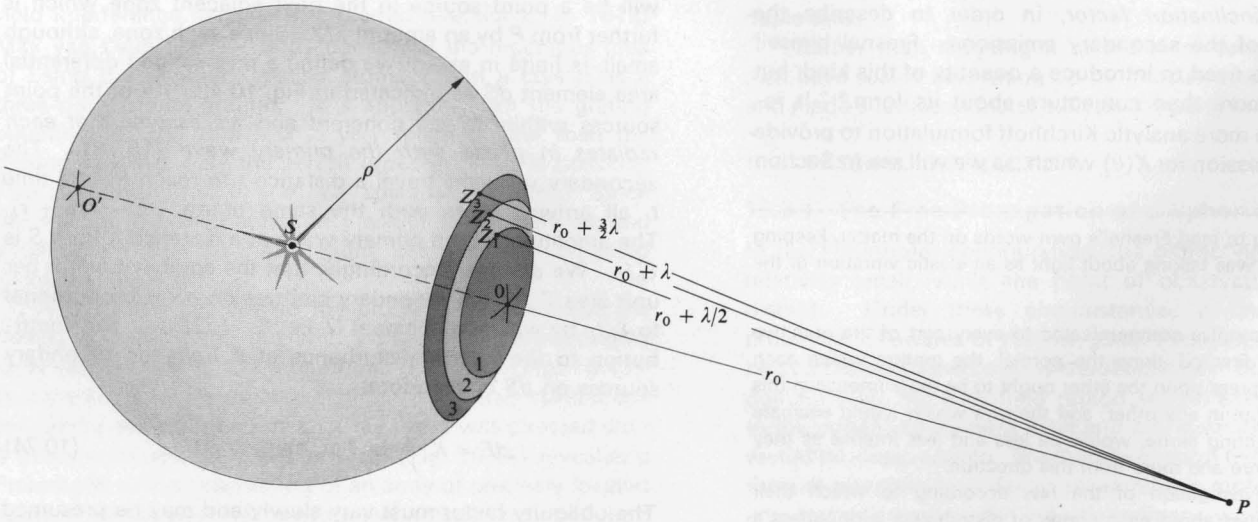


Fig. 10.47 Propagation of a spherical wavefront.

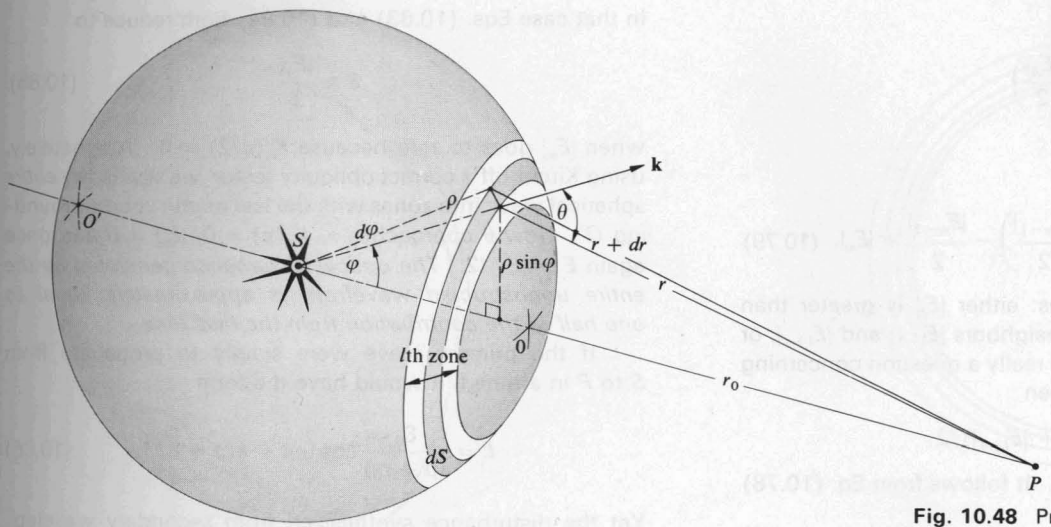


Fig. 10.48 Propagation of a spherical wavefront.

of r begin with

$$dS = \rho\, d\varphi\, 2\pi(\rho \sin \varphi),$$

applying the law of cosines get

$$r^2 = \rho^2 + (\rho + r_0)^2 - 2\rho(\rho + r_0)\cos\varphi.$$

Upon differentiation this yields

$$2r\, dr = 2\rho(\rho + r_0)\sin\varphi\, d\varphi$$

holding ρ and r_0 constant. Making use of the value of $d\varphi$ we find that the area of the element is therefore

$$dS = 2\pi \frac{\rho}{(\rho + r_0)} r\, dr. \tag{10.75}$$

The disturbance arriving at P from the lth zone is

$$E_l = K_l 2\pi \frac{\mathcal{E}_A \rho}{(\rho + r_0)} \int_{r_{l-1}}^{r_l} \cos\left[\omega t - k(\rho + r)\right] dr$$

Hence

$$E_l = \frac{-K_l \mathcal{E}_A \rho \lambda}{(\rho + r_0)} \left[\sin\left(\omega t - k\rho - kr\right)\right]_{r=r_{l-1}}^{r=r_l}.$$

Upon the introduction of $r_{l-1} = r_0 + (l-1)\lambda/2$ and $r_l = r_0 + l\lambda/2$ the expression reduces (Problem 10.10) to

$$E_l = (-1)^{l+1} \frac{2K_l \mathcal{E}_A \rho \lambda}{(\rho + r_0)} \sin\left[\omega t - k(\rho + r_0)\right]. \tag{10.76}$$

Observe that the amplitude of E_l alternates between positive and negative values depending on whether l is odd or even. This means that the contributions from adjacent zones are out of phase and tend to cancel. It is here that the obliquity factor makes a crucial difference. As l increases, θ increases and K decreases, so that successive contributions do not in fact completely cancel each other. It is interesting to note that E_l/K_l is independent of any position variables. Although the areas of each zone are almost equal, they do increase slightly as l increases, which means an increased number of emitters. But the mean distance from each zone to P also increases such that E_l/K_l is maintained constant (see Problem 10.11).

The sum of the optical disturbances from all m zones at P is

$$E = E_1 + E_2 + E_3 + \cdots + E_m$$

and since these alternate in sign, we can write

$$E = |E_1| - |E_2| + |E_3| - \cdots \pm |E_m|. \tag{10.77}$$

If m is odd, the series can be reformulated in two ways, either as

$$E = \frac{|E_1|}{2} + \left(\frac{|E_1|}{2} - |E_2| + \frac{|E_3|}{2}\right) + \left(\frac{|E_3|}{2} - |E_4| + \frac{|E_5|}{2}\right) + \cdots + \left(\frac{|E_{m-2}|}{2} - |E_{m-1}| + \frac{|E_m|}{2}\right) + \frac{|E_m|}{2}, \tag{10.78}$$

or as

$$E = |E_1| - \frac{|E_2|}{2} - \left(\frac{|E_2|}{2} - |E_3| + \frac{|E_4|}{2}\right) - \left(\frac{|E_4|}{2} - |E_5| + \frac{|E_6|}{2}\right) + \cdots + \left(\frac{|E_{m-3}|}{2} - |E_{m-2}| + \frac{|E_{m-1}|}{2}\right) - \frac{|E_{m-1}|}{2} + |E_m|. \quad (10.79)$$

There are now two possibilities: either $|E_l|$ is greater than the arithmetic mean of its two neighbors $|E_{l-1}|$ and $|E_{l+1}|$, or it is less than that mean. This is really a question concerning the rate of change of $K(\theta)$. When

$$|E_l| > (|E_{l-1}| + |E_{l+1}|)/2$$

each bracketed term is negative. It follows from Eq. (10.78) that

$$E < \frac{|E_1|}{2} + \frac{|E_m|}{2} \quad (10.80)$$

and from Eq. (10.79) that

$$E > |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m|. \quad (10.81)$$

Since the obliquity factor goes from 1 to 0 over a great many zones, we can neglect any variation between adjacent zones, i.e. $|E_1| \approx |E_2|$ and $|E_{m-1}| \approx |E_m|$. Expression (10.81), to the same degree of approximation, becomes

$$E > \frac{|E_1|}{2} + \frac{|E_m|}{2}. \quad (10.82)$$

We conclude from (10.80) and (10.82) that

$$E \approx \frac{|E_1|}{2} + \frac{|E_m|}{2}. \quad (10.83)$$

This same result is obtained in the case when

$$|E_l| < (|E_{l-1}| + |E_{l+1}|)/2.$$

When the last term $|E_m|$, in the series of Eq. (10.77), corresponds to an even m the same procedure (Problem 10.12) leads to

$$E \approx \frac{|E_1|}{2} - \frac{|E_m|}{2}. \quad (10.84)$$

Fresnel conjectured that the obliquity factor was such that the last contributing zone occurred at $\theta = 90°$, i.e.

$$K(\theta) = 0 \text{ for } \pi/2 \leq |\theta| \leq \pi.$$

In that case Eqs. (10.83) and (10.84) both reduce to

$$E \approx \frac{|E_1|}{2} \quad (10.85)$$

when $|E_m|$ goes to zero because $K_m(\pi/2) = 0$. Alternatively, using Kirchhoff's correct obliquity factor, we divide the *entire* spherical wave into zones with the last or mth zone surrounding O'. Now θ approaches π, $K_m(\pi) = 0$, $|E_m| = 0$ and once again $E \approx |E_1|/2$. *The optical disturbance generated by the entire unobstructed wavefront is approximately equal to one half of the contribution from the first zone.*

If the primary wave were simply to propagate from S to P in a time t, it would have the form

$$E = \frac{\mathcal{E}_0}{(\rho + r_0)} \cos[\omega t - k(\rho + r_0)]. \quad (10.86)$$

Yet the disturbance synthesized from secondary wavelets, Eqs. (10.76) and (10.85), is

$$E = \frac{K_1 \mathcal{E}_A \rho \lambda}{(\rho + r_0)} \sin[\omega t - k(\rho + r_0)]. \quad (10.87)$$

These two equations must however be exactly equivalent, and we interpret the constants in Eq. (10.87) to make them so. Note that there is some latitude in how we do this. We prefer to have the obliquity factor equal to one in the forward direction, i.e. $K_1 = 1$ (rather than $1/\lambda$), from which it follows that Q must be equal to $1/\lambda$. In that case, $\mathcal{E}_A \rho \lambda = \mathcal{E}_0$ which is fine dimensionally. Keep in mind that $\mathcal{E}_A$ is the secondary wavelet source strength per unit area over the primary wavefront of radius ρ, and $\mathcal{E}_0/\rho$ is the amplitude of that primary wave $E_0(\rho)$. Thus $\mathcal{E}_A = E_0(\rho)/\lambda$. There is one other problem and that is the $\pi/2$ phase difference between Eqs. (10.86) and (10.87). This can be accounted for if we are willing to assume that the secondary sources radiate one quarter of a wavelength out of phase with the primary wave (see Section 3.3.2).

We have found it necessary to modify the initial statement of the Huygens–Fresnel principle but this should not distract us from our rather pragmatic reasons for using it, and these are twofold: (1) The Huygens–Fresnel theory can be shown to be an approximation of the Kirchhoff formulation and as such is no longer merely a contrivance and (2) it yields, in a fairly simple way, many predictions which are in fine agreement with experimental observations. Don't forget that it worked quite well in the Fraunhofer approximation.

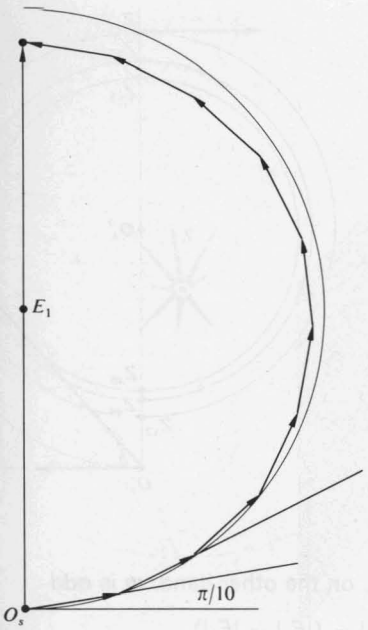


Fig. 10.49 Phasor addition.

10.3.2 The Vibration Curve

We now develop a graphical method for qualitatively analyzing a number of diffraction problems which arise predominantly from circularly symmetric configurations.

Imagine that the first, or polar Fresnel zone in Fig. 10.47 is divided into N subzones by the intersection of spheres, centered on P, of radii

$$r_0 + \lambda/2N, r_0 + \lambda/N, r_0 + 3\lambda/2N, \ldots, r_0 + \lambda/2.$$

Each subzone contributes to the disturbance at P, the resultant of which is of course just E_1. Since the phase difference across the entire zone, from O to its edge, is π rad (corresponding to $\lambda/2$) each subzone is shifted by π/N rad. Figure 10.49 depicts the vector addition of the subzone phasors, where, for convenience, $N = 10$. The chain of phasors deviates very slightly from the dashed circle, because the obliquity factor shrinks each successive amplitude. When the number of subzones is increased to infinity, i.e. $N \to \infty$, the polygon of vectors blends into a segment of a smooth spiral called a *vibration curve*. For each additional Fresnel zone, the vibration curve swings through *one half-turn* and a phase of π as it spirals inward. As shown in Fig. 10.50,

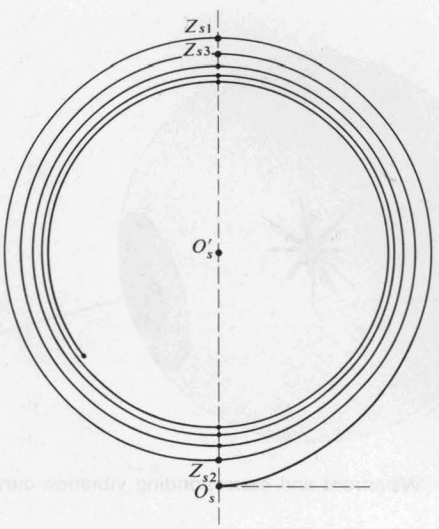


Fig. 10.50 The vibration curve.

the points $O_s, Z_{s1}, Z_{s2}, Z_{s3}, \ldots, O'_s$ on the spiral correspond to points $O, Z_1, Z_2, Z_3, \ldots, O'$ respectively, on the wavefront in Fig. 10.47. Each point $Z_1, Z_2, \ldots, Z_m$ lies on the periphery of a zone and so each point $Z_{s1}, Z_{s2}, \ldots, Z_{sm}$ is separated by a half-turn. We will see later, Eq. (10.91), that the radius of each zone is proportional to the square root of its numerical designation, m. The radius of the hundredth zone will be only ten times that of the first zone. Initially, therefore, the angle θ increases rapidly, and thereafter it gradually slows down as m becomes larger. Accordingly, $K(\theta)$ decreases rapidly only for the first few zones. The result is that as the spiral circulates around with increasing m, it becomes tighter and tighter, deviating from a circle by less for each revolution.

Keep in mind that the spiral is made up of an infinite number of phasors, each shifted by a small phase angle. The relative phase between any two disturbances at P, coming from two points on the wavefront, say O and A, can be depicted as shown in Fig. 10.51. The angle made by the tangents to the vibration curve, at points O_s and A_s, is β and this is the desired phase difference. If the point A is considered to lie on the boundary of a cap-shaped region of the wavefront, the resultant at P from the whole region is $\overrightarrow{O_sA_s}$ at an angle δ.

The total disturbance arriving at P from an unimpeded wave is the sum of the contributions from all of the zones between O and O'. The length of the vector from O_s to

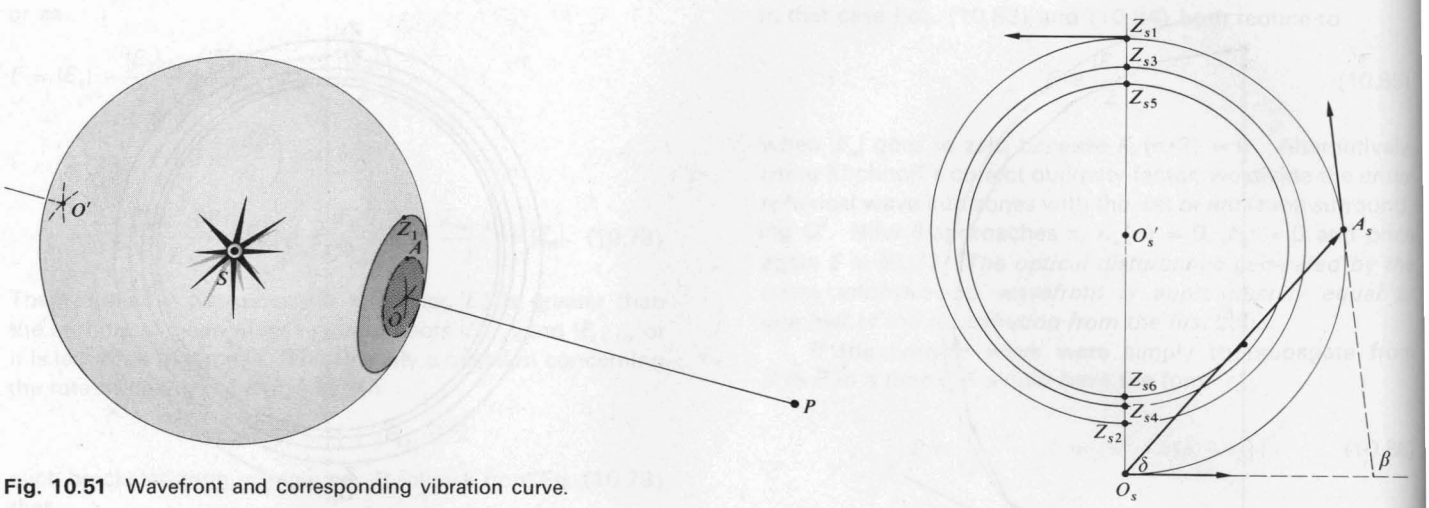


Fig. 10.51 Wavefront and corresponding vibration curve.

O_s' is therefore precisely that amplitude. Note that as expected, the amplitude O_sO_s' is just about one half of the contribution from the first zone, O_sZ_{s1}. Observe that $\overrightarrow{O_sO_s'}$ has a phase of 90° with respect to the wave arriving at P from O. A wavelet emitted at O in phase with the primary excitation gets to P still in phase with the primary wave. This means that $\overrightarrow{O_sO_s'}$ is 90° out of phase with the unobstructed primary wave. This, as we have seen, is one of the shortcomings of the Fresnel formulation.

10.3.3 Circular Apertures

i) Spherical Waves

Fresnel's procedure applied to a point source can be used as a semiquantitative method with which to study diffraction at a circular aperture. Envision a monochromatic spherical wave impinging on a screen containing a small hole, as illustrated in Fig. 10.52. We first record the irradiance arriving at a very small sensor placed at point P on the symmetry axis. Our intent is to move the sensor around in space and so get a point-by-point map of the irradiance of the region beyond Σ.

Let us assume that the sensor at P "sees" an integral number of zones m, filling the aperture. In actuality, the sensor merely records the irradiance at P; the zones having no reality. If m is even, then since $K_m \neq 0$

$$E = (|E_1| - |E_2|) + (|E_3| - |E_4|) + \cdots + (|E_{m-1}| - |E_m|).$$

Because each adjacent contribution is nearly equal

$$E \approx 0$$

and $I \approx 0$. If, on the other hand, m is odd

$$E = |E_1| - (|E_2| - |E_3|) - (|E_4| - |E_5|) - \cdots - (|E_{m-1}| - |E_m|)$$

and

$$E \approx |E_1|$$

which is roughly twice the amplitude of the unobstructed wave. This is truly an amazing result. By inserting a screen in the path of the wave, thereby blocking out most of the wavefront, we have increased the irradiance at P by a factor of four. Conservation of energy clearly demands that there be other points where the irradiance has decreased. And, because of the complete symmetry of the set-up, we can expect a circular ring pattern. If m is not an integer, i.e. a fraction of a zone appears in the aperture, the irradiance at P is somewhere between zero and its maximum value. You might see this all a bit more clearly if you imagine that the aperture is expanding smoothly from an initial value of nearly zero. The amplitude at P can be gotten from the vibration curve where A is any point on the edge of the hole. The phasor magnitude O_sA_s is the desired amplitude of the optical field. Return to Fig. 10.51; as the hole increases, A_s moves counterclockwise around the spiral toward Z_{s1} and a maximum. Allowing the second zone in reduces O_sA_s to O_sZ_{s2}, which is nearly zero, and P becomes a dark spot. As the aperture increases, O_sA_s oscillates in length from near zero to a number of successive maxima, which themselves gradually decrease. Finally, when the hole is fairly large,

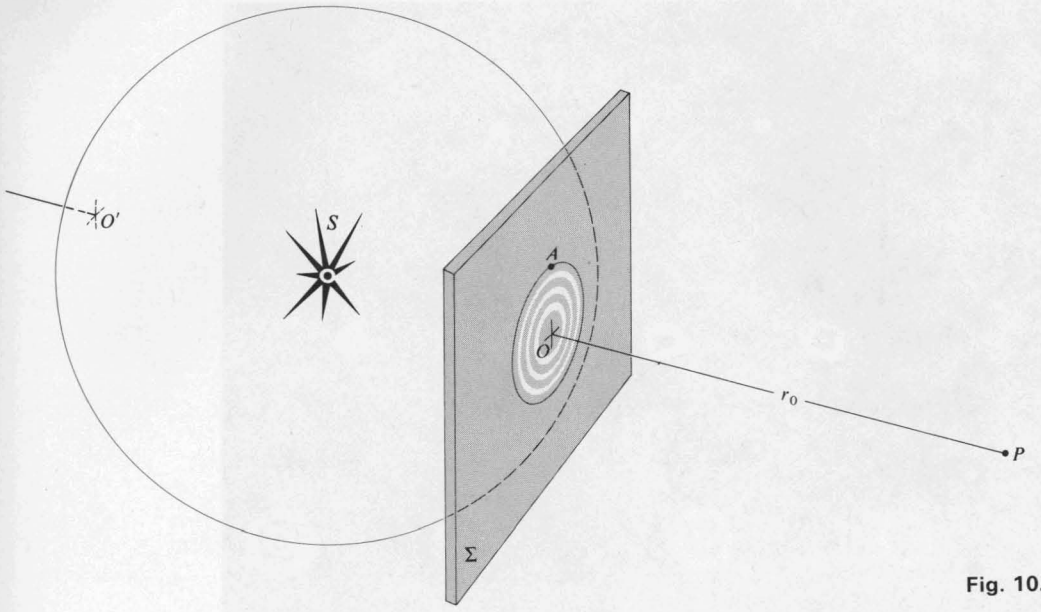


Fig. 10.52 A circular aperture.

the wave is essentially unobstructed, A_s approaches O'_s, and further changes in O_sA_s are imperceptible.

To map the rest of the pattern, we now move the sensor along any line perpendicular to the axis as shown in Fig. 10.53. At P, we assume that two complete zones fill the aperture and $E \approx 0$. At P_1 the second zone has been partially obscured and the third begins to show; E is no longer zero. At P_2 a good fraction of the second zone is hidden, while the third is even more evident. Since the contributions from the first and third zones are in phase, the sensor, placed anywhere on the dotted circle passing through P_2, records a bright spot. As it moves radially outward and portions of successive zones are uncovered, the sensor detects a series of relative maxima and minima. Figure 10.54 shows the diffraction patterns for a number of holes ranging in diameter from 1 mm to 4 mm as they appeared on a screen 1 m away. Starting from the top left and moving right, the first four holes are so small that only a fraction of the first zone is uncovered. The sixth hole uncovers the first and second zones and is therefore black at its center. The ninth hole uncovers the first three zones and is once again bright at its center. Notice that even slightly beyond the geometric shadow at P_3, in Fig. 10.53, the first zone is partially uncovered. Each of the last few contributing segments is only a small fraction of its respective zone and as such is negligible. The sum of all of the amplitudes of the fractional zones, although small, is therefore still finite. Further into the geometric shadow, however, the entire first zone is obscured, the last terms are again negligible and this time the series does indeed go to zero and darkness.

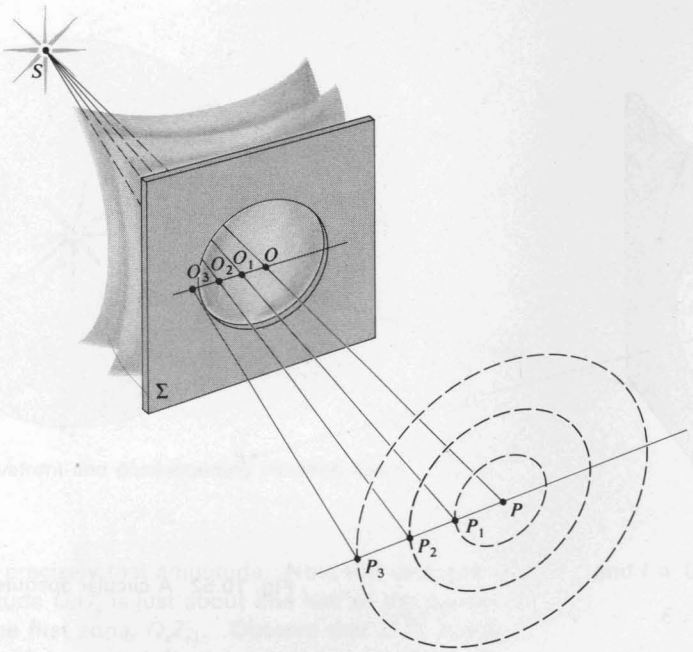
We can gain a better appreciation of the actual size of the things we are dealing with by computing the number of zones in a given aperture. The area of each zone (from Problem 10.11) is given by

$$A = \frac{\rho}{(\rho + r_0)} \pi r_0 \lambda. \tag{10.88}$$

If the aperture has a radius R, to a good approximation the number of zones within it is simply

$$\frac{\pi R^2}{A} = \frac{(\rho + r_0)R^2}{\rho r_0 \lambda}. \tag{10.89}$$

For example, with a point source 1 m behind the aperture ($\rho \approx 1$ m), a plane of observation 1 m in front of it ($r_0 = 1$ m) and $\lambda = 500$ nm, there are 4 zones when $R = 1$ mm and 400 zones when $R = 1$ cm. When both ρ and r_0 are increased to the point where only a small fraction of a zone appears in the aperture, Fraunhofer diffraction occurs. This is essentially a restatement of the Fraunhofer condition of Section 10.1.2; see Problem 10.1 as well.



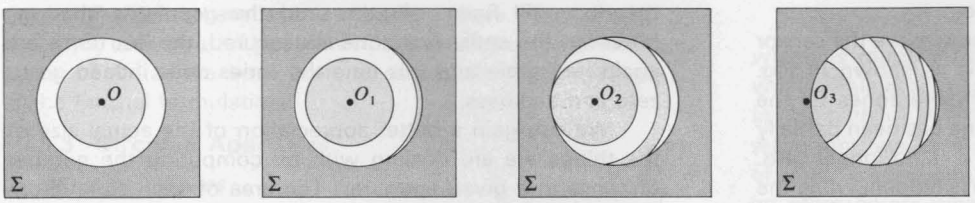


Fig. 10.53 Zones in a circular aperture.

It follows from Eq. (10.89) that the number of zones filling the aperture depends on the distance r_0 from P to O. As P moves in either direction along the central axis, the number of uncovered zones, whether increasing or decreasing, oscillates between odd and even integers. As a result, the irradiance goes through a series of maxima and minima. Clearly, this does not occur in the Fraunhofer configuration where, by definition, more than one zone cannot appear in the aperture.

ii) Plane Waves

Suppose now that the point source has been moved so far from the diffracting screen that the incoming light can be regarded as a plane wave ($\rho \to \infty$). Referring to Fig. 10.55, we now derive an expression for the radius of the mth zone, R_m. Since $r_m = r_0 + m\lambda/2$ then

$$R_m^2 = (r_0 + m\lambda/2)^2 - r_0^2$$

and so

$$R_m^2 = mr_0\lambda + m^2\lambda^2/4. \tag{10.90}$$

Under most circumstances the second term in Eq. (10.90) is negligible as long as m is not extremely large, consequently

$$R_m^2 = mr_0\lambda \tag{10.91}$$

and *the radii are proportional to the square roots of integers.* Using a collimated He–Ne laser ($\lambda = 6328$ Å), the radius of the first zone is 1 mm when viewed from a distance of 1.58 m. Under these particular conditions Eq. (10.91) is applicable as long as $m \ll 10^7$, in which case $R_m = \sqrt{m}$ in millimeters. Figure 10.53 requires a slight modification in

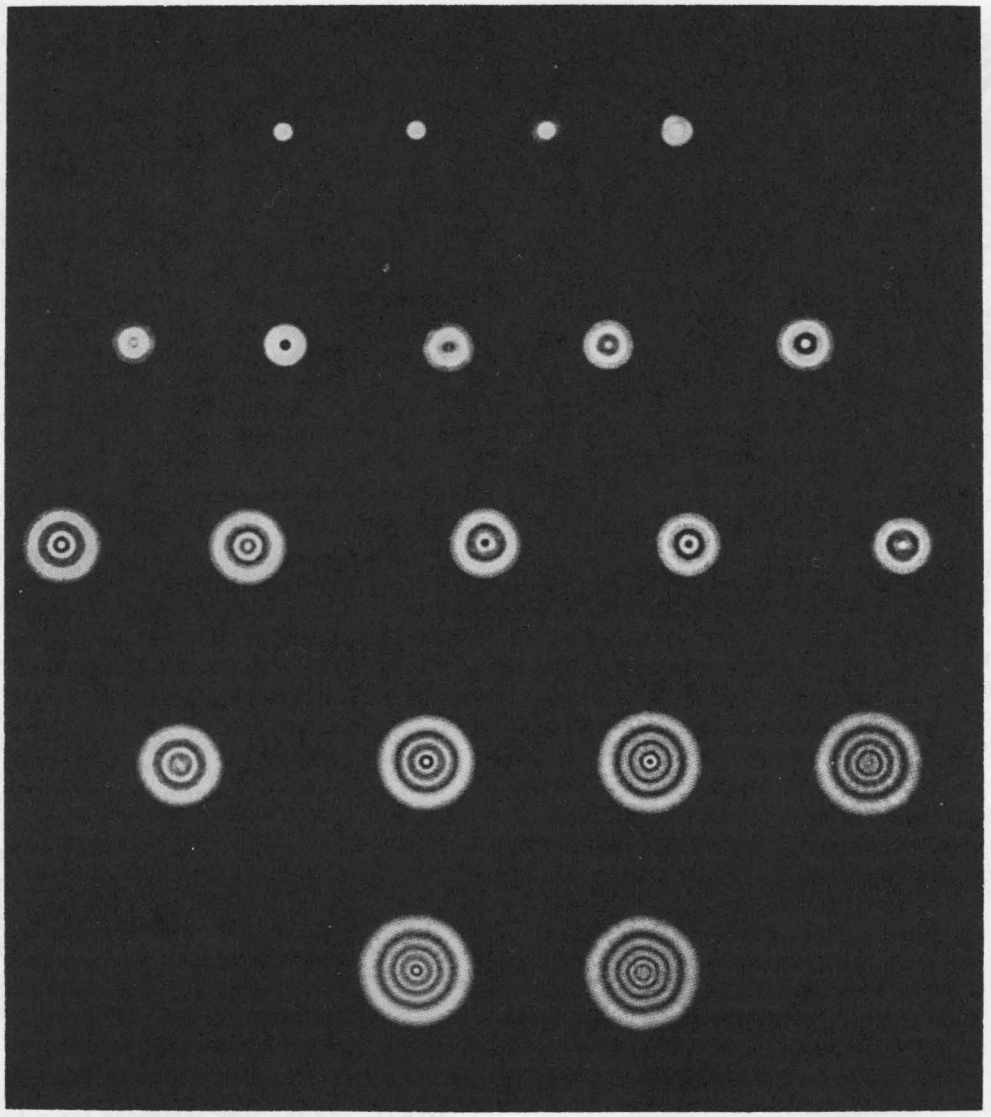


Fig. 10.54 Diffraction patterns for circular apertures of increasing size.

that now the lines $\overline{O_1P_1}$, $\overline{O_2P_2}$, and $\overline{O_3P_3}$ are perpendiculars dropped from the points of observation to Σ.

10.3.4 Circular Obstacles

In 1818 Fresnel entered a competition sponsored by the French Academy. His paper on the theory of diffraction ultimately won first prize and the title *Mémoire Courronné*, but not until it had provided the basis for a rather interesting story. The judging committee consisted of Pierre Laplace, Jean B. Biot, Siméon D. Poisson, Dominique F. Arago and Joseph L. Gay-Lussac, a formidable group indeed. Poisson, who was an ardent protagonist against the wave description of light, deduced a remarkable and seemingly untenable conclusion from Fresnel's theory. He showed that a bright

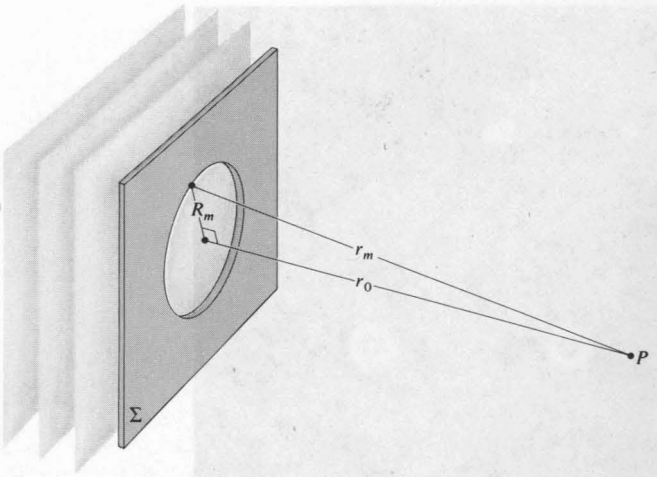


Fig. 10.55 Plane waves incident on a circular hole.

spot would be visible at the center of the shadow of a circular opaque obstacle; a result which he felt proved the absurdity of Fresnel's treatment. We can come to the same conclusion by considering the following somewhat oversimplified argument. Recall that an unobstructed wave yields a disturbance (10.85) given by $E \approx |E_1|/2$. If some sort of obstacle precisely covers the first Fresnel zone, so that its contribution of $|E_1|$ is subtracted out, then $E \approx -|E_1|/2$. It is, therefore, possible that at some point P on the axis, the irradiance will be unaltered by the insertion of that obstruction. This surprising prediction, fashioned by Poisson as the death blow to the wave theory, was almost immediately verified experimentally by Arago; the spot actually existed. Amusingly enough, Poisson's spot, as it is now called, had been observed many years earlier (1723) by Maraldi but this work had long gone unnoticed.

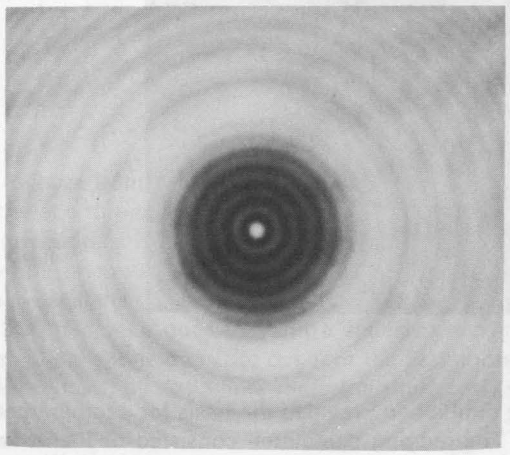


Fig. 10.56 Shadow of a $\frac{1}{8}$ inch diameter ball bearing. The bearing was glued to an ordinary microscope slide and illuminated with a He–Ne laser beam. There are some faint extraneous nonconcentric fringes arising from both the microscope slide and a lens in the beam. [Photo by E.H.]

We now examine the problem a bit more closely, since it is quite evident from Fig. 10.56 that there is a good deal of structure in the actual shadow pattern. If the opaque obstacle, be it a disk or sphere, obscures the first ℓ zones, then

$$E = |E_{\ell+1}| - |E_{\ell+2}| + \cdots + |E_m|$$

(where, as before, there is no absolute significance to the signs other than that alternate terms must subtract). Unlike the analysis for the circular aperture, E_m now approaches zero because $K_m \to 0$. The series must be evaluated in the same manner as was that of the unobstructed wave (10.78 and 10.79). Repeating that procedure yields

$$E \approx \frac{|E_{\ell+1}|}{2} \tag{10.92}$$

and the irradiance on the central axis is generally only slightly less than that of the unobstructed wave. *There is a bright spot everywhere along the central axis except immediately behind the circular obstacle.* The wavelets propagating beyond the disk's circumference meet in phase on the central axis. Notice that as P moves close to the disk, θ increases, $K_{\ell+1} \to 0$ and the irradiance gradually falls off to zero. If the disk is large, the $(\ell + 1)$th zone is very narrow and any irregularities in the obstacle's surface may seriously obscure that zone. For Poisson's spot to be readily observable, the obstacle must be smooth and circular.

If A is a point on the periphery of the disk or sphere, A_s is the corresponding point on the vibration curve (Fig. 10.57). As the disk increases for a fixed P, A_s spirals in counterclockwise toward O'_s and the amplitude $A_sO'_s$ gradually decreases. The same thing happens as P moves towards a disk of constant size.

Off the axis, the zones covered in Fig. 10.53 for the circular aperture will now be exposed and vice versa. Accordingly, a whole series of concentric bright and dark rings will surround the central spot.

The opaque disk images S at P and would similarly form a crude image of every point in an extended source. R. W.

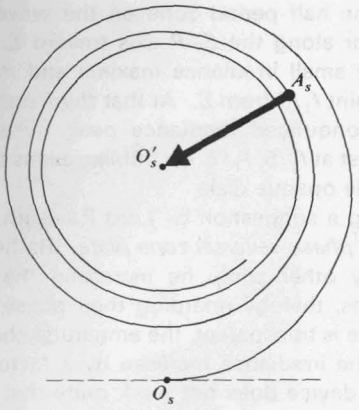


Fig. 10.57 The vibration curve applied to a circular obstruction.

Pohl has shown that a small disk can therefore be used as a crude positive lens.

The diffraction pattern can be seen with little difficulty, but you need a telescope or binoculars. Glue a small ball bearing ($\approx\frac{1}{8}$ or $\frac{1}{4}$ inch diameter) to a microscope slide which then serves as a handle. Place the bearing a few meters beyond the point source and observe it from three or four meters away. Position it directly in front of and completely obscuring the source. You will need the telescope to magnify the image since r_0 is so large. If you can hold the telescope steady, the ring system should be quite clear.

10.3.5 The Fresnel Zone Plate

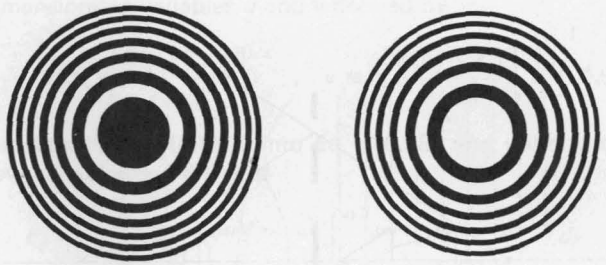
In our previous considerations we utilized the fact that successive Fresnel zones tended to nullify each other. This suggests that we would observe a tremendous increase in irradiance at P if we remove all of either the even or odd zones. A screen which alters the light, either in amplitude or phase, coming from every other half-period zone is called a *zone plate.**

Suppose that we construct a zone plate which passes only the first 20 odd zones and obstructs the even zones.

$$E = E_1 + E_3 + E_5 + \cdots + E_{39}$$

and each of these terms is approximately equal. For an unobstructed wavefront, the disturbance at P would be $E_1/2$

* Lord Rayleigh seems to have invented the zone plate as witnessed by this entry of April 11, 1871 in his notebook. "The experiment of blocking out the odd Huygens zones so as to increase the light at centre succeeded very well . . ."



(a) (b)

Fig. 10.58 Zone plates.

while with the zone plate in place $E \approx 20E_1$. The irradiance has been increased by a factor of 1600 times. The same result would obviously be true if the even zones were passed instead.

To calculate the radii of the zones shown in Fig. 10.58, refer to Fig. 10.59. The outer edge of the mth zone is marked with the point A_m. By definition, a wave which travels the path S–A_m–P must arrive out of phase by $m\lambda/2$ with a wave which traverses the path S–O–P, that is

$$(\rho_m + r_m) - (\rho_0 + r_0) = m\lambda/2. \qquad (10.93)$$

Clearly

$$\rho_m = (R_m^2 + \rho_0^2)^{1/2}$$

and

$$r_m = (R_m^2 + r_0^2)^{1/2}.$$

Expand both these expressions using the binomial series. Since R_m is comparatively small, retaining only the first two terms yields

$$\rho_m = \rho_0 + \frac{R_m^2}{2\rho_0}$$

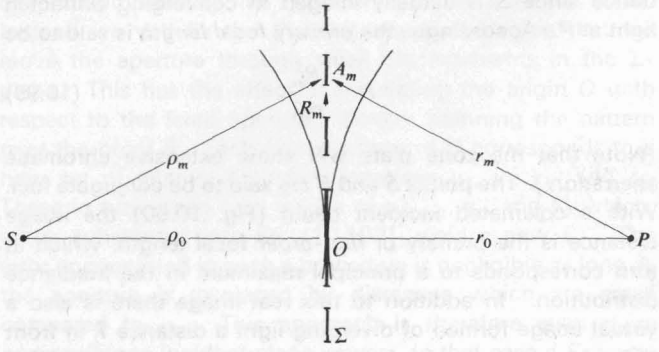


Fig. 10.59 Zone-plate geometry.

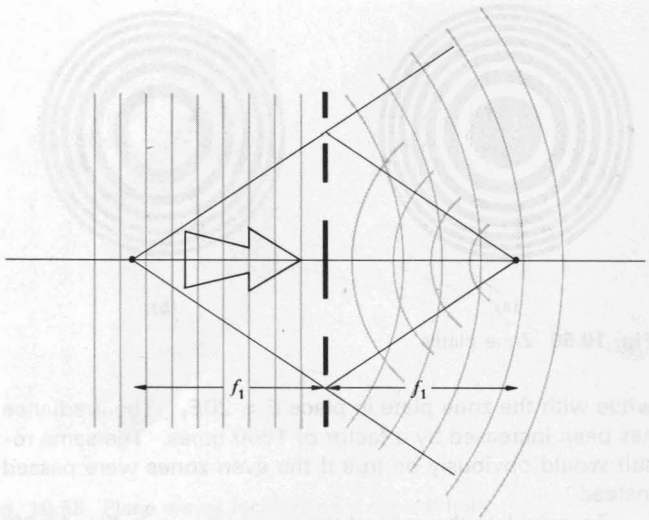


Fig. 10.60 Zone-plate foci.

and

$$r_m = r_0 + \frac{R_m^2}{2r_0}.$$

Finally then, substituting into Eq. (10.93) it becomes

$$\left(\frac{1}{\rho_0} + \frac{1}{r_0}\right) = \frac{m\lambda}{R_m^2}. \tag{10.94}$$

Under plane-wave illumination ($\rho_0 \to \infty$) and Eq. (10.94) reduces to

$$R_m^2 = mr_0\lambda, \tag{10.91}$$

which is an approximation of the exact expression stated by Eq. (10.90). Equation (10.94) has an identical form to that of the thin-lens equation, a fact which is not merely a coincidence since S is actually imaged in converging diffracted light at P. Accordingly, the *primary focal length* is said to be

$$f_1 = \frac{R_m^2}{m\lambda}. \tag{10.95}$$

(Note that the zone plate will show extensive chromatic aberration.) The points S and P are said to be conjugate foci. With a collimated incident beam (Fig. 10.60) the image distance is the primary or *first-order* focal length, which in turn corresponds to a principal maximum in the irradiance distribution. In addition to this real image there is also a virtual image formed of diverging light a distance f_1 in front of Σ. At a distance of f_1 from Σ each ring on the plate is filled by exactly one half-period zone on the wavefront. If we move a sensor along the S–P axis toward Σ, it registers a series of very small irradiance maxima and minima until it arrives at a point $f_1/3$ from Σ. At that *third-order focal point*, there is a pronounced irradiance peak. Additional focal points will exist at $f_1/5$, $f_1/7$, etc., unlike a lens but even more unlike a simple opaque disk.

Following a suggestion by Lord Rayleigh, R. W. Wood constructed a *phase-reversal zone plate*. Rather than blocking out every other zone, he increased the thickness of alternate zones, thereby retarding their phase by π. Since the entire plate is transparent, the amplitude should therefore double and the irradiance increase by a factor of four. In actuality, the device does not work quite that well because the phase is not really constant over each zone. Ideally, the retardation should be made to vary gradually over a zone, jumping back by π at the start of the next zone.*

The usual way to make an optical zone plate is to draw a large-scale version and then photographically reduce it. Plates with hundreds of zones can be made by photographing a Newton's ring pattern, in collimated quasimonochromatic light. Rings of aluminum foil on cardboard work very well for microwaves.

Zone plates can be made of metal with a self-supporting spoked structure so that the transparent regions are devoid of any material. These will function as lenses in the range from ultraviolet to soft x-rays where ordinary glass is opaque.

10.3.6 Fresnel Integrals and the Rectangular Aperture

We now consider a class of problems within the domain of Fresnel diffraction which no longer have the circular symmetry of the previously studied configurations. Consider Fig. 10.61 where dS is an area element situated at some arbitrary point A whose coordinates are (y, z). The location of the origin O is determined by a perpendicular drawn to Σ from the position of the monochromatic point source. The contribution to the optical disturbance at P from the secondary sources on dS has the form given by Eq. (10.74). Making use of what we learned from the freely propagating wave ($\mathcal{E}_A\rho\lambda = \mathcal{E}_0$) we can rewrite that equation as

$$dE_p = \frac{K(\theta)\,\mathcal{E}_0}{\rho r\lambda}\cos\,[k(\rho + r) - \omega t]\,dS. \tag{10.96}$$

* See Ditchburn, *Light*, 2nd edn., p. 232, M. Sussman, "Elementary Diffraction Theory of Zone Plates," *Am. J. Phys.* **28**, 394 (1960), or Ora E. Myers, Jr., "Studies of Transmission Zone Plates," *Am. J. Phys.* **19**, 359 (1951).

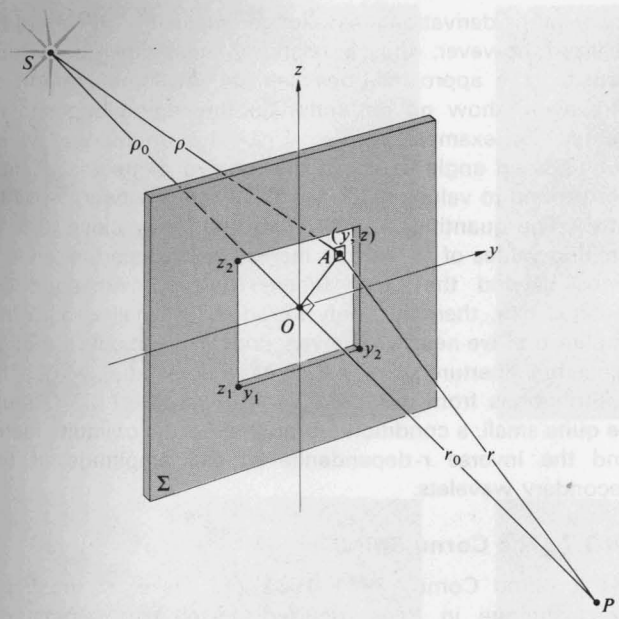


Fig. 10.61 Fresnel diffraction at a rectangular aperture.

The sign of the phase has changed from that of Eq. (10.74) and is written in this way to conform with traditional treatment. *In the case where the dimensions of the aperture are small* in comparison to ρ_0 and r_0, we can set $K(\theta) = 1$ and let $1/\rho r$ equal $1/\rho_0 r_0$ in the amplitude coefficient. Being more careful about approximations introduced into the phase, apply the Pythagorean theorem to triangles *SOA* and *POA* to get

$$\rho = (\rho_0^2 + y^2 + z^2)^{1/2}$$

and

$$r = (r_0^2 + y^2 + z^2)^{1/2}.$$

Expand these using the binomial series and form

$$\rho + r \approx \rho_0 + r_0 + (y^2 + z^2)\frac{\rho_0 + r_0}{2\rho_0 r_0}. \tag{10.97}$$

Observe that this is a more sensitive approximation than that used in the Fraunhofer analysis (10.40) where the term quadratic in the aperture variables was neglected. The disturbance at P in the complex representation is

$$E_p = \frac{\mathcal{E}_0 e^{-i\omega t}}{\rho_0 r_0 \lambda}\int_{y_1}^{y_2}\int_{z_1}^{z_2} e^{ik(\rho+r)}\,dy\,dz. \tag{10.98}$$

Following the usual form of derivation, we introduce the dimensionless variables u and v defined by

$$u \equiv y\left[\frac{2(\rho_0 + r_0)}{\lambda\rho_0 r_0}\right]^{1/2}, \quad v \equiv z\left[\frac{2(\rho_0 + r_0)}{\lambda\rho_0 r_0}\right]^{1/2}. \tag{10.99}$$

Substituting Eq. (10.97) into Eq. (10.98) and utilizing the new variables, we arrive at

$$E_p = \frac{\mathcal{E}_0}{2(\rho_0 + r_0)} e^{i[k(\rho_0 + r_0) - \omega t]}\int_{u_1}^{u_2} e^{i\pi u^2/2}\,du\int_{v_1}^{v_2} e^{i\pi v^2/2}\,dv. \tag{10.100}$$

The term in front of the integral represents the unobstructed disturbance at P divided by 2; let us call it $E_u/2$. The integral itself can be evaluated using two functions, $\mathscr{C}(w)$ and $\mathscr{S}(w)$, where w represents either u or v. These quantities, which are known as the *Fresnel integrals*, are defined by

$$\mathscr{C}(w) \equiv \int_0^w \cos(\pi w'^2/2)\,dw', \qquad \mathscr{S}(w) \equiv \int_0^w \sin(\pi w'^2/2)\,dw'. \tag{10.101}$$

Both functions have been extensively studied and their numerical values are well tabulated. Their interest to us at this point derives from the fact that

$$\int_0^w e^{i\pi w'^2/2}\,dw' = \mathscr{C}(w) + i\mathscr{S}(w),$$

and this, in turn, has the form of the integrals in Eq. (10.100). The disturbance at P is then

$$E_p = \frac{E_u}{2}[\mathscr{C}(u) + i\mathscr{S}(u)]_{u_1}^{u_2}[\mathscr{C}(v) + i\mathscr{S}(v)]_{v_1}^{v_2}, \tag{10.102}$$

which can be evaluated using the tabulated values of $\mathscr{C}(u_1)$, $\mathscr{C}(u_2)$, $\mathscr{S}(u_1)$ etc. The mathematics becomes rather involved if we compute the disturbance at all points of the plane of observation leaving the position of the aperture fixed. Instead we will fix the S–O–P line and imagine that we move the aperture through small displacements in the Σ-plane. This has the effect of translating the origin O with respect to the fixed aperture, thereby scanning the pattern over the point P. Each new position of O corresponds to a new set of relative boundary locations y_1, y_2, z_1, and z_2. These in turn mean new values of u_1, u_2, v_1, and v_2 which, when substituted into Eq. (10.102), yield a new E_p. The error encountered in such a procedure is negligible as long as the aperture is displaced by distances which are small compared to ρ_0. This approach is therefore even more appropriate to incident plane waves. In that case if E_0 is the

amplitude of the incoming plane wave at Σ, Eq. (10.96) becomes simply

$$dE_p = \frac{E_0 K(\theta)}{r\lambda} \cos(kr - \omega t)\, dS$$

where, as before, $\mathcal{E}_A = E_0/\lambda$. This time with

$$u = y\left(\frac{2}{\lambda r_0}\right)^{1/2}, \qquad v = z\left(\frac{2}{\lambda r_0}\right)^{1/2}, \tag{10.103}$$

where we have divided the numerator and denominator in Eq. (10.99) by ρ_0 and then let it go to infinity, E_p takes the same form as Eq. (10.102) where E_u is again the unobstructed disturbance. The irradiance at P is $E_p E_p^*/2$, keeping in mind that E_u is complex; hence

$$I_p = \frac{I_0}{4}\{[\mathscr{C}(u_2) - \mathscr{C}(u_1)]^2 + [\mathscr{S}(u_2) - \mathscr{S}(u_1)]^2\}$$
$$\times \{[\mathscr{C}(v_2) - \mathscr{C}(v_1)]^2 + [\mathscr{S}(v_2) - \mathscr{S}(v_1)]^2\}, \tag{10.104}$$

where I_0 is the unobstructed irradiance at P.

As a simple example, envision a square hole 2 mm on each side under plane-wave illumination at 500 nm. If P is 4 m away and directly opposite point O at the center of the aperture, $u_2 = 1.0$, $u_1 = -1.0$, $v_2 = 1.0$, and $v_1 = -1.0$. The Fresnel integrals are both odd functions, i.e.

$$\mathscr{C}(w) = -\mathscr{C}(-w) \text{ and } \mathscr{S}(w) = -\mathscr{S}(-w);$$

consequently

$$I_p = \frac{I_0}{4}\{[2\mathscr{C}(1)]^2 + [2\mathscr{S}(1)]^2\}^2$$

and a numerical value is easily obtained. To find the irradiance somewhere else in the pattern, e.g. 0.1 mm to the left of center, move the aperture relative to the OP-line accordingly, whereupon $u_2 = 1.1$, $u_1 = -0.9$, $v_2 = 1.0$, and $v_1 = -1.0$. The resultant I_p will also be equal to that found at 0.1 mm to the right of center. Indeed, because the aperture is square, the same value obtains 0.1 mm directly above and below center as well (Fig. 10.62).

We can approach the limiting case of free propagation by allowing the aperture dimensions to increase indefinitely. Making use of the fact that $\mathscr{C}(\infty) = \mathscr{S}(\infty) = \frac{1}{2}$ and $\mathscr{C}(-\infty) = \mathscr{S}(-\infty) = -\frac{1}{2}$ the irradiance at P, opposite the center of the aperture, is

$$I_p = I_0$$

which is exactly correct. This is rather remarkable considering that when the length $\overline{OA}$ is large, all of the approximations made in the derivation are no longer applicable. It should be realized however, that a relatively small aperture which satisfies the approximations can still be large enough to effectively show no diffraction in the region opposite its center. For example, with $\rho_0 = r_0 = 1$ m an aperture which subtends an angle of about one or two degrees at P may correspond to values of $|u|$ and $|v|$ of roughly twenty-five to fifty. The quantities $\mathscr{C}$ and $\mathscr{S}$ are then very close to their limiting values of $\frac{1}{2}$. Further increasing the aperture dimensions beyond the point where the approximations are violated can, therefore, only introduce a small error. This implies that we need not be very concerned about restricting the actual aperture size (as long as $r_0 \gg \lambda$ and $\rho_0 \gg \lambda$). The contributions from wavefront regions remote from O must be quite small, a condition attributable to the obliquity factor and the inverse r-dependence of the amplitude of the secondary wavelets.

10.3.7 The Cornu Spiral

Marie Alfred Cornu (1841–1902), professor at the École Polytechnique in Paris, devised an elegant geometrical depiction of the Fresnel integrals, akin to the vibration curve already considered. Figure 10.63, which is known as the *Cornu spiral*, is a plot in the complex plane of the points $B(w) \equiv \mathscr{C}(w) + i\mathscr{S}(w)$ as w takes on all possible values from 0 to $\pm\infty$. This just means that we plot $\mathscr{C}(w)$ on the horizontal or real axis and $\mathscr{S}(w)$ on the vertical or imaginary axis. The appropriate numerical values are taken from the tables. If $d\ell$ be an element of arc length measured along the curve, then

$$d\ell^2 = d\mathscr{C}^2 + d\mathscr{S}^2.$$

From the definitions (10.101)

$$d\ell^2 = (\cos^2 \pi w^2/2 + \sin^2 \pi w^2/2)\, dw^2$$

and

$$d\ell = dw.$$

Values of w correspond to the arc length and are marked off along the spiral in Fig. 10.63. As w approaches $\pm\infty$ the curve spirals into its limiting values at $B^+ = \frac{1}{2} + i\frac{1}{2}$ and $B^- = -\frac{1}{2} - i\frac{1}{2}$. The slope of the spiral is

$$\frac{d\mathscr{S}}{d\mathscr{C}} = \frac{\sin \pi w^2/2}{\cos \pi w^2/2} = \tan \frac{\pi w^2}{2} \tag{10.105}$$

and so the angle between the tangent to the spiral at any point and the $\mathscr{C}$-axis is $\beta = \pi w^2/2$.

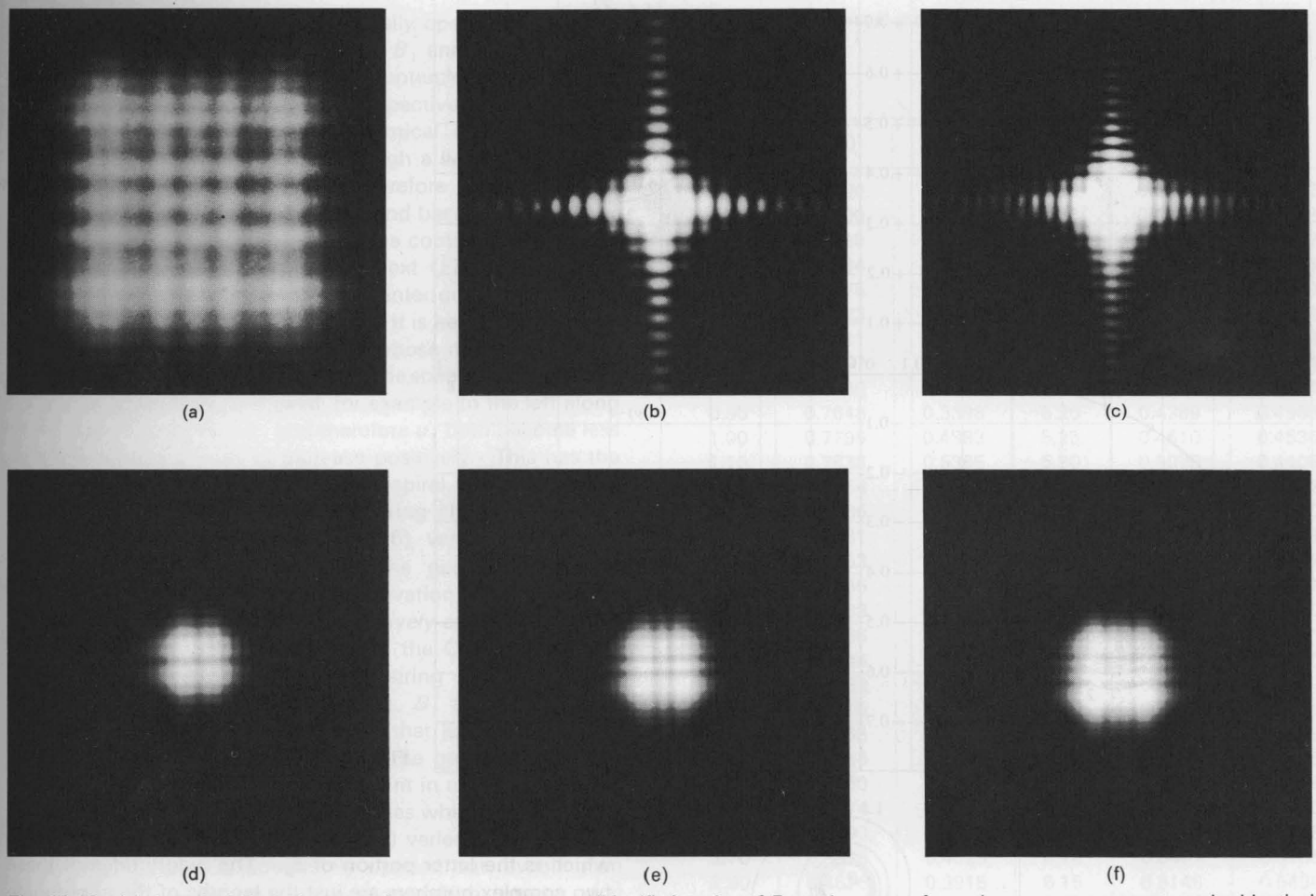


Fig. 10.62 (a) A typical Fresnel pattern for a square aperture. (b)–(f) A series of Fresnel patterns for various square apertures under identical conditions. Note that as the hole increases in size the pattern changes from a spread-out Fraunhofer-like distribution to a far more localized structure. [Photos by E.H.]

The Cornu spiral can be used either as a convenient tool for quantitative determinations or as an aid to gaining a qualitative picture of a diffraction pattern (as was the case with the vibration curve). As an example of its quantitative uses, reconsider the problem of a 2 mm square hole, dealt with in the previous section ($\lambda = 500$ nm, $r_0 = 4$ m and plane-wave illumination). We wish to find the irradiance at P directly opposite the aperture's center where, in this case, $u_1 = -1.0$ and $u_2 = 1.0$. The variable u is measured along the arc, i.e. w is replaced by u on the spiral. Place two points on the spiral at distances from O_s equal to u_1 and u_2; (these are symmetric with respect to O_s because P is now opposite the aperture's center). Label the two points $B_1(u)$ and $B_2(u)$ respectively, as in Fig. 10.64. The phasor $\mathbf{B}_{12}(u)$ drawn from $B_1(u)$ to $B_2(u)$ is just the complex number $B_2(u) - B_1(u)$,

$$\mathbf{B}_{12}(u) = [\mathscr{C}(u) + i\mathscr{S}(u)]_{u_1}^{u_2},$$

and is the first term in the expression (10.102) for E_p. Similarly for $v_1 = -1.0$ and $v_2 = 1.0$, $B_2(v) - B_1(v)$ is

$$\mathbf{B}_{12}(v) = [\mathscr{C}(v) + i\mathscr{S}(v)]_{v_1}^{v_2},$$

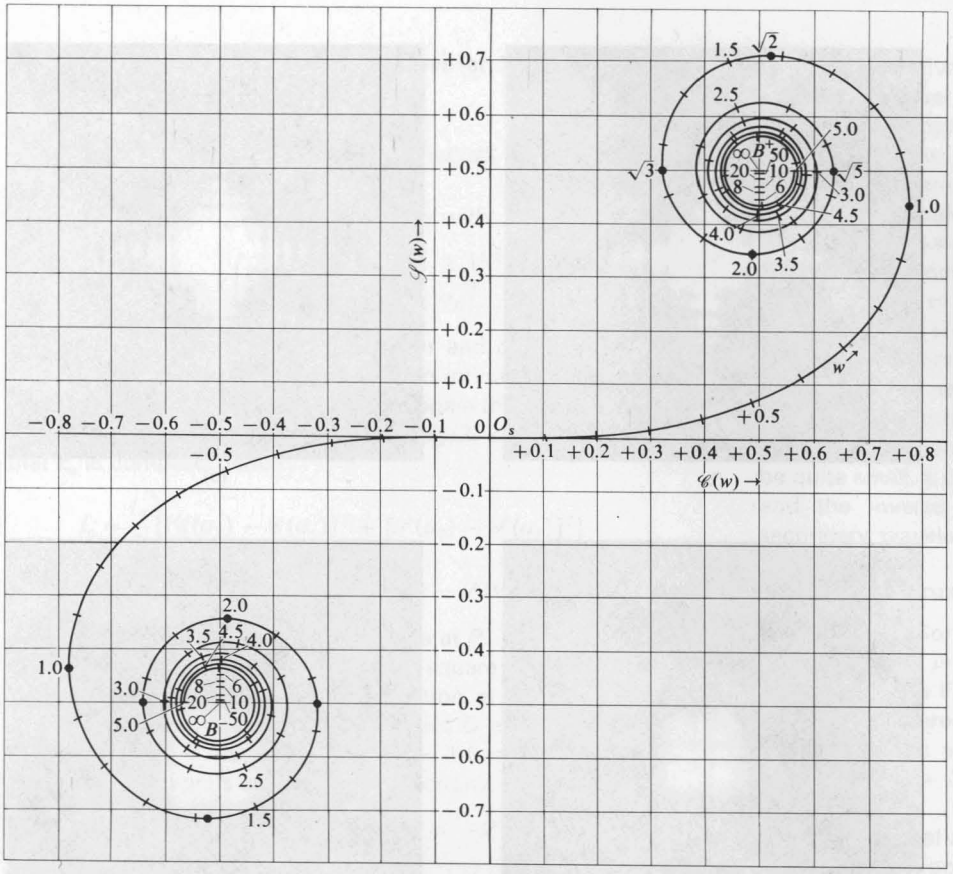


Fig. 10.63 The Cornu spiral.

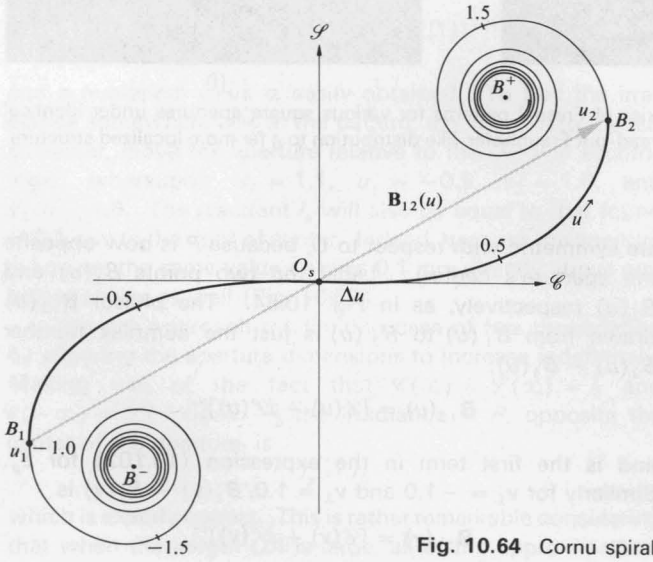


Fig. 10.64 Cornu spiral.

which is the latter portion of E_p. The magnitudes of these two complex numbers are just the lengths of the appropriate $\mathbf{B}_{12}$-phasors which can be read off the curve with a ruler, using either axis as a scale. The irradiance is then simply

$$I_p = \frac{I_0}{4}|\mathbf{B}_{12}(u)|^2|\mathbf{B}_{12}(v)|^2 \qquad (10.106)$$

and the problem is solved. Notice that the arc lengths along the spiral, i.e. $\Delta u = u_2 - u_1$ and $\Delta v = v_2 - v_1$, are proportional to the aperture's overall dimensions in the y- and z-directions, respectively. *The arc lengths are therefore constant regardless of the position of P in the plane of observation.* On the other hand, the phasors $\mathbf{B}_{12}(u)$ and $\mathbf{B}_{12}(v)$ which span the arc lengths are not constant, and they do indeed depend on the location of P.

Maintaining the position of P opposite the center of the diffracting hole, we now suppose the aperture size to be ad-

justable. As the square hole is gradually opened, Δv and Δu increase accordingly. The endpoints B_1 and B_2 of either of these arc lengths spiral around counterclockwise toward their limiting values of B^- and B^+ respectively. The phasors $\mathbf{B}_{12}(u)$ and $\mathbf{B}_{12}(v)$, which are identical in this instance because of the symmetry, pass through a series of extrema. The central spot in the pattern therefore gradually shifts from relative brightness to darkness and back. All the while, the entire irradiance distribution varies continually from one beautifully intricate display to the next (Fig. 10.62). For any particular aperture size the off-center diffraction pattern can be computed by repositioning P. It is helpful to visualize the arc length as a piece of string, whose measure is equal to either Δv or Δu. Imagine it lying on the spiral, with O_s initially at its midpoint. As P is moved, for example to the left along the y-axis (Fig. 10.61), y_1 and therefore u_1 both become less negative, while y_2 and u_2 increase positively. This has the effect that our Δu-string slides up the spiral. As the distance between the endpoints of the Δu-string changes, $|\mathbf{B}_{12}(u)|$ changes, and the irradiance (10.106) varies accordingly. When P is at the left edge of the geometric shadow $y_1 = u_1 = 0$. As the point of observation moves into the geometric shadow u_1 increases *positively* and the Δu-string is now entirely on the upper half of the Cornu spiral. As u_1 and u_2 continue to increase, the string winds ever more tightly about the B^+-limit. Its ends, B_1 and B_2, become closer to each other with the result that $|\mathbf{B}_{12}(u)|$ becomes quite small and I_p decreases within the geometric shadow region (we will come back to this point in more detail in the next section). The same process applies when we scan in the z-direction; Δv is constant and $B_{12}(v)$ varies.

If the aperture is completely opened out revealing an unobstructed wave, $u_1 = v_1 = -\infty$ which means that $B_1(u) = B_1(v) = B^-$ and $B_2(u) = B_2(v) = B^+$. The B^-B^+-line makes a 45° angle with the $\mathscr{C}$-axis and has a length equal to $\sqrt{2}$. Consequently, the phasors $\mathbf{B}_{12}(u)$ and $\mathbf{B}_{12}(v)$ each have magnitude $\sqrt{2}$ and phase $\pi/4$, i.e. $\mathbf{B}_{12}(u) = \sqrt{2}\exp(i\pi/4)$ and $\mathbf{B}_{12}(v) = \sqrt{2}\exp(i\pi/4)$. It follows from Eq. (10.102) that

$$E_p = E_u e^{i\pi/2}, \tag{10.107}$$

and as in Section 10.3.1 we have the unobstructed amplitude except for a $\pi/2$ phase discrepancy.* Finally, using (10.106), $I_p = I_0$.

* The phase discrepancy will be resolved by the Kirchhoff theory of Section 10.4.

Table 10.1 Fresnel integrals.

w	$\mathscr{C}(w)$	$\mathscr{S}(w)$	w	$\mathscr{C}(w)$	$\mathscr{S}(w)$
0.00	0.0000	0.0000	4.50	0.5261	0.4342
0.10	0.1000	0.0005	4.60	0.5673	0.5162
0.20	0.1999	0.0042	4.70	0.4914	0.5672
0.30	0.2994	0.0141	4.80	0.4338	0.4968
0.40	0.3975	0.0334	4.90	0.5002	0.4350
0.50	0.4923	0.0647	5.00	0.5637	0.4992
0.60	0.5811	0.1105	5.05	0.5450	0.5442
0.70	0.6597	0.1721	5.10	0.4998	0.5624
0.80	0.7230	0.2493	5.15	0.4553	0.5427
0.90	0.7648	0.3398	5.20	0.4389	0.4969
1.00	0.7799	0.4383	5.25	0.4610	0.4536
1.10	0.7638	0.5365	5.30	0.5078	0.4405
1.20	0.7154	0.6234	5.35	0.5490	0.4662
1.30	0.6386	0.6863	5.40	0.5573	0.5140
1.40	0.5431	0.7135	5.45	0.5269	0.5519
1.50	0.4453	0.6975	5.50	0.4784	0.5537
1.60	0.3655	0.6389	5.55	0.4456	0.5181
1.70	0.3238	0.5492	5.60	0.4517	0.4700
1.80	0.3336	0.4508	5.65	0.4926	0.4441
1.90	0.3944	0.3734	5.70	0.5385	0.4595
2.00	0.4882	0.3434	5.75	0.5551	0.5049
2.10	0.5815	0.3743	5.80	0.5298	0.5461
2.20	0.6363	0.4557	5.85	0.4819	0.5513
2.30	0.6266	0.5531	5.90	0.4486	0.5163
2.40	0.5550	0.6197	5.95	0.4566	0.4688
2.50	0.4574	0.6192	6.00	0.4995	0.4470
2.60	0.3890	0.5500	6.05	0.5424	0.4689
2.70	0.3925	0.4529	6.10	0.5495	0.5165
2.80	0.4675	0.3915	6.15	0.5146	0.5496
2.90	0.5624	0.4101	6.20	0.4676	0.5398
3.00	0.6058	0.4963	6.25	0.4493	0.4954
3.10	0.5616	0.5818	6.30	0.4760	0.4555
3.20	0.4664	0.5933	6.35	0.5240	0.4560
3.30	0.4058	0.5192	6.40	0.5496	0.4965
3.40	0.4385	0.4296	5.45	0.5292	0.5398
3.50	0.5326	0.4152	6.50	0.4816	0.5454
3.60	0.5880	0.4923	6.55	0.4520	0.5078
3.70	0.5420	0.5750	6.60	0.4690	0.4631
3.80	0.4481	0.5656	6.65	0.5161	0.4549
3.90	0.4223	0.4752	6.70	0.5467	0.4915
4.00	0.4984	0.4204	6.75	0.5302	0.5362
4.10	0.5738	0.4758	6.80	0.4831	0.5436
4.20	0.5418	0.5633	6.85	0.4539	0.5060
4.30	0.4494	0.5540	6.90	0.4732	0.4624
4.40	0.4383	0.4622	6.95	0.5207	0.4591

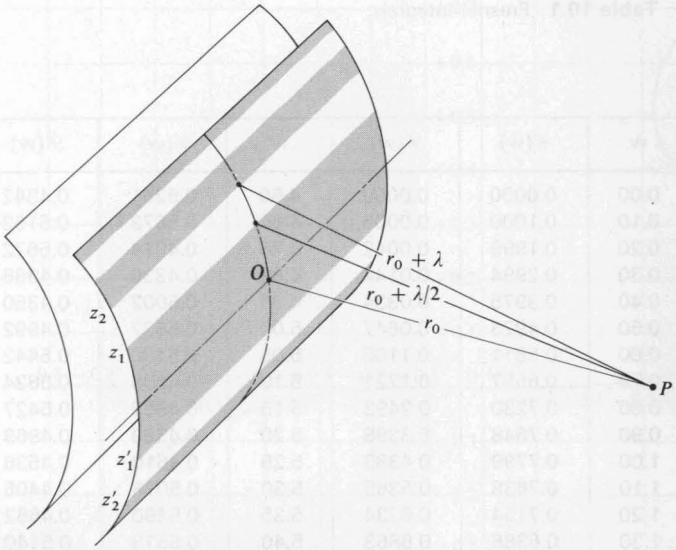


Fig. 10.65 Cylindrical wavefront zones.

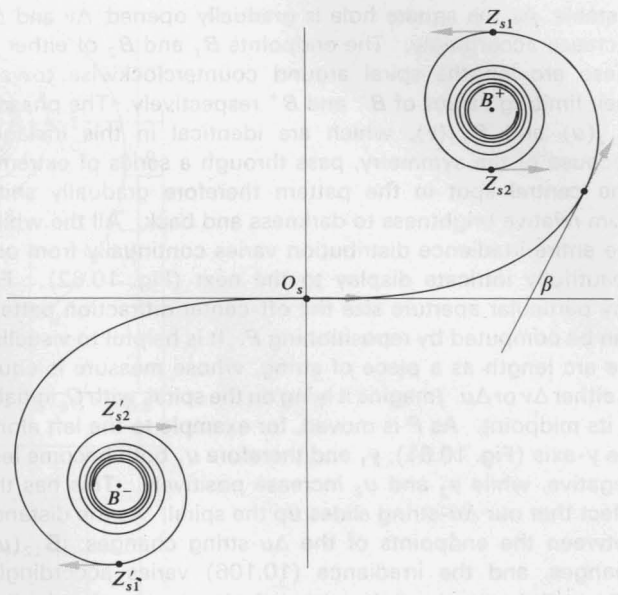


Fig. 10.66 Cornu spiral related to the cylindrical wavefront.

We can construct a more palpable picture of what the Cornu spiral represents by considering Fig. 10.65 which depicts a cylindrical wavefront propagating from a coherent line source. The present procedure is exactly the same as that used in deriving the vibration curve and the reader is referred back to Section 10.3.2 for a more leisurely discussion. Suffice it to say that the wavefront is divided into *half-period strip zones* by its intersection with a family of cylinders having a common axis and radii of $r_0 + \lambda/2$, $r_0 + \lambda$, $r_0 + 3\lambda/2$ etc. *The contributions from these strip zones are proportional to their areas, which decrease rapidly*. This is in contrast to the circular zones whose radii increase, thereby keeping the areas nearly constant. Each strip zone is in turn similarly divided into N subzones which have a relative phase difference of π/N. The vector sum of all of the amplitude contributions from zones above the center line is a spiraling polygon. Letting N go to ∞ and including the contributions generated by the strip zones below the center line causes the polygon to smooth out into a continuous Cornu spiral. This is not surprising since the coherent line source generates an infinite number of overlapping point-source patterns.

Figure 10.66 shows a number of unit tangent vectors at various positions along the spiral. The vector at O_s corresponds to the contribution from the central axis passing through O on the wavefront. The points associated with the boundaries of each strip zone can be located on the spiral since at those positions the relative phase, β, is either an even or odd multiple of π. For example the point Z_{s1} on the spiral (Fig. 10.66), which is related to z_1 (Fig. 10.65) on the wavefront is, by definition, 180° out of phase with O_s. Therefore Z_{s1} must be located at the top of the spiral where $w = \sqrt{2}$, inasmuch as there $\beta = \pi w^2/2 = \pi$.

It will be helpful as we go along in the treatment to visualize the blocking out of these strip zones when analyzing the effects of obstructions. Quite obviously one could even make an appropriate zone plate which would accomplish this to some advantage and such devices are in use.

10.3.8 Fresnel Diffraction by a Slit

We can treat Fresnel diffraction at a long slit as an extension of the rectangular-aperture problem. We need only elongate the rectangle by allowing y_1 and y_2 to move very far from O, as shown in Fig. 10.67. As the point of observation moves along the y-axis, so long as the vertical boundaries at either end of the slit are still essentially at infinity, $u_2 \approx \infty$, $u_1 \approx -\infty$ and $\mathbf{B}_{12}(u) \approx \sqrt{2}e^{i\pi/4}$. From Eq. (10.106), for either point-source or plane-wave illumination,

$$I_p = \frac{I_0}{2}|\mathbf{B}_{12}(v)|^2, \tag{10.108}$$

and the pattern is independent of y. The values of z_1 and

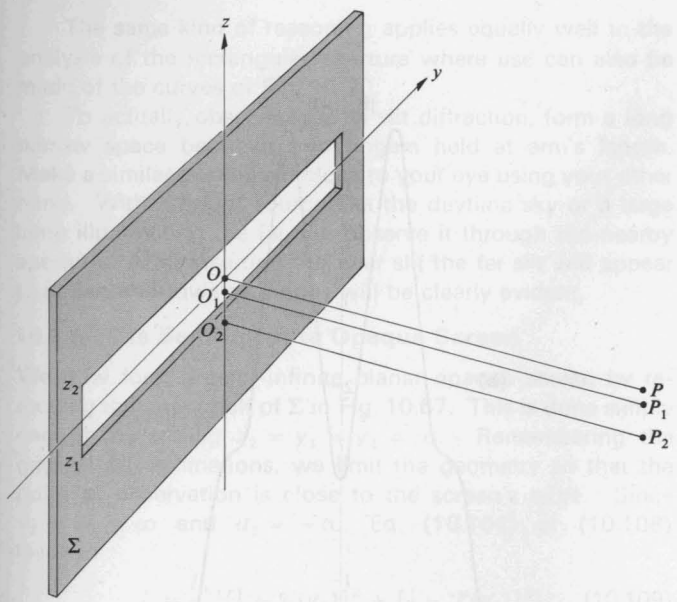


Fig. 10.67 Single-slit geometry.

z_2 which fix the slit width determine the important parameter $\Delta v = v_2 - v_1$, which in turn governs $\mathbf{B}_{12}(v)$. Imagine once again that we have a string of length Δv lying along the spiral. At P, opposite point O, the aperture is symmetric and the string would be centered on O_s (Fig. 10.68). The chord $|\mathbf{B}_{12}(v)|$ need only be measured and substituted into Eq. (10.108) to find I_p. At point P_1, z_1 and therefore v_1 are smaller negative numbers while z_2 and v_2 have increased positively. The arc length Δv (the string) moves up the spiral (Fig. 10.68) and the chord decreases. As the point of observation moves down into the geometric shadow, the string winds about B^+ and the chord goes through a series of relative extrema. If Δv is very small, our imaginary piece of string is small and the chord $|\mathbf{B}_{12}(v)|$ decreases appreciably only when the radius of curvature of the spiral itself is small. This occurs in the vicinity of B^+ or B^- i.e. far out into the geometric shadow. There will, therefore, be light well beyond the edges of the aperture, as long as the aperture is relatively small. Note too that with small Δv there will be a broad central maximum. In fact, if Δv is much less than one, $r_0\lambda$ is much greater than the aperture width and the Fraunhofer condition prevails. This transition of Eq. (10.108) into the form of Eq. (10.17) is more plausible when we realize that for large w the Fresnel integrals have trigonometric representations (see Problem 10.15).

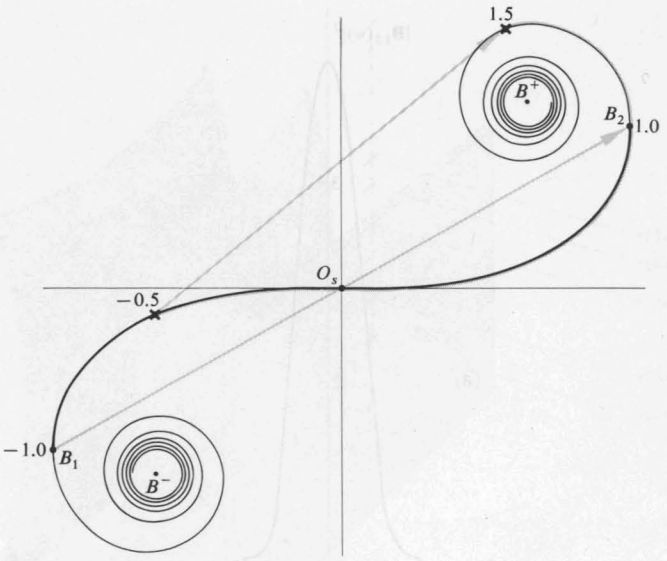


Fig. 10.68 Cornu spiral for the slit.

As the slit widens, Δv becomes larger, for a fixed r_0, until a configuration like that in Fig. 10.69 exists for a point opposite the slit's center. If the point of observation is moved vertically either up or down, Δv slides either down or up the spiral. Yet the chord increases in both cases, so that

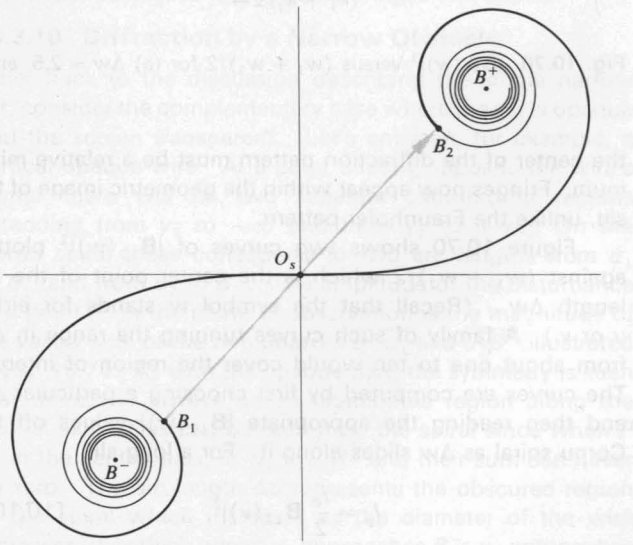


Fig. 10.69 An irradiance minimum in the slit pattern.

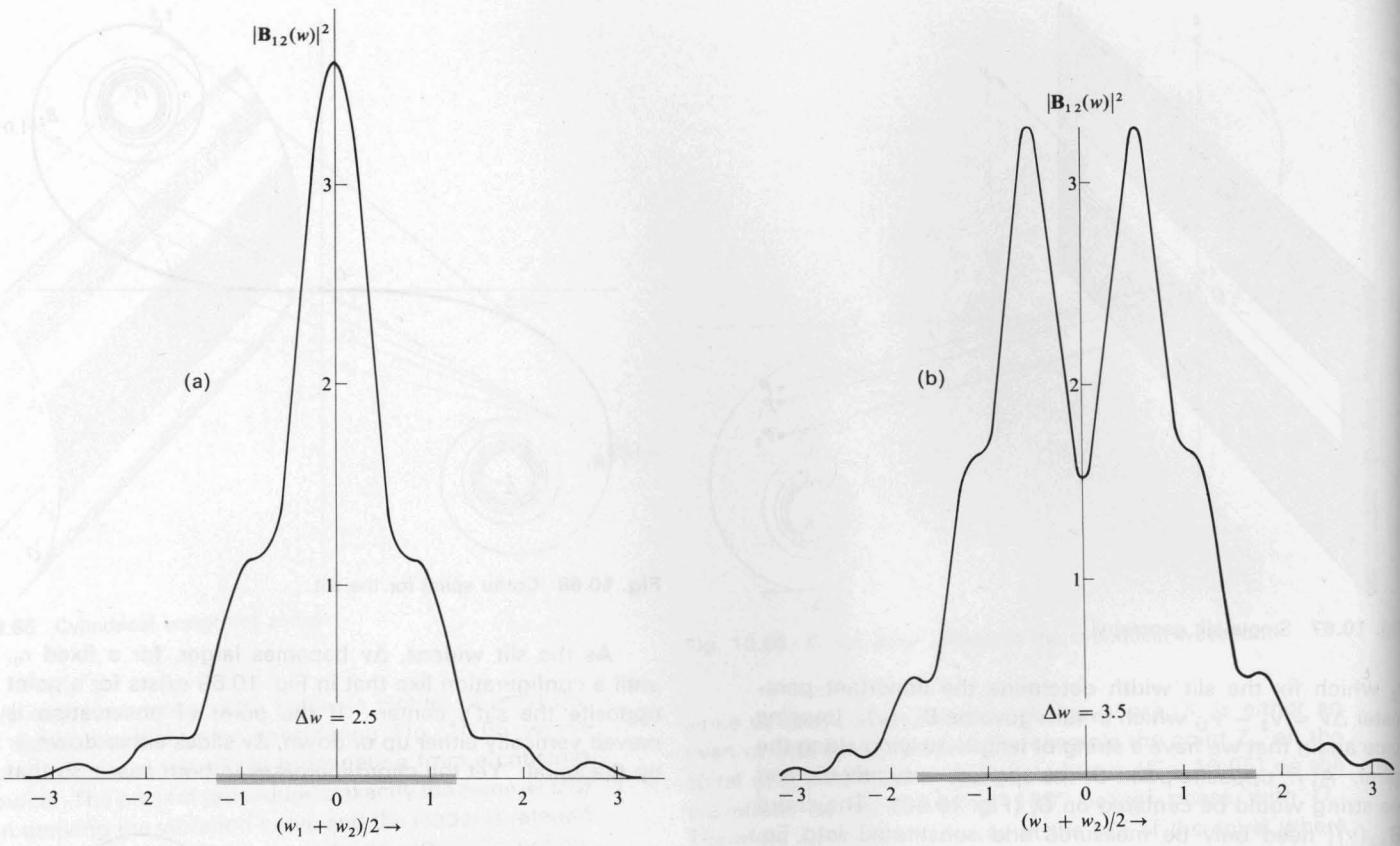


Fig. 10.70 $|\mathbf{B}_{12}(w)|^2$ versus $(w_1 + w_2)/2$ for (a) $\Delta w = 2.5$, and (b) $\Delta w = 3.5$.

the center of the diffraction pattern must be a relative minimum. Fringes now appear within the geometric image of the slit, unlike the Fraunhofer pattern.

Figure 10.70 shows two curves of $|\mathbf{B}_{12}(w)|^2$ plotted against $(w_1 + w_2)/2$ which is the center point of the arc length Δw. (Recall that the symbol w stands for either u or v.) A family of such curves running the range in Δw from about one to ten would cover the region of interest. The curves are computed by first choosing a particular Δw and then reading the appropriate $|\mathbf{B}_{12}(w)|$ values off the Cornu spiral as Δw slides along it. For a long slit

$$I_p = \frac{I_0}{2}|\mathbf{B}_{12}(v)|^2, \qquad [10.108]$$

and since Δz is the slit width which corresponds to Δv, each curve in Fig. 10.70 *is proportional to the irradiance distribution* for a given slit. For example, Fig. 10.70(a) can be read as $|\mathbf{B}_{12}(v)|^2$ versus $(v_1 + v_2)/2$ for $\Delta v = 2.5$. The abscissa relates to $(z_1 + z_2)/2$, i.e. the displacement of the point of observation from the center of the slit. In Fig. 10.70(b) $\Delta w = 3.5$, which means that a slit having a $\Delta v = 3.5$ clearly has fringes appearing within the geometric image as expected (Problem 10.13). The curves could, of course, be plotted in terms of values of Δz or Δy explicitly, but that would unnecessarily limit them to only one set of configuration parameters ρ_0, r_0 and λ.

As the slit is widened still further, Δv approaches and then surpasses 10. An increasing number of fringes appear within the geometric image and the pattern no longer extends appreciably beyond that image.

The same kind of reasoning applies equally well to the analysis of the rectangular aperture where use can also be made of the curves of Fig. 10.70.

To actually observe Fresnel slit diffraction, form a long narrow space between two fingers held at arm's length. Make a similar parallel slit close to your eye using your other hand. With a *bright* source like the daytime sky or a large lamp illuminating the far slit, observe it through the nearby aperture. After inserting the near slit the far slit will appear to widen and rows of fringes will be clearly evident.

10.3.9 The Semi-infinite Opaque Screen

We now form a semi-infinite planar opaque screen by removing the upper half of Σ in Fig. 10.67. This is done simply enough by letting $z_2 = y_1 = y_2 = \infty$. Remembering the original approximations, we limit the geometry so that the point of observation is close to the screen's edge. Since $v_2 = u_2 = \infty$ and $u_1 = -\infty$, Eq. (10.104) or (10.108) leads to

$$I_p = \frac{I_0}{2}\{[\tfrac{1}{2} - \mathscr{C}(v_1)]^2 + [\tfrac{1}{2} - \mathscr{S}(v_1)]^2\}. \quad (10.109)$$

When the point P is directly opposite the edge $v_1 = 0$, $\mathscr{C}(0) = \mathscr{S}(0) = 0$ and $I_p = I_0/4$. This was to be expected, since half of the wavefront is obstructed, the amplitude of the disturbance is halved and the irradiance drops to one quarter. This occurs at point (3) in Figs. 10.71 and 10.72. Moving into the geometric shadow region to point (2) and then on to (1) and still further, the successive chords clearly decrease monotonically (Problem 10.15). No irradiance oscillations exist within that region; the irradiance merely drops off rapidly. At any point above (3) the screen's edge will be below it, i.e. $z_1 < 0$ and $v_1 < 0$. At about $v_1 = -1.2$ the chord reaches a maximum and the irradiance is a maximum. Thereafter I_p oscillates about I_0, gradually diminishing in magnitude. With sensitive electronic techniques many hundreds of these fringes can be observed.*

It is evident that the diffraction pattern of Fig. 10.73 would appear in the vicinity of the edges of a wide *slit* (Δv greater than about 10) as a limiting case. The irradiance distribution suggested by geometrical optics is obtained only when λ goes to zero. Indeed as λ decreases, the fringes move closer to the edge and become increasingly finer in extent.

The straight-edge pattern can be observed using any kind of slit, held up in front of a broad lamp at arm's length, as a source. Introduce an opaque obstruction (e.g. a blackened microscope slide or a razor blade) very near your eye. As the edge of the obstruction passes in front of the source slit parallel to it, a series of fringes will appear.

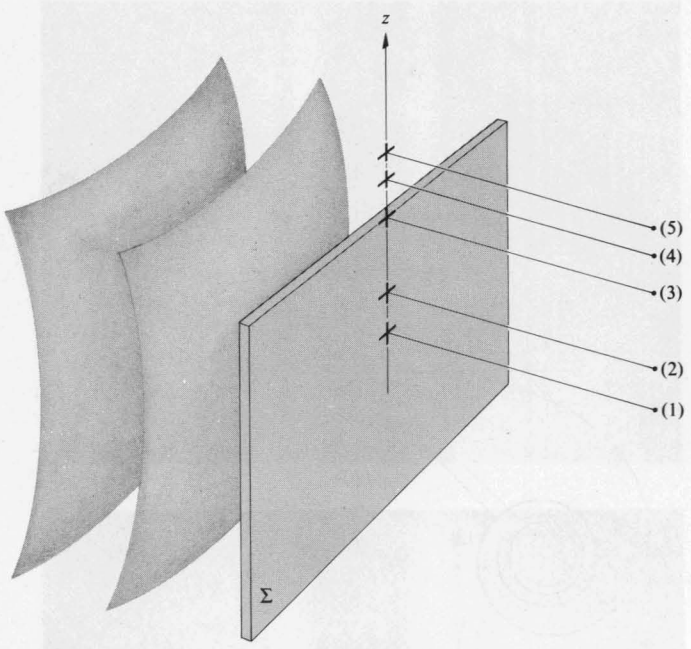


Fig. 10.71 The semi-infinite opaque screen.

10.3.10 Diffraction by a Narrow Obstacle

Refer back to the discussion describing the single narrow slit; consider the complementary case where the slit is opaque and the screen transparent. Let's envision, for example, a vertical opaque wire. At a point directly opposite the wire's center there will be two separate contributing regions extending from y_1 to $-\infty$ and from y_2 to $+\infty$. On the Cornu spiral these correspond to two arc lengths from u_1 to B^- and from u_2 to B^+. The amplitude of the disturbance at a point P on the plane of observation is the magnitude of the *vector* sum of the two phasors $\overrightarrow{B^-u_1}$ and $\overrightarrow{u_2B^+}$ illustrated in Fig. 10.74. As with the opaque disk, the symmetry is such that there will always be an illuminated region along the central axis. This can be seen from the spiral since when P is on the central axis, $\overrightarrow{B^-u_1} = \overrightarrow{u_2B^+}$ and their sum can never be zero. The arc length Δu represents the obscured region of the spiral which increases as the diameter of the wire increases. For thick wires, u_1 approaches B^-, u_2 approaches B^+, the phasors decrease in length and the irradiance on

* J. D. Barnett and F. S. Harris, Jr., *J. Opt. Soc. Amer.* **52**, 637 (1962).

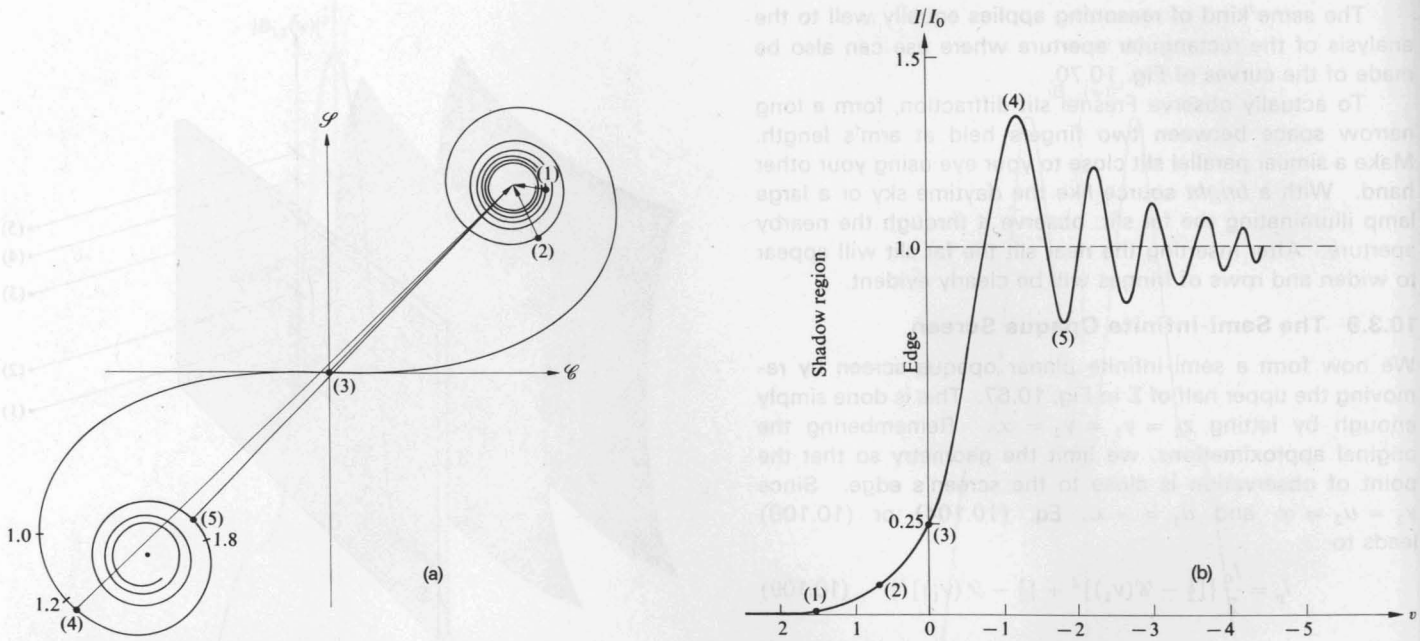


Fig. 10.72 (a) The Cornu spiral for a semi-infinite screen. (b) The corresponding irradiance distribution.

the shadow's axis drops off. This is evident in Fig. 10.75 which shows the patterns actually cast by a thin piece of lead from a mechanical pencil and by a $\frac{1}{8}$ inch diameter rod. Imagine that we have a small irradiance sensor at point P on the plane of observation (or the film plate). As P moves off the central axis to the right, y_1 and u_1 increase negatively while y_2 and u_2, which are positive, decrease. The opaque region Δu, slides down the spiral. When the sensor is at the right edge of the geometrical shadow $y_2 = 0$, $u_2 = 0$, i.e. u_2 is at O_s. Notice that if the wire is thin, i.e. if Δu is small, the sensor will record a gradual decrease in irradiance as u_2 approaches O_s. On the other hand, if the wire is thick Δu is large and u_1 and u_2 are large. As Δu slides down the spiral, the two phasors revolve through a number of complete rotations going in and out of phase in the process. The resulting additional extrema appearing within the geometrical shadow are evident in Fig. 10.75(b). In fact, the separation between internal fringes varies inversely with the width of the rod just as if the pattern arose from the interference of two waves (Young's experiment) reflected at the rod's edges.

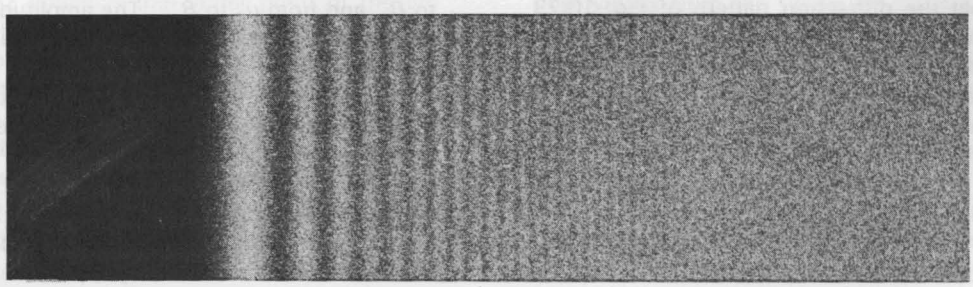


Fig. 10.73 The fringe pattern for a half-screen.

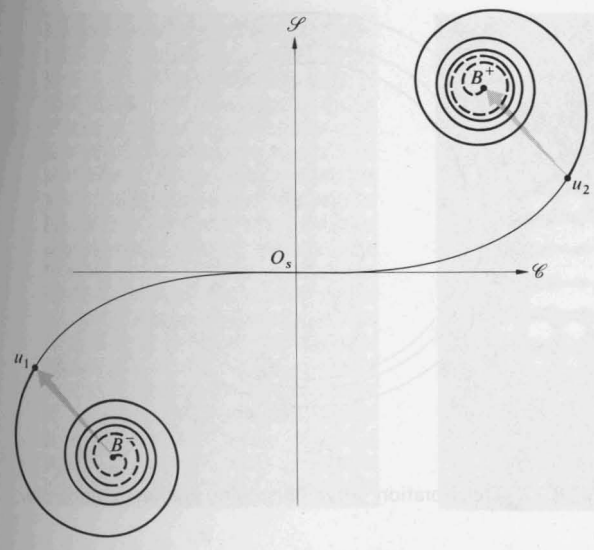


Fig. 10.74 The Cornu spiral as applied to a narrow obstacle.

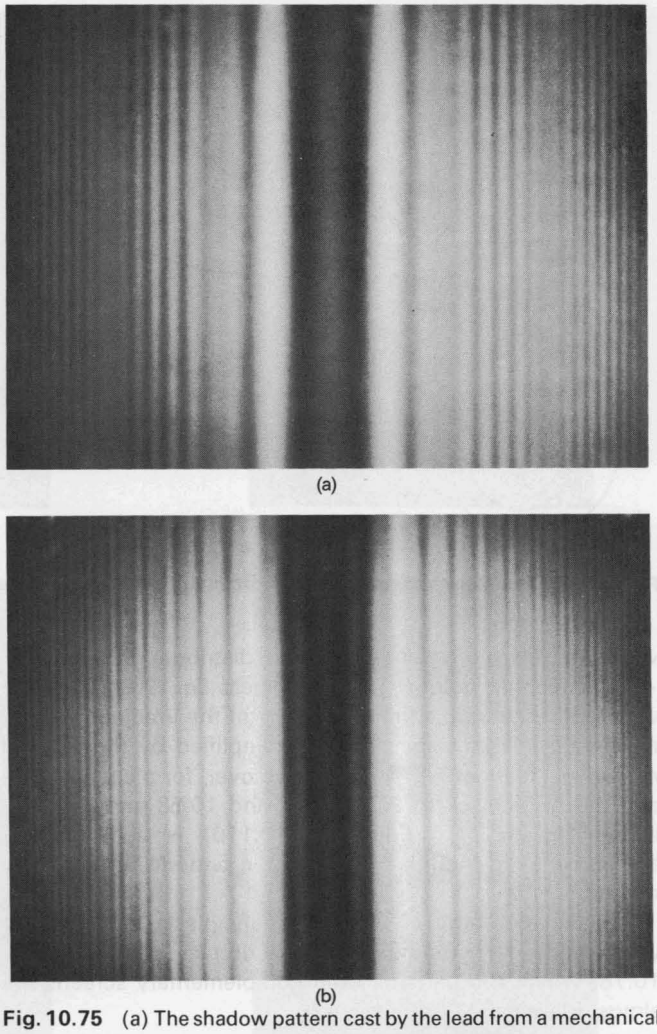
10.3.11 Babinet's Principle

Two diffracting screens are said to be *complementary* when the transparent regions on one exactly correspond to opaque regions on the other and vice versa. When two such screens are overlapped, the combination is obviously completely opaque. Now then, let E_1 or E_2 be the scalar optical disturbance arriving at P when either complementary screen Σ_1 or Σ_2, respectively, is in place. The total contribution from each aperture is determined by integrating over the area bounded by that aperture. If both *apertures* are present at once, there are no opaque regions at all; the limits of integration go to infinity and we have the unobstructed disturbance E_0, whereupon

$$E_1 + E_2 = E_0 \tag{10.110}$$

which is the statement of *Babinet's principle*. Take a close look at Figs. 10.69 and 10.74 which depict the Cornu spiral configurations for a transparent slit and a narrow opaque obstacle. If the two arrangements are made complementary, Fig. 10.76 illustrates Babinet's principle quite clearly. The phasor arising from a narrow obstacle ($\overrightarrow{B^-B_1} + \overrightarrow{B_2B^+}$) added to that from a slit $\overrightarrow{B_1B_2}$ yields the unobstructed phasor $\overrightarrow{B^-B^+}$.

The principle implies that when $E_0 = 0$, $E_1 = -E_2$, i.e. these disturbances are precisely equal in magnitude and 180° out of phase. One would, therefore, observe exactly the same irradiance distribution with either Σ_1 or Σ_2 in place; an interesting result indeed. It is evident, however, that the principle cannot be exactly true since for an unobstructed wave from a point source there are no zero-amplitude points (i.e. $E_0 \neq 0$ everywhere). Yet if the source is imaged at P_0 by perfect lenses as in Fig. 10.9 (with neither Σ_1 nor Σ_2 present), there will be a large essentially zero-amplitude region beyond the immediate vicinity of P_0 (beyond the Airy disk) in which $E_1 + E_2 = E_0 = 0$. It is therefore only for the case of Fraunhofer diffraction that complementary screens



(a)

(b)

Fig. 10.75 (a) The shadow pattern cast by the lead from a mechanical pencil. (b) The pattern cast by a $\frac{1}{8}$ inch diameter rod. [Photos by E.H.]

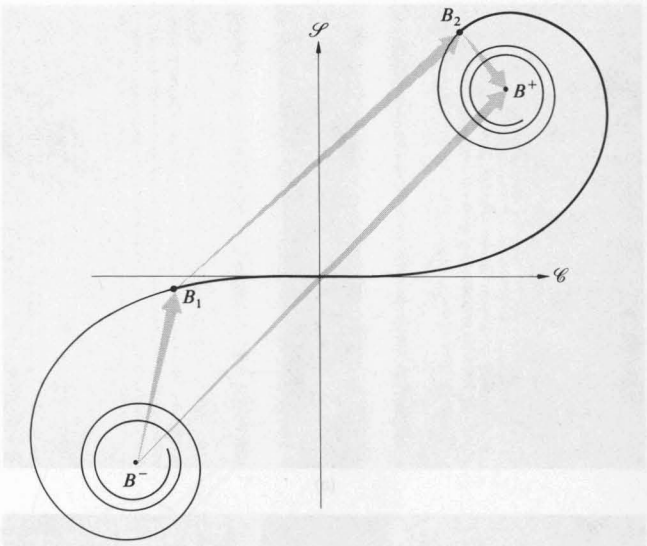


Fig. 10.76 The Cornu spiral illustrating Babinet's principle.

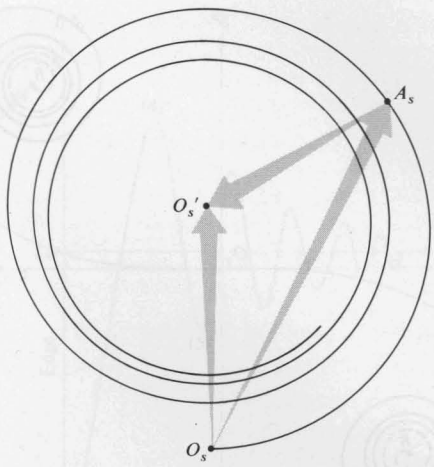


Fig. 10.77 The vibration curve illustrating Babinet's principle.

will generate equivalent irradiance distributions, i.e. $E_1 = -E_2$ (excluding point P_0). Nonetheless Eq. (10.110) is still valid in Fresnel diffraction even though the irradiances obey no simple relationship. This is exemplified by the slit and narrow obstacle of Fig. 10.76. Moreover, for a circular hole and disk refer back to Figs. 10.52 and 10.58 and then examine Fig. 10.77. Equation (10.110) is again clearly applicable even though the diffraction patterns are certainly not equivalent.

The real beauty of Babinet's principle is most evident when applied to Fraunhofer diffraction as shown in Fig. 10.78, where the patterns from complementary screens are almost identical.

10.4 KIRCHHOFF'S SCALAR DIFFRACTION THEORY

We have described a number of diffracting configurations, quite satisfactorily, within the context of the relatively simple Huygens–Fresnel theory. Yet the whole imagery of surfaces covered with fictitious point sources, which was the basis of that analysis, was merely postulated rather than derived from fundamental principles. The Kirchhoff treatment shows that these results are actually derivable from the *scalar* differential wave equation.

The discussion to follow is rather formal and involved. As such, portions of it have been relegated to an appendix where we can indulge in succinctness and risk sacrificing easy readability for rigor.

In the past, when dealing with a distribution of monochromatic point sources, we computed the resultant optical disturbance at point P, i.e. E_p, by carrying out a superposition of the individual waves. There is, however, another completely different approach which is founded in potential theory. Here one is concerned, not with the sources themselves, but rather with the scalar optical disturbance and its derivatives over an arbitrary closed surface surrounding P. We assume that a Fourier analysis can separate the constituent frequencies so that we need only deal with one such frequency at a time. The monochromatic optical disturbance E is a solution of the differential wave equation

$$\nabla^2 E = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}. \tag{10.111}$$

Without specifying the precise spatial nature of the wave it can be written as

$$E = \mathscr{E} e^{-ikct}. \tag{10.112}$$

Here $\mathscr{E}$ represents the complex space part of the disturbance. Substituting into the wave equation it then becomes

$$\nabla^2 \mathscr{E} + k^2 \mathscr{E} = 0. \tag{10.113}$$

This is known as the *Helmholtz equation* and is solved, with the aid of Green's theorem, in Appendix 2. The optical disturbance existing at a point P, expressed in terms of the optical disturbance and its gradient evaluated on an arbitrary

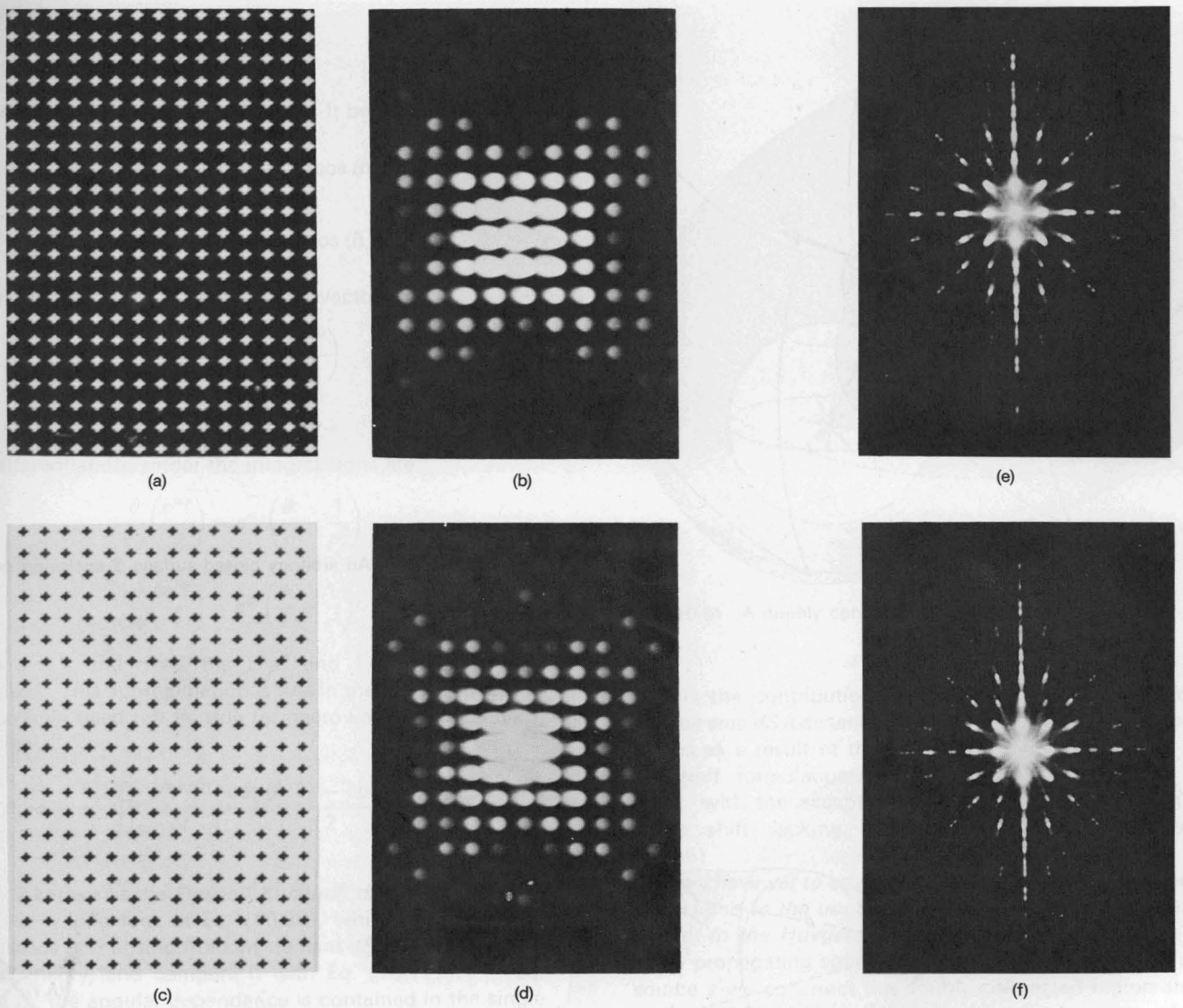


Fig. 10.78 (a)–(d) White light diffraction patterns for regular arrays of apertures and complementary obstacles in the form of rounded plus signs. (e) and (f) Diffraction patterns for a regular array of rectangular apertures and obstacles, respectively. [Photos courtesy The Ealing Corporation and Richard B. Hoover.]

closed surface S, enclosing P, is

$$\mathscr{E}_p = \frac{1}{4\pi}\left[\iint_S \frac{e^{ikr}}{r}\nabla\mathscr{E}\cdot d\mathbf{S} - \iint_S \mathscr{E}\nabla\left(\frac{e^{ikr}}{r}\right)\cdot d\mathbf{S}\right] \quad (10.114)$$

Known as the *Kirchhoff integral theorem*, Eq. (10.114) relates to the geometric configuration illustrated in Fig. 10.79.

We now apply the theorem to the specific instance of an unobstructed spherical wave originating at a point source s, as shown in Fig. 10.80. The disturbance has the form

$$E(\rho, t) = \frac{\mathscr{E}_0}{\rho} e^{i(k\rho - \omega t)}, \quad (10.115)$$

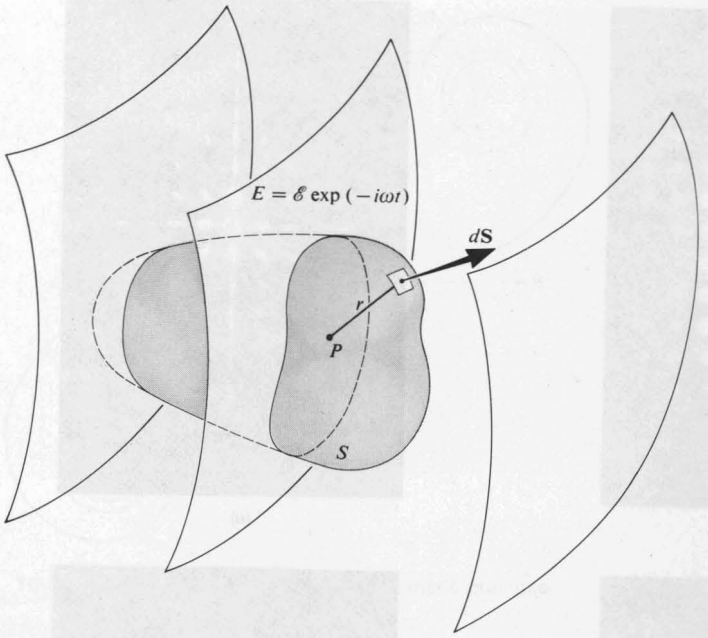
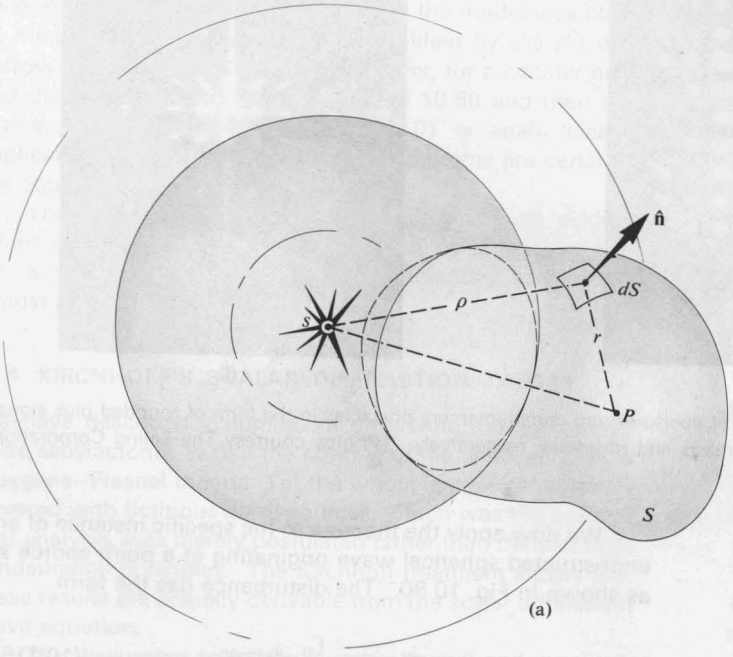


Fig. 10.79 An arbitrary closed surface S enclosing point P.



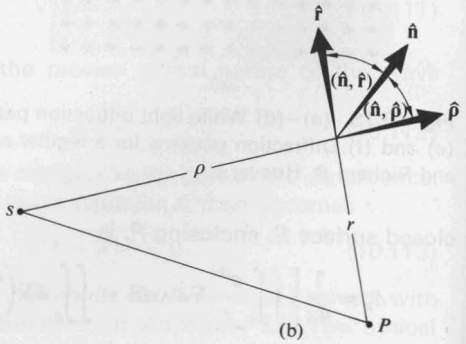


Fig. 10.80 A spherical wave emitted from point s.

in which case

$$\mathscr{E}(\boldsymbol{\rho}) = \frac{\mathcal{E}_0}{\rho} e^{ik\rho}. \tag{10.116}$$

If we substitute this into Eq. (10.114) it becomes

$$\mathscr{E}_p = \frac{1}{4\pi}\left[\iint_S \frac{e^{ikr}}{r}\frac{\partial}{\partial\rho}\left(\frac{\mathcal{E}_0}{\rho}e^{ik\rho}\right)\cos(\hat{\mathbf{n}},\hat{\boldsymbol{\rho}})\,dS - \iint_S \frac{\mathcal{E}_0}{\rho}e^{-ik\rho}\frac{\partial}{\partial r}\left(\frac{e^{ikr}}{r}\right)\cos(\hat{\mathbf{n}},\hat{\mathbf{r}})\,dS\right]$$

where $d\mathbf{S} = \hat{\mathbf{n}}\,dS$, $\hat{\mathbf{n}}$, $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\rho}}$ are unit vectors,

$$\nabla\left(\frac{e^{ikr}}{r}\right) = \hat{\mathbf{r}}\frac{\partial}{\partial r}\left(\frac{e^{ikr}}{r}\right)$$

and

$$\nabla\mathscr{E}(\boldsymbol{\rho}) = \hat{\boldsymbol{\rho}}\,\partial\mathscr{E}/\partial\rho.$$

The differentiations under the integral signs are

$$\frac{\partial}{\partial\rho}\left(\frac{e^{ik\rho}}{\rho}\right) = e^{ik\rho}\left(\frac{ik}{\rho} - \frac{1}{\rho^2}\right)$$

and

$$\frac{\partial}{\partial r}\left(\frac{e^{ikr}}{r}\right) = e^{ikr}\left(\frac{ik}{r} - \frac{1}{r^2}\right).$$

When $\rho \gg \lambda$ and $r \gg \lambda$ the $1/\rho^2$ and $1/r^2$ terms can be neglected. This approximation is fine in the optical spectrum but certainly need not be true for microwaves. Proceeding, we write

$$\mathscr{E}_p = -\frac{\mathcal{E}_0 i}{\lambda}\iint_S \frac{e^{ik(\rho+r)}}{\rho r}\left[\frac{\cos(\hat{\mathbf{n}},\hat{\mathbf{r}}) - \cos(\hat{\mathbf{n}},\hat{\boldsymbol{\rho}})}{2}\right]dS, \tag{10.117}$$

which is known as the *Fresnel–Kirchhoff diffraction formula*.

Take a long look at Eq. (10.96), which represents the disturbance at P arising from an element dS in the Huygens–Fresnel theory, and compare it with Eq. (10.117). In Eq. (10.117), the angular dependence is contained in the single term $\frac{1}{2}[\cos(\hat{\mathbf{n}},\hat{\mathbf{r}}) - \cos(\hat{\mathbf{n}},\hat{\boldsymbol{\rho}})]$ which we shall call the obliquity factor $K(\theta)$, showing it to be equivalent to Eq. (10.72) later on. Notice as well that k can be replaced by $-k$ everywhere, since we certainly could have chosen the phase of Eq. (10.115) to have been $(\omega t - k\rho)$. Now, multiply both sides of Eq. (10.117) by $\exp(-i\omega t)$; the differential element is then

$$dE_p = \frac{K(\theta)\,\mathcal{E}_0}{\rho r\lambda}\cos[k(\rho + r) - \omega t - \pi/2]\,dS. \tag{10.118}$$

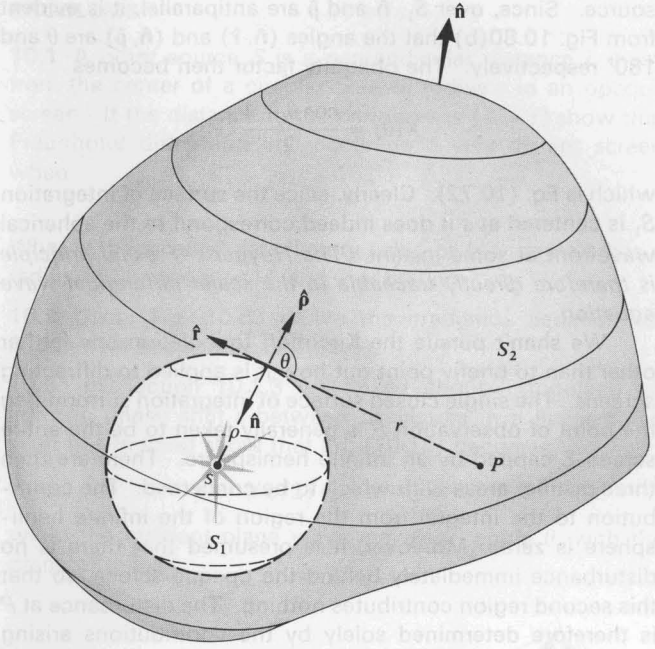


Fig. 10.81 A doubly connected region surrounding point s.

This is the contribution to E_p arising from an element of surface area dS a distance r from P. The $\pi/2$ term in the phase comes as a result of the fact that $-i = \exp(-i\pi/2)$. The Kirchhoff formulation therefore leads to the same total result, with the exception that it includes the correct $\pi/2$ phase shift, lacking in the Huygens–Fresnel treatment (10.96).

We have yet to ensure that the surface S can be made to correspond to the unobstructed portion of the wavefront as it does in the Huygens–Fresnel theory. For the case of a freely propagating spherical wave emanating from the point source s we construct the doubly connected region shown in Fig. 10.81. The surface S_2 completely surrounds the small spherical surface S_1. At $\rho = 0$ the disturbance $E(\rho, t)$ has a singularity and is therefore properly excluded from the volume V between S_1 and S_2. The integral must now include both surfaces S_1 and S_2. But we can have S_2 increase outward indefinitely by requiring its radius to go to infinity. In that case, the contribution to the surface integral vanishes. (This is true whatever the form of the incoming disturbance as long as it drops off at least as rapidly as a spherical wave.) The remaining surface S_1 is a sphere centered at the point

source. Since, over S_1, $\hat{\mathbf{n}}$ and $\hat{\boldsymbol{\rho}}$ are antiparallel, it is evident from Fig. 10.80(b) that the angles $(\hat{\mathbf{n}}, \hat{\mathbf{r}})$ and $(\hat{\mathbf{n}}, \hat{\boldsymbol{\rho}})$ are θ and 180° respectively. The obliquity factor then becomes

$$K(\theta) = \frac{\cos\theta + 1}{2},$$

which is Eq. (10.72). Clearly, since the surface of integration S_1 is centered at s it does indeed correspond to the spherical wavefront at some instant. *The Huygens–Fresnel principle is therefore directly traceable to the scalar differential wave equation.*

We shan't pursue the Kirchhoff formulation any farther other than to briefly point out how it is applied to diffracting screens. The single closed surface of integration surrounding the point of observation P is generally taken to be the entire screen Σ capped by an infinite hemisphere. There are then three distinct areas with which to be concerned. The contribution to the integral from the region of the infinite hemisphere is zero. Moreover, it is presumed that there is no disturbance immediately behind the opaque screen, so that this second region contributes nothing. The disturbance at P is therefore determined solely by the contributions arising from the aperture and one need only integrate Eq. (10.117) over that area.

The very fine results obtained by using the Huygens–Fresnel principle are now quite justified theoretically, the main limitations being that $\rho \gg \lambda$ and $r \gg \lambda$.

10.5 BOUNDARY DIFFRACTION WAVES

In Section 10.1.1 we took the view that the diffracted wave could be envisioned as arising from a fictitious distribution of secondary emitters spread across the unobstructed portion of the wavefront, i.e. the Huygens–Fresnel principle. There is, however, another completely different and rather appealing possibility. Suppose that an incoming wave sets the electrons on the rear of the diffracting screen Σ into oscillation and these in turn radiate. We anticipate a twofold effect. First, all of the oscillators which are remote from the edge of the aperture radiate back toward the source in such a fashion as to cancel the incoming wave at all points, except within the projection of the aperture itself. In other words, if this were the only contributing mechanism, a perfect geometrical image of the aperture would appear on the plane of observation. There is, however, an additional contribution arising from those oscillators in the vicinity of the aperture's edge. A portion of the energy radiated by these secondary sources propagates in the forward direction. The superposition of this scattered wave (known as the *boundary diffraction wave*) and the unobstructed portion of the primary wave (known as the *geometrical wave*) yields the diffraction pattern. A rather cogent reason for contemplating such a scheme appears when one examines the following arrangement. Tear a small hole ($\approx \frac{1}{2}$ cm dia.) of arbitrary shape in a piece of paper and, holding it at arm's length, view an ordinary light bulb some meters distant. Even with your eye in the shadow region, the edges of the aperture will be brightly illuminated. The ripple-tank photograph of Fig. 10.82 also illustrates the process. Notice how each edge of the slit seems to serve as a center for a circular disturbance which then propagates beyond the aperture. There are no electron-oscillators here, which rather implies that these ideas have a certain generality, being applicable to elastic waves as well.

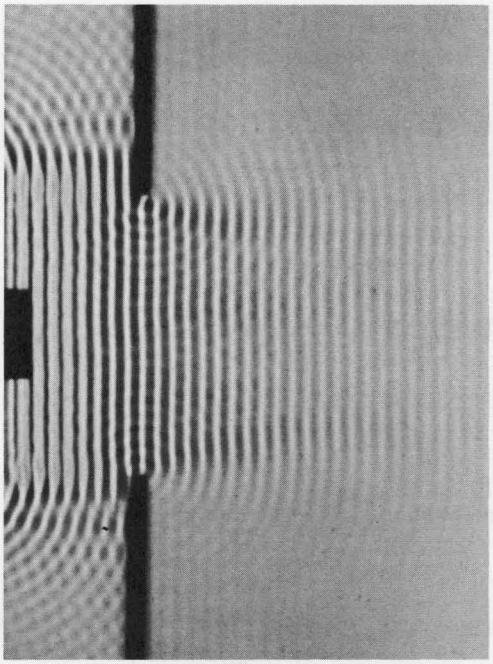


Fig. 10.82 Ripple tank waves passing through a slit. [Photo courtesy PSSC *Physics*, D. C. Heath, Boston 1960.]

The formulation of diffraction in terms of the interference of a scattered edge wave and a geometrical wave is perhaps more physically appealing than the fictitious emitters of the Huygens–Fresnel principle. It is not, however,

a new concept. Indeed it was first propounded by the ubiquitous Thomas Young even before Fresnel's celebrated memoir on diffraction. But in time Fresnel's brilliant successes unfortunately convinced Young to reject his own ideas, and he finally did so in a letter to Fresnel in 1818. Strengthened by Kirchhoff's work, the Fresnel conception of diffraction became generally accepted and has persisted (right up to Section 10.3). The very beginnings of a resurrection of Young's theory came in 1888. At that time Gian Antonio Maggi proved that Kirchhoff's analysis, for a point source at least, was equivalent to two contributing terms. One of these was a geometrical wave but the other, unhappily, was an integral which allowed no clear physical interpretation at the time. In his doctoral thesis (1893) Eugen Maey showed that an edge wave could indeed be extracted from a modified Kirchhoff formulation for a semi-infinite half-plane. Arnold Sommerfeld's rigorous solution of the half-plane problem (see Section 10.1) showed that a cylindrical wave actually does proceed from the screen's edge. It propagates into both the geometrical shadow region and the illuminated region. In the latter, the boundary diffraction wave combines with the geometrical wave, in complete accord with Young's theory. Adalbert (Wojciech) Rubinowicz in 1917, was able to prove that Kirchhoff's formula for a plane or spherical wave can be appropriately decomposed into the two desired waves, thereby revealing the basic correctness of Young's ideas. He also later established that the boundary diffraction wave, to a first approximation, was generated by reflection of the primary wave from the aperture's edge. In 1923, Friedrich Kottler pointed out the equivalence of the solutions of Maggi and Rubinowicz and one now speaks of the Young–Maggi–Rubinowicz theory. Most recently, Kenro Miyamoto and Emil Wolf (1962) have extended the boundary-diffraction theory to the case of arbitrary incident waves.* A very useful contemporary approach to the problem has been devised by Joseph B. Keller. He has developed a geometrical theory of diffraction which is closely related to Young's edge wave picture. Along with the usual rays of geometrical optics, he hypothesizes the existence of diffracted rays. Rules governing these diffracted rays, which are analogous to the laws of reflection and refraction, are employed to determine the resultant fields.

* A fairly complete bibliography can be found in the article by A. Rubinowicz in *Progress in Optics*, Vol. 4, p. 199.

PROBLEMS

10.1 A point source S is a perpendicular distance L away from the center of a circular hole of radius r in an opaque screen. If the distance to the periphery is $(L + \ell)$ show that Fraunhofer diffraction will occur on a very distant screen when

$$\lambda L \gg r^2/2.$$

What is the smallest satisfactory value of L if the hole has a radius of 1 mm, $\ell \leq \lambda/10$ and $\lambda = 500$ nm?

10.2 Using Fig. 10.83 derive the irradiance equation for N coherent oscillators, Eq. (10.5).

10.3* In Section 10.1.3 we talked about introducing an intrinsic phase shift ε between oscillators in a linear array. With this in mind show that Eq. (10.18) becomes

$$\beta = (kb/2)(\sin\theta - \sin\theta_i)$$

when the incident plane wave makes an angle θ_i with the plane of the slit.

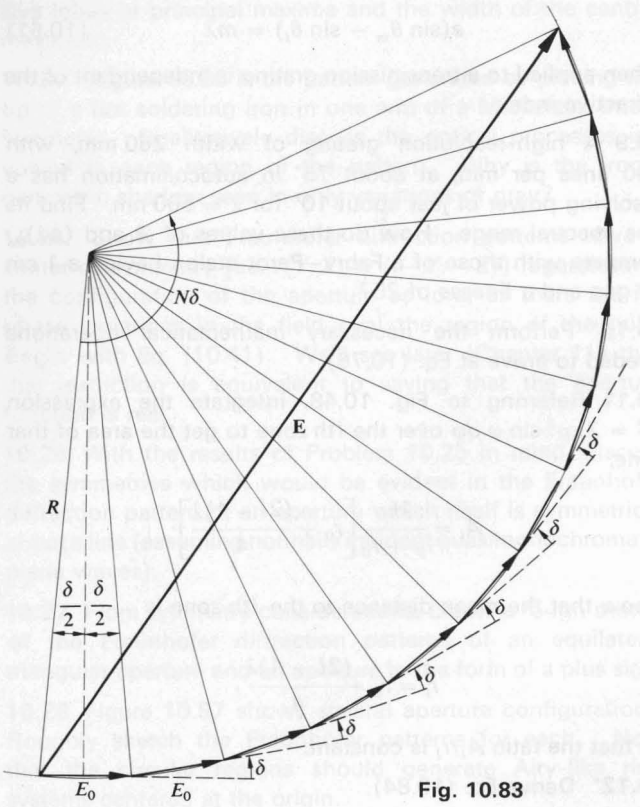


Fig. 10.83

10.4 The angular distance between the center and the first minimum of a single-slit Fraunhofer diffraction pattern is called the *half-angular breadth*; write an expression for it. Find the corresponding *half-linear width* when (a) no focusing lens is present and the slit-viewing screen distance is L, (b) when a lens of focal length f_2 is very close to the aperture. Notice that the half-linear width is also the distance between the successive minima.

10.5 Show that for a double-slit Fraunhofer pattern, if $a = mb$, the number of bright fringes (or parts thereof) within the central diffraction maximum will be equal to $2m$.

10.6 What is the relative irradiance of the subsidiary maxima in a three slit Fraunhofer diffraction pattern? Draw a graph of the irradiance distribution, when $a = 2b$, for two and then three slits.

10.7* Starting with the irradiance expression for a finite slit, shrink the slit down to a minuscule area element and show that it emits equally in all directions.

10.8 Prove that the equation

$$a(\sin\theta_m - \sin\theta_i) = m\lambda \qquad [10.61]$$

when applied to a transmission grating is independent of the refractive index.

10.9 A high-resolution grating of width 260 mm, with 300 lines per mm, at about 75° in autocollimation has a resolving power of just about 10^6 for $\lambda = 500$ nm. Find its free spectral range. How do these values of $\mathscr{R}$ and $(\Delta\lambda)_{\text{fsr}}$ compare with those of a Fabry–Perot etalon having a 1 cm air gap and a finesse of 25?

10.10* Perform the necessary mathematical operations needed to arrive at Eq. (10.76).

10.11 Referring to Fig. 10.48, integrate the expression $dS = 2\pi\rho^2 \sin\varphi\, d\varphi$ over the lth zone to get the area of that zone,

$$A_l = \frac{\lambda\pi\rho}{\rho + r_0}\left[r_0 + \frac{(2l-1)\lambda}{4}\right].$$

Show that the mean distance to the lth zone is

$$r_l = r_0 + \frac{(2l-1)\lambda}{4},$$

so that the ratio A_l/r_l is constant.

10.12* Derive Eq. (10.84).

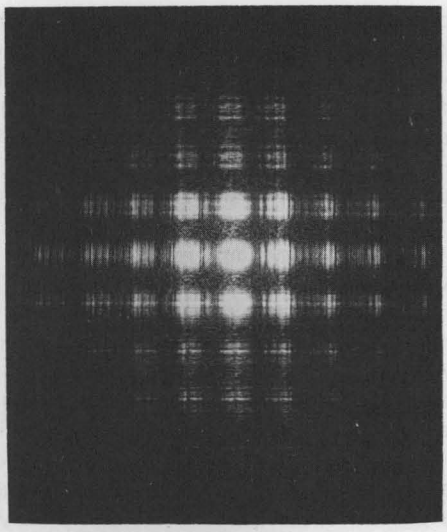


Fig. 10.84

10.13 Use the Cornu spiral to make a rough sketch of $|\mathbf{B}_{12}(w)|^2$ versus $(w_1 + w_2)/2$ for $\Delta w = 5.5$. Compare your results to those of Fig. 10.70.

10.14 The neo-impressionist painter Georges Seurat was a member of the pointillist school. His paintings consisted of an enormous number of closely spaced small dots ($\approx \frac{1}{10}$ inch) of pure pigment. The illusion of color mixing was produced only in the eye of the observer. How far from such a painting should one stand in order to achieve the desired blending of color?

10.15 The Fresnel integrals have the asymptotic forms (corresponding to large values of w) given by

$$\mathscr{C}(w) \approx \tfrac{1}{2} + \left(\frac{1}{\pi w}\right)\sin\left(\frac{\pi w^2}{2}\right),$$

$$\mathscr{S}(w) \approx \tfrac{1}{2} - \left(\frac{1}{\pi w}\right)\cos\left(\frac{\pi w^2}{2}\right).$$

Using this fact, show that the irradiance in the shadow of a semi-infinite opaque screen decreases in proportion to the inverse square of the distance to the edge, as z_1 and therefore v_1 become large.

10.16* Suppose then that we have a laser emitting a diffraction-limited beam ($\lambda_0 = 632.84$ nm) having a 2 mm diameter. How big a light spot would be produced on the surface of the moon a distance of 376×10^3 km away from such a device? Neglect any effects of the earth's atmosphere.

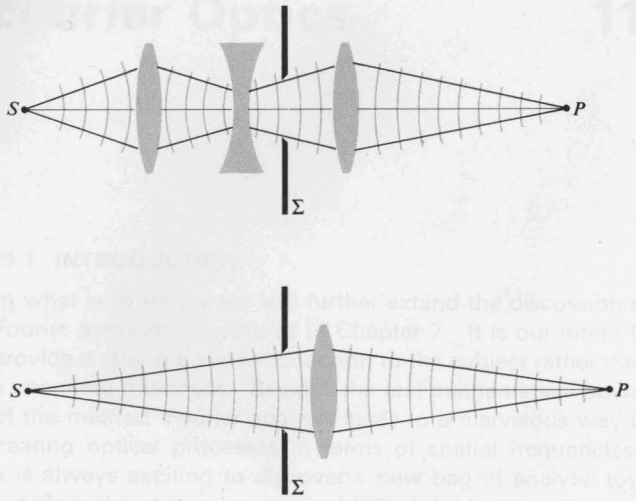


Fig. 10.85

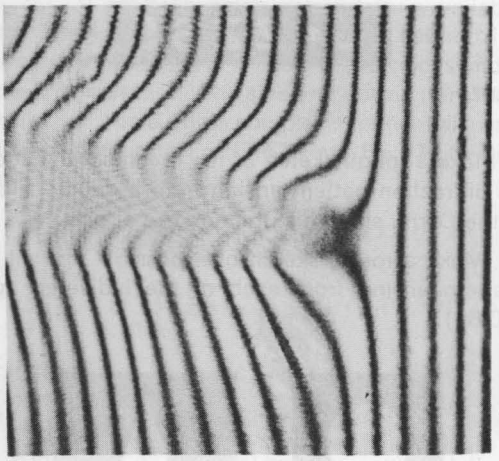


Fig. 10.86

10.17 What would you expect to see on the plane of observation if the half-plane Σ in Fig. 10.71 were semitransparent?

10.18 Imagine that you are looking through a piece of square woven cloth at a point source ($\lambda_0 = 600$ nm) 20 meters away. If you see a square arrangement of bright spots located about the point source (Fig. 10.84), each separated by an apparent nearest-neighbor distance of 12 cm, how close together are the strands of cloth?

10.19 Examine the set-up of Fig. 10.5 in order to determine what is happening in the image space of the lenses, i.e. locate the exit pupil and relate it to the diffraction process. Show that the configurations in Fig. 10.85 are equivalent to that of Fig. 10.5 and will therefore result in Fraunhofer diffraction. Design at least one more such arrangement.

10.20* Imagine an opaque screen containing thirty randomly located circular holes. The light source is such that every aperture is coherently illuminated by its own plane wave. Each wave in turn is completely incoherent with respect to all the others. Describe the resulting far-field diffraction pattern.

10.21 What should be the total number of lines a grating must have in order to just separate the sodium doublet ($\lambda_1 = 5895.9$ Å, $\lambda_2 = 5890.0$ Å) in the third order?

10.22 Plane waves from a collimated He–Ne laser beam ($\lambda_0 = 632.8$ nm) impinge on a 2.5 mm diameter steel rod. Draw a rough *graphical* representation of the diffraction pattern which would be seen on a screen 3.16 m from the rod.

10.23 Referring back to the multiple antenna system of Fig. 10.7, compute the angular separation between successive lobes or principal maxima and the width of the central maximum.

10.24* Figure 10.86 is the pattern generated by placing the tip of a hot soldering iron in one arm of a Michelson interferometer. Qualitatively discuss the optical processes involved in each region of the pattern. Why is the iron's geometric shadow seen in only variations of gray?

10.25* Show that Fraunhofer diffraction patterns have a center of symmetry [i.e. $I(Y, Z) = I(-Y, -Z)$] regardless of the configuration of the aperture as long as there are no phase variations in the field over the region of the hole. Begin with Eq. (10.41). We'll see later (Chapter 11) that this restriction is equivalent to saying that the aperture function is real.

10.26 With the results of Problem 10.25 in mind, discuss the symmetries which would be evident in the Fraunhofer diffraction pattern of an aperture which itself is symmetrical about a line (assuming normally incident quasimonochromatic plane waves).

10.27 From symmetry considerations, create a rough sketch of the Fraunhofer diffraction patterns of an equilateral triangular aperture and an aperture in the form of a plus sign.

10.28 Figure 10.87 shows several aperture configurations. Roughly sketch the Fraunhofer patterns for each. Note that the circular regions should generate Airy-like ring systems centered at the origin.

Fig. 10.87

10.29 Make a rough sketch of the irradiance function for a Fresnel diffraction pattern arising from a double slit. What would the Cornu spiral picture look like at point P_0?

10.30* Make a rough sketch of a possible Fresnel diffraction pattern arising from each of the indicated apertures (Fig. 10.88).

Fig. 10.88

10.31* Suppose the slit in Fig. 10.67 is made very wide. What would the Fresnel diffraction pattern look like?